

Report prepared for the WHO Consultation on the Composition of Influenza Virus Vaccines for the Southern Hemisphere 2025

Worldwide Influenza Centre, WHO CC for Reference and Research on Influenza
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Vaccine Composition Meeting: Melbourne, 23th - 26th September 2024



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Laboratories involved in Global Influenza Surveillance

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Additional Personnel from the Francis Crick Institute

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Influenza Epidemiology in the WHO European Region: The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and accessed through the European Respiratory Virus Surveillance Summary ([ERVISS](#)).

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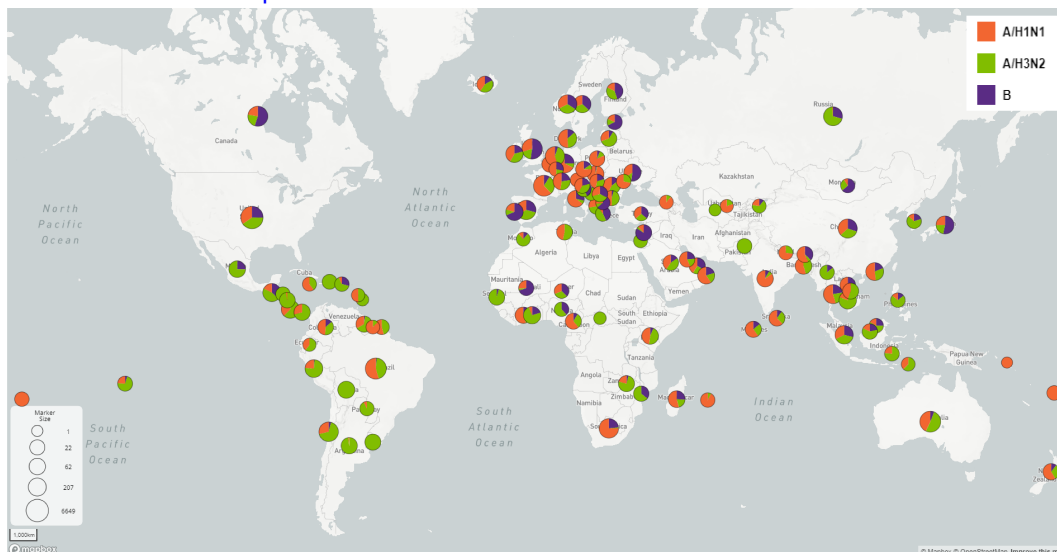
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Global influenza activity

Influenza by type/subtype

Global map for influenza viruses with collection dates from 1st February 2024 through to 31st August 2024 as deposited in GISAID, coloured by Type/subtype and downloaded on 12th September 2024. Map prepared with [Microreact](#). Geographic markers scaled to detection proportions. Full length HA. Overview of influenza activity summarised from [GISRS Influenza updates](#)



As of September 2024, in the Northern hemisphere influenza activity in temperate countries remained at interepidemic levels. Activity continued to increase in a few countries in Middle Africa.

In the Southern hemisphere, influenza activity remained elevated in some countries in South America, Eastern Africa, Southern Africa, and Oceania. Activity declined or was similar compared with the prior report across the Southern hemisphere.

The relative proportions of A/H1N1, A/H3N2 and B/Victoria varied by geographic region, with predominance of A/H3N2 in the Americas, Africa, South East Asia, Western Pacific region and Russian Federation, and some predominance of A/H1N1 in Europe, Middle East, South Africa, Madagascar, India and Brazil. Some countries showed predominance of B/Victoria such as Canada, UK, Portugal, Syria, Ukraine, Mali, Japan and Mongolia, as indicated by the different colours in the pie charts by country.

Table of shipments to the WIC

Country	Number of samples received	Influenza Type		Subtype/lineage			Number of shipments
		A	B	A/H1N1	A/H3N2	B/Victoria	
Albania	110	1	37	41	31		1
Argentina	39			2	37		1
Belgium	18			9	9		1
Bosnia and Herzegovina	14				13	1	2
Bulgaria	78			46	25	7	1
Burkina Faso	50			2	47	1	1
Cameroon	40		10	15	15		1
Central African Republic	26		7	9	10		2
Croatia	15			5	5	5	1
Cyprus	30			28		2	1
Denmark	10			5	5		1
Estonia	59			17	13	29	1
Egypt	10		4	6			1
France	91			32	41	18	3
Germany	40			17	12	11	3
Ghana	33	1		2	14	16	2
Greece	16			11	3	2	2
Hong Kong	30			10	10	10	1
Hungary	10	1		5	2	2	1
Iceland	50			20	20	10	1
Ireland	40		8	13	17	2	1
Israel	61			2	59		1
Italy	32			19	3	10	1
Kyrgyzstan	40			2	37	1	2
Lebanon	121	1		8	6	106	2
Lithuania	30			6	19	5	1
Luxembourg	17	2	7	5	3		1
Macedonia	20			15	4	1	1
Madagascar	71	5		38	7	21	1
Mauritius	40			35	5		1
Moldova	12		1	7	5		1
Morocco	20		1	8	11		1
Mozambique	30			2	18	10	1
Netherlands	30			23	7		2
Norway	8			1	7		2
Oman	19			7	4	8	1
Palestine	25			1	24		1
Poland	66	5	3	53	5		2
Portugal	22			4	8	10	1
Romania	40			21	19		1
Russia	16			2	9	5	1
Saudi Arabia	20			7	8	5	1
Senegal	30				28	2	1
Serbia	60		5	23	29	3	2
Slovakia	13		1	11	1		1
Slovenia	44			25	14	5	2
South Africa	35		11	23	1		1
South Sudan	79		23	14	41		2
Spain	133	1	8	51	42	31	6
Sweden	36			11	12	13	1
Switzerland	21			8	7	6	1

(continued)

Country	Number of samples received	Influenza Type		Subtype/lineage			Number of shipments
		A	B	A/H1N1	A/H3N2	B/Victoria	
Tajikistan	15	13		1	1		1
Tunisia	90			44	46		1
Turkey	12		4	4	4		1
Turkmenistan	6				5	1	1
Ukraine	60	12	7	11	2	28	2
United Kingdom	34	2		5	18	9	3
Yemen	5		1	4			1
TOTALS	2222	44	138	796	848	396	82

Epidemiology

WHO European Region Summary, Week 36/2024 (2–8 September 2024)

Overview

No countries or areas of the WHO European Region reporting data, reported rates of influenza-like illness (ILI) and/or acute respiratory infection (ARI) elevated above baseline levels on week 36/2024.

SARS-CoV2 continues to circulate in the region with a moderate percent positive and low but notable number of infections, which vary by country.

At the Regional level, hospital and ICU admissions for influenza and RSV are stable at low levels. Admissions for SARS-CoV-2, appear to have stabilized, although relatively few countries reported this week and there may be reporting delays.

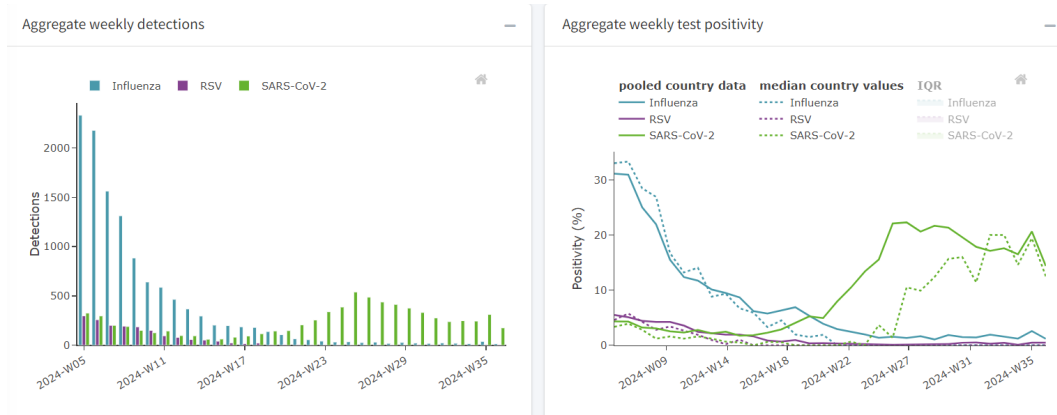
Respiratory virus activity

Influenza remained below the 10% epidemic threshold at 1% compared to 3% in the prior week. The median country positivity rate for 15 countries and areas was 0% (range: 0%–4%) and no country or area reported at least 10%. SARS-CoV-2 decreased to 14% compared to 21% in the prior week. The median positivity rate for 15 countries and areas was 10% (range: 0%–37%).

RSV remained at 0% compared to the prior week. The median positivity rate for 13 countries and areas was 0% (range: 0%–3%).

Influenza virus detections with known type reported from sentinel primary care for the past week (n=13) were mainly influenza type A viruses (n = 9) and, among the A viruses subtyped (n=6), there was a mix of A(H3) (n = 5) and A(H1)pdm09 (n = 1).

Aggregate weekly sentinel detections and test positivity from week 5 (1 February) to week 36



Severity of infection

Across the Region, the percentage of all specimens from patients presenting to sentinel SARI sites that tested positive for:

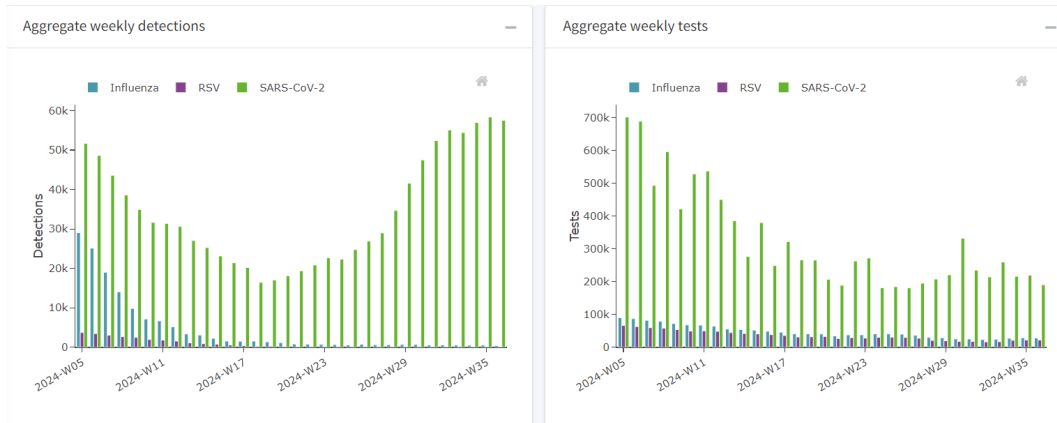
Influenza virus was 1% compared to 3% in the prior week (range: 0%-16%). Among the SARI specimens, there were fewer than 10 influenza virus detections reported from the network this week, all of which were type A (n = 8).

SARS-CoV-2 was 10% compared to 16% in the prior week (range = 0%-31%).

RSV was 0% compared to 0% in the prior week (range = 0%-1%).

COVID-19 death rates remained low among the 41 countries reporting data.

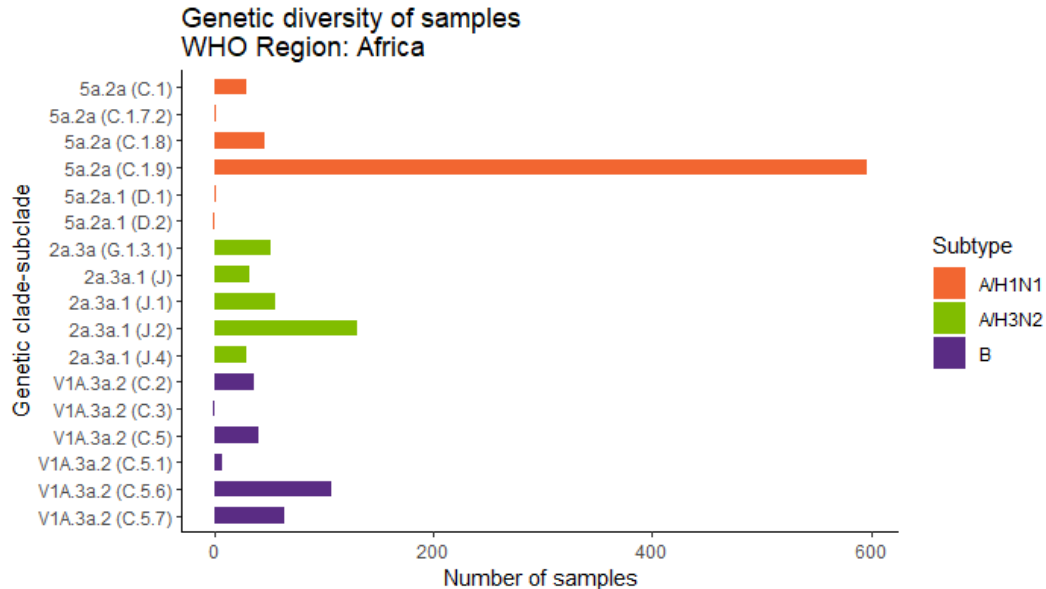
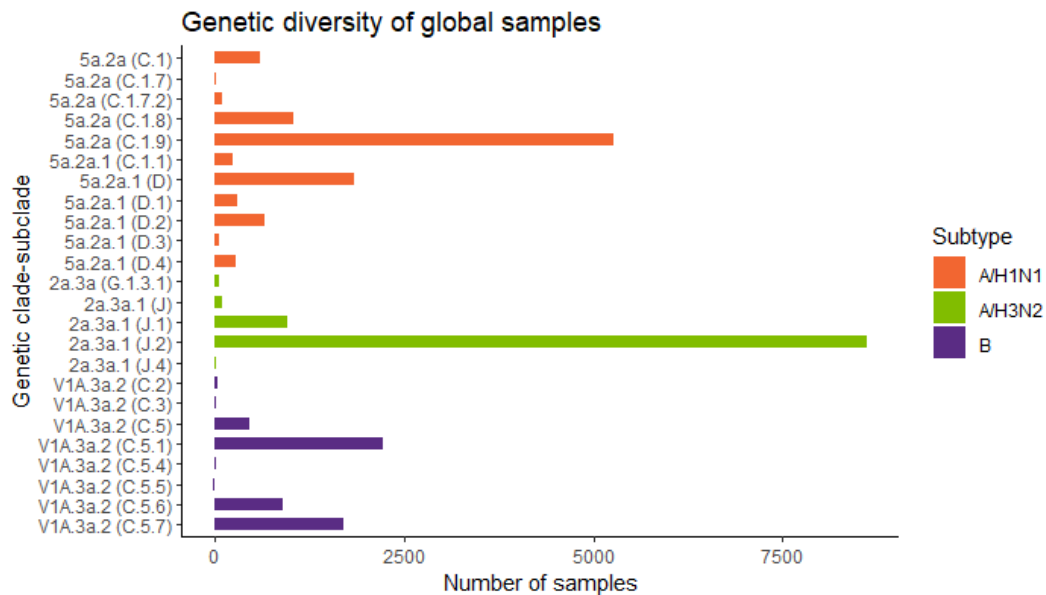
Aggregate weekly non-sentinel detections and tests from week 5 (1 February) to week 36

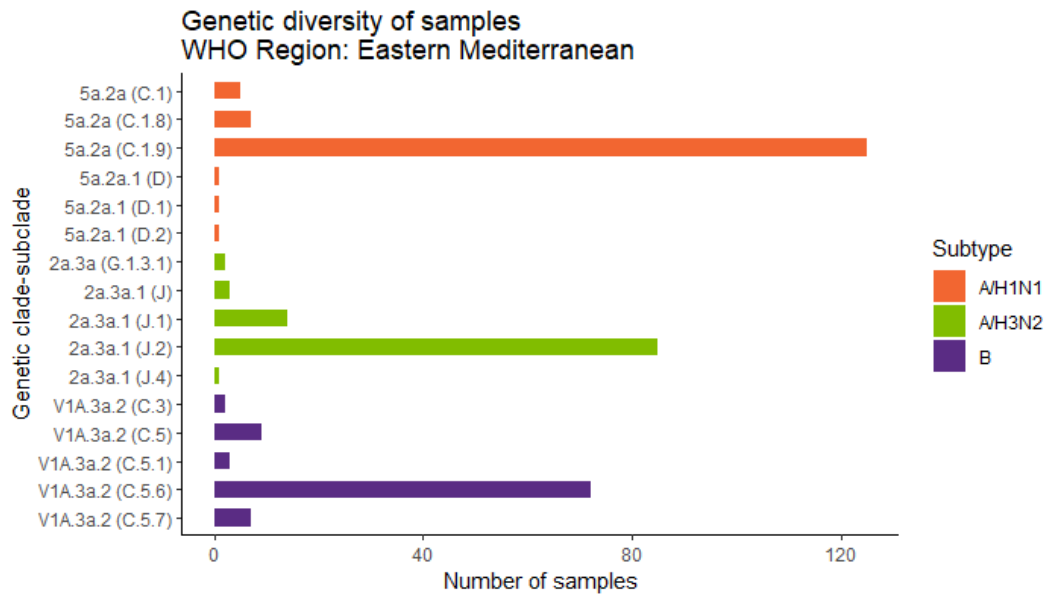
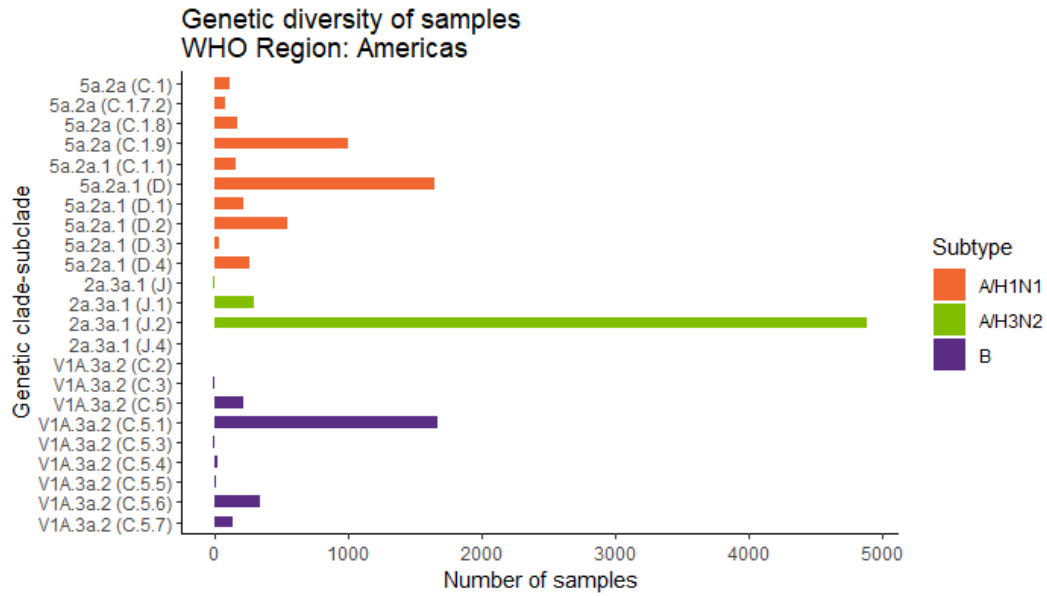


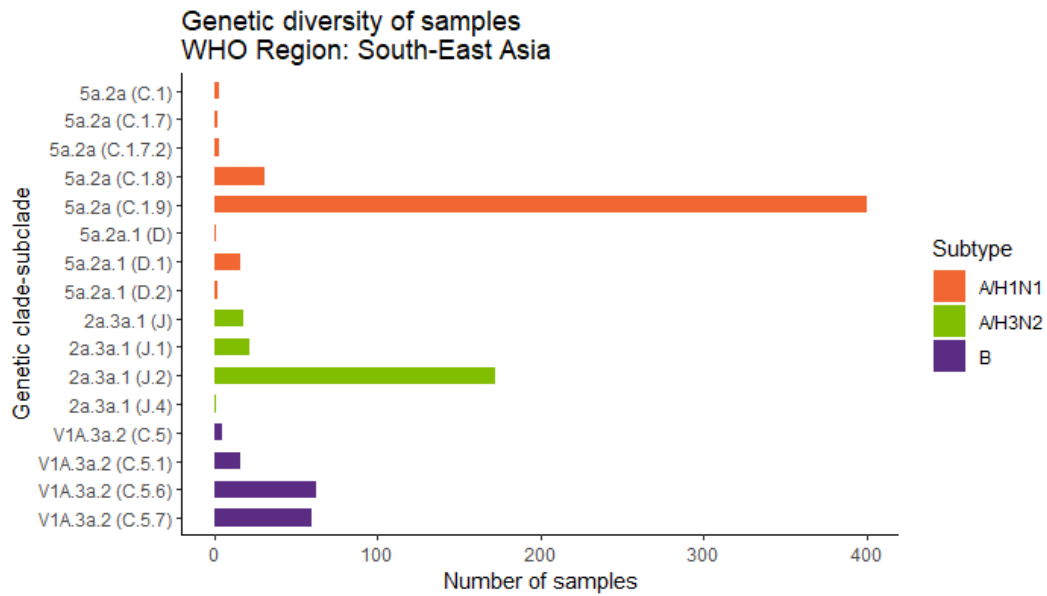
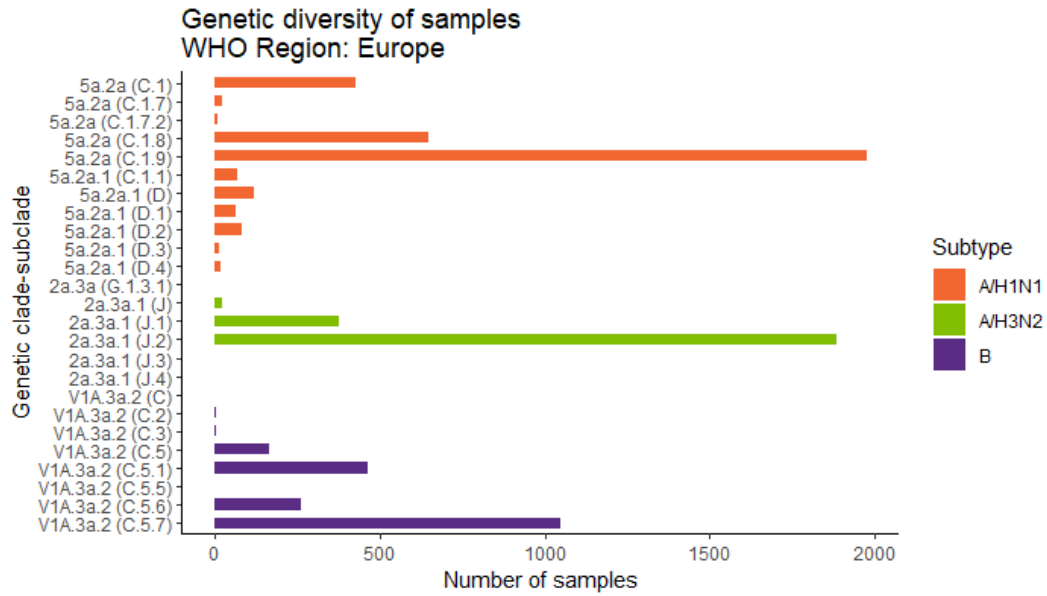
Influenza genetic diversity by WHO region

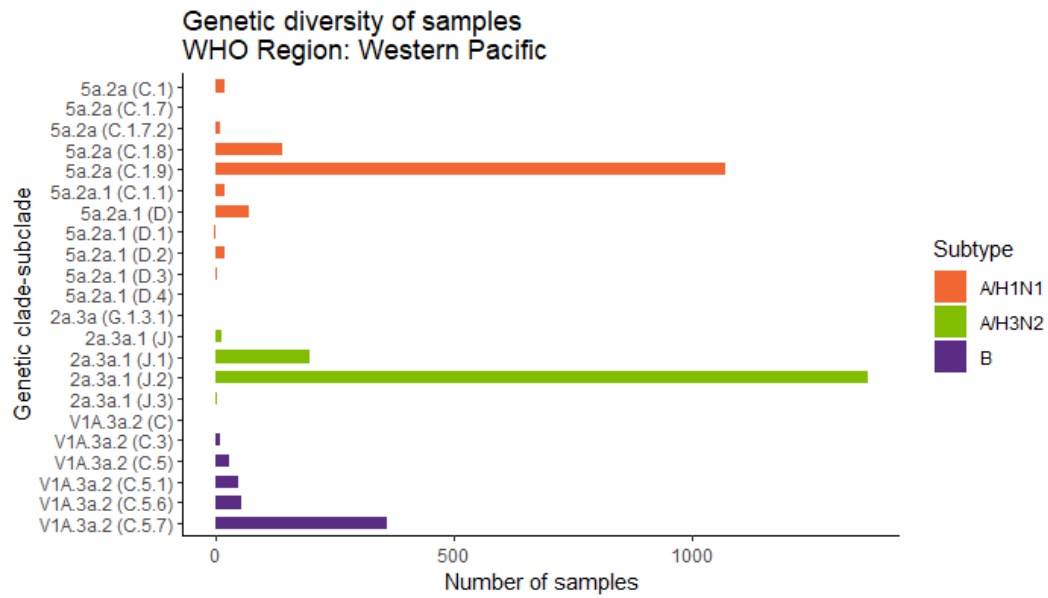
Genetic diversity by Type/Lineage and clade

Global and regional genetic diversity by number of samples genetically characterised as belonging to each clade (subclade). Please note differences in X axis scale across figures.







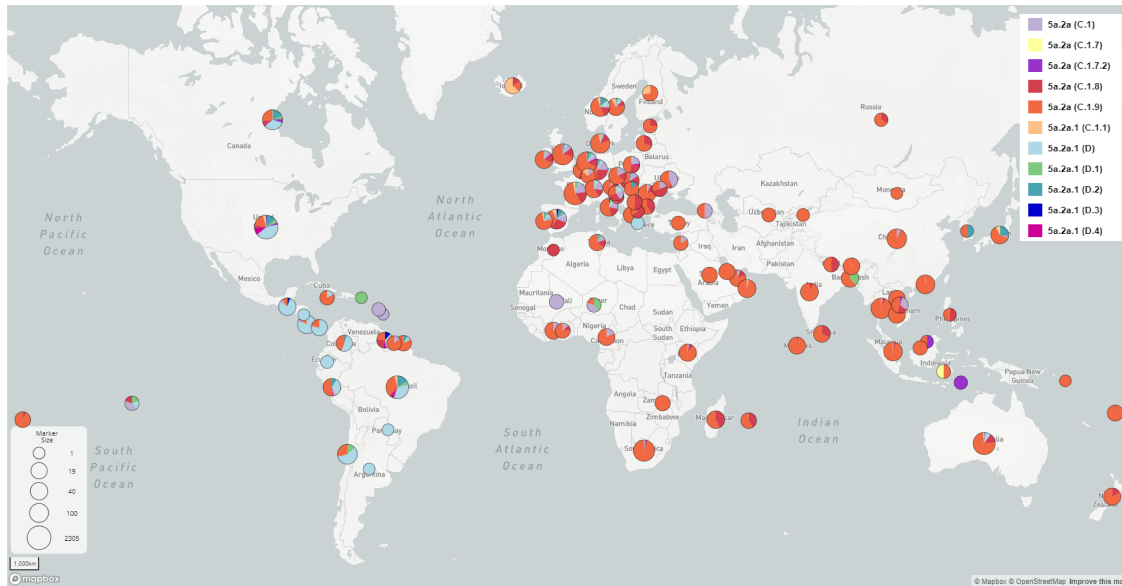


Influenza A/H1N1

Genetic analyses

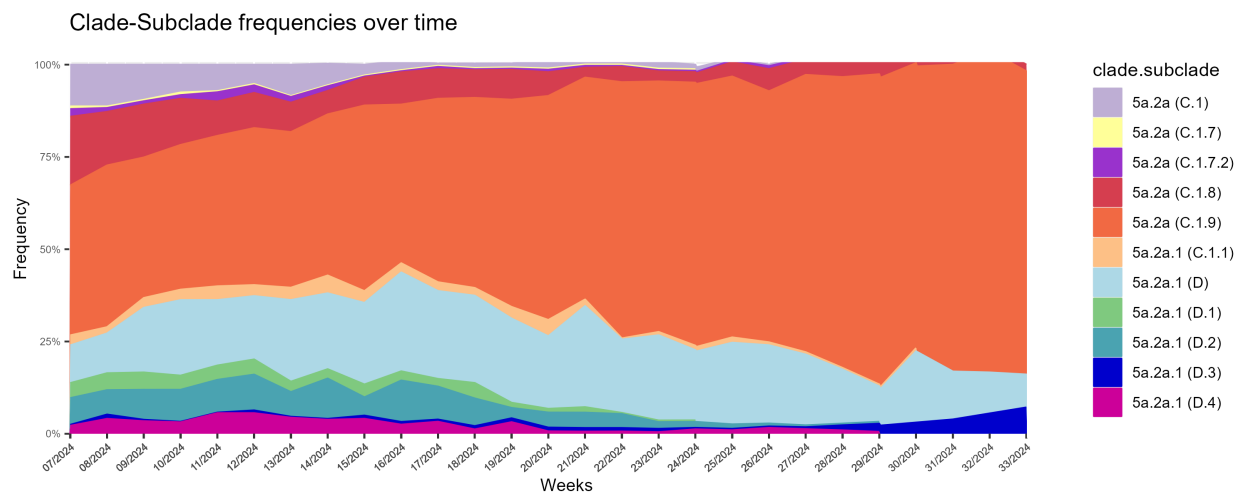
Global distribution of A/H1N1 subclades

Global geographical distribution of influenza A/H1N1 genetic clades (subclades) viruses with collection dates from 1st February 2024 to 31st August 2024, obtained with full HA sequences as deposited in GISAID, downloaded on 12th September 2024 and classified using [Nextclade](#). Map prepared with [Microreact](#). Geographic markers scaled to detection proportions.



Global time-dependent variation of A/H1N1 subclades

Global time-dependent variation in frequencies of genetic clades-subclades of A/H1N1 viruses collected from 1st February 2024 to 31st August 2024.



Maximum likelihood phylogenetic tree inferred using IQTree2 with HA sequence data obtained from GISAID, collection dates from 1st February 2024 to 31st August 2024, manually curated and then downsampled using Treemmer to retain a representative tree topology of 200 sequences, with emphasis on sequences generated at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

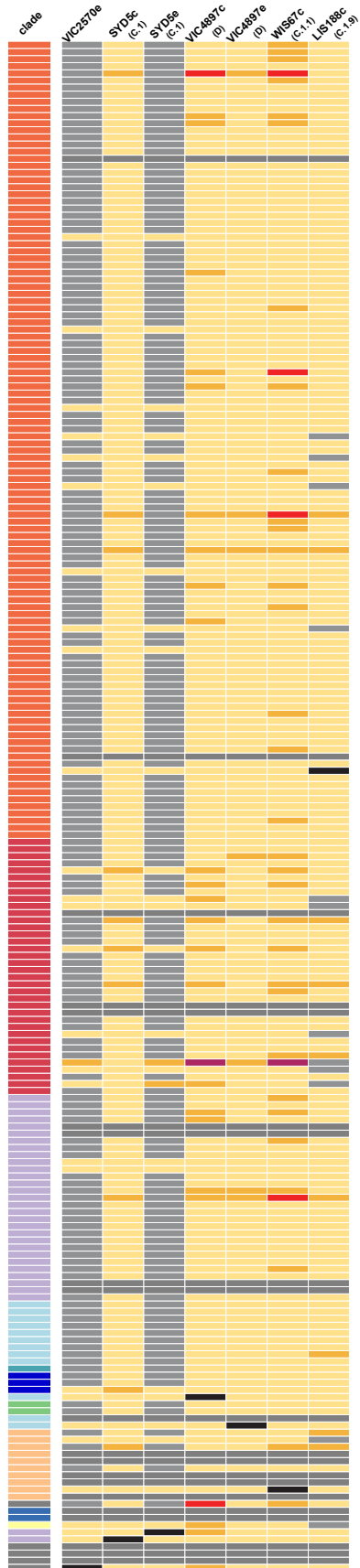
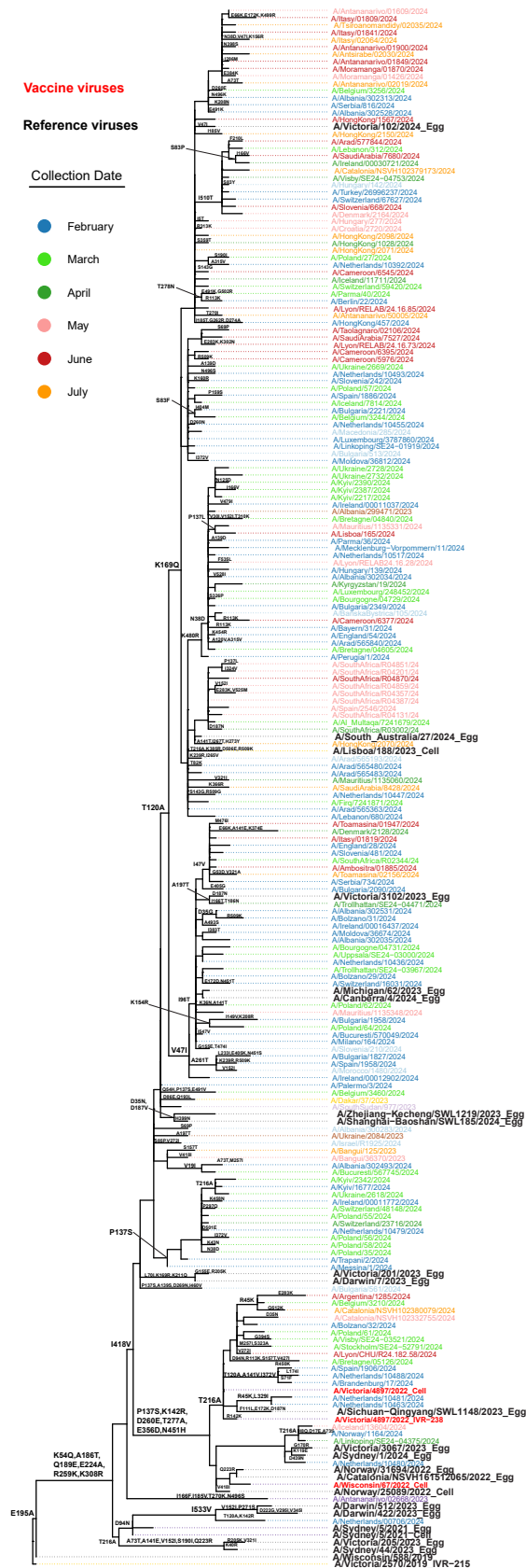


Vaccine viruses

Reference viruses

Collection Date

- February
- March
- April
- May
- June
- July



HI fold change

- < 4
- 4
- 8
- > 8
- N/A

Clade-Subclade

- 5a.2a (C.1)
- 5a.2a (C.1.7)
- 5a.2a (C.1.7.1)
- 5a.2a (C.1.8)
- 5a.2a (C.1.9)
- 5a.2a.1 (C.1.1)
- 5a.2a.1 (D)
- 5a.2a.1 (D.1)
- 5a.2a.1 (D.2)
- 5a.2a.1 (D.3)

Table H1-02. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-05-01)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
							F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
							5a.2 (C)	5a.2a (C.1)	5a.2a (C.1)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES												
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2 (C)	2018-11-22	E4/D7/E3	1280	640	640	640	640	640	640	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	3CK3/MDCK4	1280	1280	1280	1280	1280	1280	1280	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223)	5a.2a (C.1)	2022-10-31	E3/E3	2560	2560	2560	2560	2560	2560	2560	640
A/Victoria/4897/2022	P137S, K142R, T216A, D280E, T277A	5a.2a.1 (D)	2022-10-02	3IAT2/MDCK2	2560	2560	2560	5120	2560	2560	1280	1280
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D280E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	1280	1280	2560	640	640
A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	1280	1280	1280	1280	1280	1280	1280
TEST VIRUSES												
242624	A/BurkinaFaso/3075/2023	5a.2a (C.1)	2023-11-02	MDCK1	2560	2560	2560	2560	2560	2560	1280	1280
242623	A/BurkinaFaso/171/2023	5a.2a (C.1)	2023-12-05	MDCK1	5120	2560	5120	5120	5120	5120	5120	2560
242567	A/Morocco/1211/2024	5a.2a (C.1)	2024-01-16	MDCK2	2560	2560	2560	2560	2560	2560	2560	1280
242804	A/Slovenia/196/2024	5a.2a (C.1)	2024-01-16	CKF1/MDCK1	2560	2560	2560	2560	2560	2560	2560	1280
242800	A/Slovenia/241/2024	5a.2a (C.1)	2024-01-22	CKF1/MDCK1	2560	2560	2560	2560	2560	2560	2560	1280
242730	A/Norway/1164/2024	5a.2a.1 (C.1.1)	2024-02-01	MDCK1	2560	2560	2560	5120	2560	2560	2560	2560
242837	A/Netherlands/00706/2024 (240008461)	5a.2a (C.1.7)	2024-02-09	hCK1/MDCK1	640	640	1280	1280	1280	1280	640	640
242602	A/Serbia/16652/2023	5a.2a (C.1.8)	2023-12-27	MDCK1	5120	2560	2560	2560	2560	2560	2560	1280
242569	A/Morocco/1480/2024	5a.2a (C.1.8)	2024-01-04	MDCK1	1280	1280	640	1280	1280	1280	640	640
242601	A/Serbia/35/2024	5a.2a (C.1.8)	2024-01-05	MDCK1	2560	2560	5120	5120	5120	5120	2560	2560
242599	A/Serbia/113/2024	5a.2a (C.1.8)	2024-01-11	MDCK1	2560	2560	2560	2560	2560	2560	2560	1280
242393	A/Israel/R939/2024	5a.2a (C.1.8)	2024-01-15	MDCK1	2560	2560	2560	1280	2560	2560	2560	1280
242802	A/Slovenia/210/2024	5a.2a (C.1.8)	2024-01-18	CKF1/MDCK1	320	640	640	320	640	640	80	80
242794	A/Slovenia/262/2024	5a.2a (C.1.8)	2024-01-22	CKF1/MDCK1	5120	2560	5120	5120	5120	5120	1280	1280
242590	A/Serbia/280/2024	5a.2a (C.1.8)	2024-01-23	MDCK1	1280	1280	1280	1280	1280	1280	640	640
242588	A/Serbia/381/2024	5a.2a (C.1.8)	2024-01-30	MDCK2	5120	2560	2560	2560	2560	2560	2560	1280
242494	A/Moldova/36791/2024	5a.2a (C.1.8)	2024-02-05	MDCK2	5120	2560	2560	2560	2560	2560	2560	1280
242586	A/Serbia/619/2024	5a.2a (C.1.8)	2024-02-09	MDCK1	2560	2560	2560	2560	2560	2560	2560	1280
242585	A/Serbia/734/2024	5a.2a (C.1.8)	2024-02-17	MDCK1	1280	1280	1280	1280	1280	1280	640	640
242806	A/Slovenia/188/2024	5a.2a (C.1.9)	2024-01-14	CKF1/MDCK1	2560	1280	1280	1280	1280	1280	1280	640
242597	A/Serbia/153/2024	5a.2a (C.1.9)	2024-01-15	MDCK1	5120	2560	2560	5120	2560	2560	2560	2560
242565	A/Morocco/1262/2024	5a.2a (C.1.9)	2024-01-16	MDCK1	2560	2560	2560	2560	2560	2560	2560	1280
242594	A/Serbia/181/2024	5a.2a (C.1.9)	2024-01-16	MDCK1	1280	640	640	1280	1280	1280	640	640
242798	A/Slovenia/243/2024	5a.2a (C.1.9)	2024-01-22	CKF1/MDCK1	2560	1280	1280	1280	2560	2560	640	640
242796	A/Slovenia/250/2024	5a.2a (C.1.9)	2024-01-22	CKF1/MDCK1	5120	2560	2560	2560	2560	2560	2560	1280
242490	A/Moldova/36812/2024	5a.2a (C.1.9)	2024-02-07	MDCK2	5120	2560	5120	5120	5120	5120	2560	2560

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND =

Vaccine
 SH 2021
 NH 2021-22
 SH 2022
 NH 2022-23

Vaccine
 SH 2023

Vaccine
 NH 2023-24
 SH 2024
 NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table H1-03. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-05-14)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
							F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
							5a.2 (C)	5a.2a (C.1)	5a.2a (C.1)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES												
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)			5a.2 (C)	2018-11-22	E4/D7/E3	1280	640	640	1280	1280	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	3CK3/MDCK4	1280	1280	2560	2560	1280	1280
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223			5a.2a (C.1)	2022-10-31	E3/E3	1280	1280	1280	1280	2560	640
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (D)	2022-10-02	3IAT2/MDCK2	2560	1280	2560	5120	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	1280	2560	640
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	1280	1280	2560	1280	1280
TEST VIRUSES												
242872 A/Bulgaria/1684/2024				5a.2a (C.1.8)	2024-02-01	3CK1/MDCK1	2560	1280	1280	1280	1280	1280
242871 A/Bulgaria/1827/2024				5a.2a (C.1.8)	2024-02-02	3CK2/MDCK1	2560	2560	2560	2560	2560	1280
242866 A/Bulgaria/1968/2024				5a.2a (C.1.8)	2024-02-07	3CK1/MDCK1	2560	2560	1280	2560	2560	1280
242865 A/Bulgaria/2090/2024				5a.2a (C.1.8)	2024-02-08	3CK2/MDCK1	2560	2560	2560	2560	2560	640
242863 A/Bulgaria/2219/2024				5a.2a (C.1.8)	2024-02-13	3CK2/MDCK1	1280	1280	1280	1280	1280	640
242323 A/Macedonia/285/2024				5a.2a (C.1.9)	2024-01-04	MDCK1	2560	2560	2560	2560	2560	640
242322 A/Macedonia/295/2024				5a.2a (C.1.9)	2024-01-04	MDCK1	2560	2560	2560	2560	2560	1280
242321 A/Macedonia/14/2024				5a.2a (C.1.9)	2024-01-05	MDCK1	2560	2560	2560	2560	2560	1280
242317 A/Macedonia/316/2024				5a.2a (C.1.9)	2024-01-05	MDCK1	2560	2560	2560	2560	2560	1280
242868 A/Bulgaria/1922/2024				5a.2a (C.1.9)	2024-02-06	3CK2/MDCK1	5120	2560	2560	2560	2560	1280
242867 A/Bulgaria/1925/2024				5a.2a (C.1.9)	2024-02-06	3CK2/MDCK1	2560	2560	2560	2560	2560	1280
242864 A/Bulgaria/2180/2024				5a.2a (C.1.9)	2024-02-09	3CK2/MDCK1	2560	2560	2560	2560	2560	1280
242862 A/Bulgaria/2221/2024				5a.2a (C.1.9)	2024-02-13	3CK2/MDCK1	2560	2560	2560	2560	2560	1280
242860 A/Bulgaria/2348/2024				5a.2a (C.1.9)	2024-02-19	3CK2/MDCK1	2560	2560	2560	2560	2560	640
242859 A/Bulgaria/2349/2024				5a.2a (C.1.9)	2024-02-19	3CK2/MDCK1	5120	2560	2560	5120	2560	1280
242855 A/Bulgaria/2387/2024				5a.2a (C.1.9)	2024-02-27	3CK2/MDCK1	2560	2560	2560	2560	2560	1280
< relates to the lowest dilution of antiserum used							Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23	Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25	
1 hyperimmune sheep serum; ND =												

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-04. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-05-21)

Viruses	Other information	Haemagglutination inhibition titre										
		Passage	Post-infection ferret antiserum									
			A/Rydney s/21 MDCK	A/Lisboa/ MDCK	A/Victoria/ MDCK	IVR-238 4897/2022 VLC4897/22 Egg	IVR-238 4897/2022 VLC4897/22 Egg	F07/23 MDCK	F07/23 MDCK	F17/23 MDCK	F17/23 MDCK	
			Ferret number	F46/22	F09/24	F05/23	F07/23	F17/23	F17/23	F17/23		
			Genetic group	Collection Date	Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a (D.1)	5a.2a (D.1)	5a.2a (D.1)	5a.2a (D.1)	
REFERENCE VIRUSES												
A/Rydney/2021	K54Q, A196T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-18	MDCK/MDCK4	2560	2560	2560	2560	2560	2560		
A/Lisboa/188/2023	K169D, T216A, D506E	5a.2a (C.1.9)	2023-11-22	BIAT1/MDCK2	2560	2560	2560	2560	2560	2560		
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a (D.1)	2022-10-02	BIAT1/MDCK3	2560	2560	2560	2560	2560	2560		
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A	5a.2a (D.3)	2022-10-02	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
A/Wisconsin/67/2022	RI142Q, RI142Q, Q223R	5a.2a (C.1.1)	2022-10-25	MDCK2	2560	2560	2560	2560	2560	2560		
TEST VIRUSES												
240649 A/Dakar/37/2023		5a.2a (C.1)	2023-11-02	MDCK2/BIAT1	1280	2560	1280	1280	1280	1280		
241274 A/Spain/15/2023		5a.2a (C.1.9)	2023-11-18	MDCK1	2560	2560	2560	2560	2560	2560		
241248 A/Spain/540/2023		5a.2a (C.1)	2023-12-09	MDCK1	2560	2560	2560	2560	2560	2560		
241201 A/Spain/683/2023		5a.2a (C.1)	2023-12-13	MDCK1	2560	2560	2560	2560	2560	2560		
241236 A/Spain/643/2023		5a.2a (C.1)	2023-12-14	MDCK1	1280	2560	1280	2560	2560	2560		
242884 A/Bulgaria/561/2024		5a.2a (C.1)	2024-01-16	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242882 A/Bulgaria/671/2024		5a.2a (C.1)	2024-01-17	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242879 A/Bulgaria/1016/2024		5a.2a (C.1.9)	2024-01-19	MDCK2/MDCK1	1280	1280	2560	2560	2560	2560		
242874 A/Bulgaria/1323/2024		5a.2a (C.1)	2024-01-23	MDCK2/MDCK1	1280	1280	2560	2560	2560	2560		
242410 A/Israel/19325/2024		5a.2a (C.1)	2024-01-29	BIAT1	2560	2560	2560	2560	2560	2560		
241287 A/Spain/135/2023		5a.2a (C.1.7)	2023-11-14	MDCK1	2560	2560	2560	2560	2560	2560		
241293 A/Spain/121/2023		5a.2a (C.1.8)	2023-11-10	MDCK1	2560	2560	2560	2560	2560	2560		
241230 A/Spain/484/2023		5a.2a (C.1.8)	2023-11-27	MDCK1	1280	1280	2560	2560	2560	2560		
241206 A/Spain/900/2023		5a.2a (C.1.8)	2023-12-19	MDCK1	2560	2560	2560	2560	2560	2560		
242884 A/Bulgaria/1290/2024		5a.2a (C.1.8)	2024-01-10	MDCK2/MDCK1	1280	2560	2560	2560	2560	2560		
242893 A/Bulgaria/311/2024		5a.2a (C.1.8)	2024-01-10	MDCK2/MDCK1	2560	2560	1280	1280	1280	1280		
242890 A/Bulgaria/430/2024		5a.2a (C.1.8)	2024-01-10	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242888 A/Bulgaria/564/2024		5a.2a (C.1.8)	2024-01-15	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242885 A/Bulgaria/501/2024		5a.2a (C.1.8)	2024-01-16	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242883 A/Bulgaria/916/2024		5a.2a (C.1.8)	2024-01-16	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560		
242878 A/Morocco/1284/2024		5a.2a (C.1.8)	2024-01-18	BIAT1	1280	1280	1280	1280	1280	1280		
242878 A/Bulgaria/1016/2024		5a.2a (C.1.9)	2024-01-19	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242877 A/Bulgaria/1017/2024		5a.2a (C.1.8)	2024-01-19	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242876 A/Bulgaria/1327/2024		5a.2a (C.1.8)	2024-01-22	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242873 A/Bulgaria/1328/2024		5a.2a (C.1.8)	2024-01-23	MDCK2/MDCK1	1280	2560	2560	2560	2560	2560		
241260 A/Spain/200/2023		5a.2a (C.1.9)	2023-11-23	MDCK1	2560	2560	2560	2560	2560	2560		
241205 A/Spain/915/2023		5a.2a (C.1.9)	2023-12-19	MDCK1	2560	2560	2560	2560	2560	2560		
242826 A/Malaysia/2056/2023		5a.2a (C.1.9)	2023-12-29	MDCK1	2560	1120	2560	2560	2560	2560		
242886 A/Bulgaria/302/2024		5a.2a (C.1.9)	2024-01-09	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242885 A/Bulgaria/238/2024		5a.2a (C.1.9)	2024-01-10	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560		
242892 A/Bulgaria/112/2024		5a.2a (C.1.9)	2024-01-10	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242891 A/Bulgaria/391/2024		5a.2a (C.1.9)	2024-01-12	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242889 A/Bulgaria/513/2024		5a.2a (C.1.9)	2024-01-15	MDCK2/MDCK1	1280	2560	2560	2560	2560	2560		
242891 A/Bulgaria/87/2024		5a.2a (C.1.9)	2024-01-16	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242880 A/Bulgaria/674/2024		5a.2a (C.1.9)	2024-01-18	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
242875 A/Bulgaria/151/2024		5a.2a (C.1.9)	2024-01-23	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560		
241216 A/Spain/971/2023		5a.2a (D.1)	2023-12-16	MDCK1	1280	2560	2560	2560	2560	2560		
241289 A/Spain/138/2023		5a.2a (D.1)	2023-11-12	MDCK1	2560	2560	2560	2560	2560	2560		
241271 A/Spain/173/2023		5a.2a (D.1)	2023-11-20	MDCK1	1280	2560	2560	2560	2560	2560		
241286 A/Spain/187/2023		5a.2a (D.1)	2023-11-21	MDCK1	2560	2560	2560	2560	2560	2560		
242897 A/Bulgaria/575/2024		5a.2a (D.1)	2024-01-15	MDCK2/MDCK1	1280	1280	2560	2560	2560	2560		
241288 A/Spain/462/2023		5a.2a (D.2)	2023-11-12	MDCK1	2560	2560	2560	2560	2560	2560		
241277 A/Spain/461/2023		5a.2a (D.2)	2023-11-17	MDCK1	2560	2560	2560	2560	2560	2560		
241262 A/Spain/196/2023		5a.2a (D.2)	2023-11-23	MDCK1	2560	2560	2560	2560	2560	2560		
241261 A/Spain/186/2023		5a.2a (D.2)	2023-11-23	MDCK1	1280	1280	2560	2560	2560	2560		
241232 A/Spain/601/2023		5a.2a (D.2)	2023-12-13	MDCK1	1280	1280	2560	2560	2560	2560		
241226 A/Spain/733/2023		5a.2a (D.2)	2023-12-14	MDCK1	2560	2560	2560	2560	2560	2560		
241221 A/Spain/683/2023		5a.2a (D.2)	2023-12-16	MDCK1	2560	2560	2560	2560	2560	2560		
241212 A/Spain/643/2023		5a.2a (D.2)	2023-12-17	MDCK1	1280	1280	2560	2560	2560	2560		
241210 A/Spain/996/2023		5a.2a (D.3)	2023-12-15	MDCK1	1280	2560	2560	2560	2560	2560		
241215 A/Spain/979/2023		5a.2a (D.3)	2023-12-16	MDCK1	1280	2560	2560	2560	2560	2560		
241213 A/Spain/581/2023		5a.2a (D.3)	2023-12-17	MDCK1	640	1280	2560	2560	2560	2560		
< relates to the lowest dilution of antiserum used † hyperimmune swine serum; ND =												
Vaccine SH 2023 NH 2023-24 NH 2023-25 SH 2024 NH 2024-25												

[†] hyperimmune sheep serum: ND =

Vaccine
SH 2023Vaccine
NH 2023-24
2023-24SH 2024
NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-05. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-05-29)

Viruses		Other information	Haemagglutination inhibition titre										
			Passage	Post-infection ferret antiserum					Genetic group	Collection Date			
				A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/67/2022 MDCK					
				F46/22	F09/24	F05/23	F07/23	F17/23					
								5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)	
REFERENCE VIRUSES													
A/Sydney/5/2021		K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280			
A/Lisboa/188/2023		K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	2560	2560	2560	1280			
A/Victoria/4897/2022		P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	2560			
IVR-238 (A/Victoria/4897/2022)		P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	1280	2560	2560	1280			
A/Wisconsin/67/2022			5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560			
TEST VIRUSES													
242484	A/SouthSudan/977/2023		5a.2a (C.1)	2023-09-03	MDCK2	1280	2560	1280	1280	1280			
242477	A/SouthSudan/1000/2023		5a.2a (C.1)	2023-11-23	MDCK2	2560	2560	2560	2560	1280			
241351	A/Ukraine/2057/2023		5a.2a (C.1)	2023-12-18	SIAT1	2560	2560	2560	2560	1280			
241987	A/Poland/PL05/2023		5a.2a (C.1)	2023-12-19	MDCK1	1280	640	1280	1280	640			
241349	A/Ukraine/2059/2023		5a.2a (C.1)	2023-12-19	SIAT1	2560	2560	2560	2560	1280			
241984	A/Poland/PL10/2023		5a.2a (C.1)	2023-12-20	MDCK1	2560	2560	2560	2560	1280			
241346	A/Ukraine/2066/2023		5a.2a (C.1)	2023-12-21	SIAT1	1280	1280	1280	1280	640			
241343	A/Ukraine/2083/2023		5a.2a (C.1)	2023-12-26	SIAT1	2560	2560	2560	2560	1280			
241342	A/Ukraine/2084/2023		5a.2a (C.1)	2023-12-26	SIAT1	2560	2560	2560	2560	1280			
242897	A/Bulgaria/279/2024		5a.2a (C.1)	2024-01-09	MDCK2/MDCK1	2560	2560	2560	2560	1280			
241975	A/Poland/PL22/2024		5a.2a (C.1)	2024-01-10	MDCK1	2560	2560	2560	2560	1280			
242036	A/Antananarivo/02668/2023		5a.2a (C.1.5)	2023-10-25	SIAT1	1280	1280	640	1280	640			
241965	A/Greece/ILI4955/2023		5a.2a (C.1.8)	2023-12-13	MDCK1	2560	2560	2560	2560	1280			
241347	A/Ukraine/2051/2023		5a.2a (C.1.8)	2023-12-20	SIAT1	1280	1280	1280	1280	640			
241983	A/Poland/PL20/2023		5a.2a (C.1.8)	2023-12-21	MDCK1	2560	2560	2560	2560	1280			
241979	A/Poland/PL06/2024		5a.2a (C.1.8)	2024-01-03	MDCK1	2560	2560	2560	2560	1280			
242900	A/Bulgaria/149/2024		5a.2a (C.1.8)	2024-01-08	MDCK2/MDCK1	1280	2560	2560	1280	640			
241960	A/Greece/ILI107/2024		5a.2a (C.1.8)	2024-01-08	MDCK1	2560	2560	2560	2560	1280			
242898	A/Bulgaria/158/2024		5a.2a (C.1.8)	2024-01-09	MDCK2/MDCK1	2560	2560	2560	2560	1280			
241348	A/Ukraine/2064/2023		5a.2a (C.1.9)	2023-12-19	SIAT1	2560	2560	2560	2560	1280			
241344	A/Ukraine/2069/2023		5a.2a (C.1.9)	2023-12-25	SIAT1	1280	1280	1280	1280	640			
241964	A/Greece/ILI5061/2023		5a.2a (C.1.9)	2023-12-28	MDCK1	2560	2560	2560	2560	1280			
241982	A/Poland/PL18/2023		5a.2a (C.1.9)	2023-12-28	MDCK1	2560	2560	2560	2560	1280			
242899	A/Bulgaria/97/2024		5a.2a (C.1.9)	2024-01-08	MDCK2/MDCK1	2560	1280	1280	2560	1280			
241980	A/Poland/PL19/2023		5a.2a.1 (D.1)	2023-12-30	MDCK1	1280	1280	2560	1280	1280			
242226	A/Kyrgyzstan/65/2023		5a.2a.1 (D.2)	2023-11-27	SIAT1	1280	2560	5120	2560	2560			
241989	A/Poland/PL03/2023		5a.2a.1 (D.4)	2023-12-05	MDCK1	2560	2560	5120	2560	2560			
< relates to the lowest dilution of antiserum used ¹ hyperimmune sheep serum; ND =						Vaccine SH 2023	Vaccine NH 2023-24 SH 2024 NH 2024-25						

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-06. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-06-05)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
							Post-infection ferret antiserum				
							A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/67/2022 MDCK
							F46/22	F09/24	F05/23	F07/23	F17/23
							5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES											
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	DCCK3/MDCK4	2560	2560	2560	2560	1280
A/Lisboa/188/2023	K169Q, T216A, D506E			5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	2560	2560	2560	1280
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	2560	2560	2560	1280
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	2560	2560	2560
TEST VIRUSES											
240651 A/Dakar/35/2023				5a.2a (C.1)	2023-10-11	MDCK2/SIAT2	1280	1280	1280	1280	1280
242976 A/Albania/302489/2024				5a.2a (C.1)	2024-02-09	MDCK1	2560	2560	2560	5120	2560
242975 A/Albania/302493/2024				5a.2a (C.1)	2024-02-09	MDCK1	2560	2560	5120	2560	1280
242971 A/Albania/302516/2024				5a.2a (C.1)	2024-02-12	MDCK1	2560	2560	5120	5120	1280
242970 A/Albania/302659/2024				5a.2a (C.1)	2024-02-12	MDCK1	1280	1280	1280	1280	640
242974 A/Albania/302531/2024				5a.2a (C.1.8)	2024-02-09	MDCK1	2560	2560	2560	2560	1280
242979 A/Albania/302314/2024				5a.2a (C.1.9)	2024-02-06	MDCK1	2560	2560	2560	2560	1280
242978 A/Albania/302313/2024				5a.2a (C.1.9)	2024-02-07	MDCK1	2560	2560	2560	2560	1280
242977 A/Albania/302342/2024				5a.2a (C.1.9)	2024-02-07	MDCK1	2560	2560	5120	2560	1280
242973 A/Albania/302528/2024				5a.2a (C.1.9)	2024-02-10	MDCK1	5120	5120	5120	2560	2560
242972 A/Albania/302529/2024				5a.2a (C.1.9)	2024-02-10	MDCK1	5120	2560	5120	2560	2560
242969 A/Albania/302566/2024				5a.2a (C.1.9)	2024-02-13	MDCK1	5120	2560	5120	2560	2560
242968 A/Albania/302658/2024				5a.2a (C.1.9)	2024-02-13	MDCK1	2560	2560	2560	2560	1280
242967 A/Albania/302666/2024				5a.2a (C.1.9)	2024-02-14	MDCK1	5120	2560	5120	2560	2560
242966 A/Albania/302796/2024				5a.2a (C.1.9)	2024-02-15	MDCK1	1280	1280	1280	1280	640
243009 A/Albania/302977/2024				5a.2a (C.1.9)	2024-02-21	SIAT1	640	320	320	320	320
242144 A/Catalonia/NSBE222136865/2023				5a.2a.1 (D.1)	2023-12-29	MDCK1	1280	1280	2560	1280	1280
242141 A/Catalonia/NSVH170131271/2024				5a.2a.1 (D.1)	2024-01-03	MDCK1	2560	2560	5120	2560	2560
< relates to the lowest dilution of antiserum used							Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25	
¹ hyperimmune sheep serum; ND =											

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-07. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-06-11)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
							Post-infection ferret antiserum				
							A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 MDCK	IVR-238 A/Vici/4897/22 Egg	A/Wisconsin/67/2022 MDCK
							F46/22	F09/24	F05/23	F07/23	F17/23
							5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES											
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	DCK3/MDCK4	2560	2560	2560	2560	1280
A/Lisboa/188/2023	K169Q, T216A, D506E			5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	2560	2560	2560	1280
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	1280	2560	2560	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	2560	2560	2560	1280
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560
TEST VIRUSES											
243198	A/Poland/58/2024			5a.2a (C.1)	2024-03-08	MDCK1	1280	1280	1280	1280	1280
243196	A/Poland/35/2024			5a.2a (C.1)	2024-03-11	MDCK1	2560	2560	2560	2560	1280
243194	A/Poland/64/2024			5a.2a (C.1.8)	2024-03-11	MDCK1	2560	2560	2560	2560	2560
243191	A/Poland/62/2024			5a.2a (C.1.8)	2024-03-14	MDCK1	1280	1280	2560	2560	1280
243136	A/Lebanon/198/2024			5a.2a (C.1.9)	2024-02-13	MDCK1	2560	2560	2560	2560	1280
243135	A/Lebanon/680/2024			5a.2a (C.1.9)	2024-02-26	MDCK1	2560	2560	2560	2560	1280
243197	A/Poland/65/2024			5a.2a (C.1.9)	2024-03-10	MDCK1	2560	2560	2560	2560	1280
243193	A/Poland/54/2024			5a.2a (C.1.9)	2024-03-12	MDCK1	2560	2560	2560	2560	1280
243167	A/Lithuania/2403180025/2024			5a.2a (C.1.9)	2024-03-16	MDCK1	2560	2560	2560	2560	640
243132	A/Lebanon/312/2024			5a.2a (C.1.9)	2024-03-22	MDCK1	2560	2560	2560	2560	1280
243165	A/Lithuania/2404020029/2024			5a.2a (C.1.9)	2024-03-31	MDCK1	1280	1280	1280	1280	640
243164	A/Lithuania/2404110065/2024			5a.2a (C.1.9)	2024-04-04	MDCK1	5120	2560	2560	2560	1280
243162	A/Lithuania/2404090067/2024			5a.2a (C.1.9)	2024-04-05	MDCK1	2560	2560	2560	2560	1280
243189	A/Poland/61/2024			5a.2a.1 (D)	2024-03-24	MDCK1	1280	1280	2560	1280	1280
< relates to the lowest dilution of antiserum used							Vaccine			Vaccine	
¹ hyperimmune sheep serum; ND =							SH 2023			NH 2023-24 SH 2024 NH 2024-25	

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Wisconsin/ A/Vic/4897/22 Egg	67/2022 MDCK	
							F46/22	F09/24	F05/23	F07/23	F17/23	
							Sa.2a (C.1)	Sa.2a (C.1.9)	Sa.2a.1 (D)	Sa.2a.1 (D)	Sa.2a.1 (D)	Sa.2a.1 (C.1.1)
REFERENCE VIRUSES												
A/Sydney/5/2021	K54Q, A186T, E224A, R289K, K308R, D54N, T216A			Sa.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280	2560
A/Lisboa/188/2023	K169Q, T216A, D506E			Sa.2a (C.1.9)	2021-11-22	SIAT1/MDCK2	2560	2560	2560	2560	1280	2560
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			Sa.2a.1 (D)	2022-10-02	SIAT2/MDCK3	1280	2560	2560	2560	2560	2560
IVR-238 A/Victoria/4897/2022				Sa.2a (C.1.9)	2022-10-02	E3/6/EE1	2560	2560	2560	2560	1280	2560
A/Wisconsin/67/2022	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			Sa.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	8120	2560	2560	2560
TEST VIRUSES												
242998	A/Abania/300283/2024			Sa.2a (C.1)	2024-01-11	MDCK1	1280	1280	1280	1280	1280	640
243440	A/Netherlands/10338/2024			Sa.2a (C.1)	2024-01-31	MDCK-MIX2/MDCK1	2560	2560	2560	2560	2560	2560
243424	A/Netherlands/10479/2024			Sa.2a (C.1)	2024-02-14	MDCK-MIX2/MDCK1	2560	2560	2560	2560	2560	2560
243288	A/Iceland/6892/2024			Sa.2a (C.1.8)	2024-02-27	MDCK1	2560	1280	1280	1280	1280	640
243252	A/Iceland/10938/2024			Sa.2a (C.1.8)	2024-04-08	MDCK1	1280	1280	1280	1280	1280	640
243443	A/Netherlands/10644/2023			Sa.2a (C.1.9)	2023-12-20	MDCK-MIX2/MDCK1	1280	1280	2560	2560	1280	2560
243442	A/Netherlands/10644-T1C3/2023			Sa.2a (C.1.9)	2024-01-02	MDCK-MIX3/MDCK1	1280	1280	2560	1280	640	2560
243441	A/Netherlands/10331/2024			Sa.2a (C.1.9)	2024-01-31	MDCK-MIX2/MDCK1	2560	2560	2560	2560	1280	2560
242892	A/Abania/302024/2024			Sa.2a (C.1.9)	2024-02-05	MDCK2	2560	2560	2560	2560	1280	2560
243438	A/Netherlands/10392/2024			Sa.2a (C.1.9)	2024-02-05	MDCK-MIX1/MDCK1	2560	5120	1280	1280	1280	2560
243436	A/Netherlands/10436/2024			Sa.2a (C.1.9)	2024-02-08	MDCK-MIX1/MDCK1	5120	5120	5120	2560	2560	2560
243428	A/Netherlands/10410/2024			Sa.2a (C.1.9)	2024-02-12	MDCK-MIX3/MDCK1	2560	1280	2560	2560	1280	2560
243430	A/Netherlands/10447/2024			Sa.2a (C.1.9)	2024-02-12	MDCK-MIX1/MDCK1	2560	2560	5120	2560	2560	2560
243429	A/Netherlands/10455/2024			Sa.2a (C.1.9)	2024-02-12	MDCK-MIX1/MDCK1	5120	5120	5120	5120	5120	2560
243422	A/Netherlands/10493/2024			Sa.2a (C.1.9)	2024-02-14	MDCK-MIX2/MDCK1	5120	5120	5120	2560	2560	1280
243287	A/Turkey/26996237/2024			Sa.2a (C.1.9)	2024-02-15	MDCK1	2560	2560	2560	2560	1280	2560
243416	A/Netherlands/10517/2024			Sa.2a (C.1.9)	2024-02-19	MDCK-MIX2/MDCK1	5120	5120	5120	5120	2560	2560
243412	A/Netherlands/10555/2024			Sa.2a (C.1.9)	2024-03-01	MDCK	2560	2560	2560	2560	1280	2560
243256	A/Iceland/7814/2024			Sa.2a (C.1.9)	2024-03-07	MDCK1	2560	2560	2560	2560	1280	2560
243284	A/Turkey/2714816/2024			Sa.2a (C.1.9)	2024-03-20	MDCK1	2560	2560	2560	2560	1280	2560
243391	A/Kyrgyzstan/178/2024			Sa.2a (C.1.9)	2024-03-26	MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
243305	A/Kyrgyzstan/18/2024			Sa.2a (C.1.9)	2024-04-06	MDCK1/MDCK1	2560	2560	2560	2560	1280	2560
243249	A/Iceland/11711/2024			Sa.2a (C.1.9)	2024-04-13	MDCK1	1280	1280	1280	1280	1280	640
243248	A/Iceland/11752/2024			Sa.2a (C.1.9)	2024-04-14	MDCK1	2560	2560	2560	2560	1280	2560
243418	A/Netherlands/10480/2024			Sa.2a.1 (C.1.1)	2024-02-16	MDCK-MIX3/MDCK1	2560	2560	5120	2560	2560	2560
243258	A/Iceland/875/2024			Sa.2a.1 (C.1.1)	2024-02-18	MDCK1	1280	1280	1280	1280	1280	640
243254	A/Iceland/9716/2024			Sa.2a.1 (C.1.1)	2024-03-25	MDCK1	1280	1280	2560	2560	1280	1280
243253	A/Iceland/10430/2024			Sa.2a.1 (C.1.1)	2024-04-03	MDCK1	1280	1280	2560	1280	640	2560
243251	A/Iceland/11482/2024			Sa.2a.1 (C.1.1)	2024-04-11	MDCK1	1280	1280	2560	2560	1280	1280
243247	A/Iceland/12718/2024			Sa.2a.1 (C.1.1)	2024-04-22	MDCK2	1280	1280	1280	1280	1280	640
243246	A/Iceland/13126/2024			Sa.2a.1 (C.1.1)	2024-04-26	MDCK2	640	640	1280	1280	1280	640
243245	A/Iceland/13576/2024			Sa.2a.1 (C.1.1)	2024-04-30	MDCK2	1280	1280	5120	1280	1280	2560
243244	A/Iceland/13604/2024			Sa.2a.1 (C.1.1)	2024-05-01	MDCK1	1280	640	2560	1280	1280	1280
243243	A/Iceland/13962/2024			Sa.2a.1 (C.1.1)	2024-05-04	MDCK1	1280	1280	2560	1280	1280	2560
243242	A/Iceland/13991/2024			Sa.2a.1 (C.1.1)	2024-05-05	MDCK1	640	640	1280	1280	1280	640
243241	A/Iceland/14912/2024			Sa.2a.1 (C.1.1)	2024-05-14	MDCK1	2560	1280	5120	2560	2560	2560
243240	A/Iceland/14952/2024			Sa.2a.1 (C.1.1)	2024-05-14	MDCK1	2560	1280	2560	2560	2560	2560
243246	A/Netherlands/10463/2024			Sa.2a.1 (D.1)	2024-02-12	MDCK-MIX2/MDCK1	2560	2560	5120	2560	2560	2560
243432	A/Netherlands/10451/2024			Sa.2a.1 (D.1)	2024-02-12	MDCK-MIX1/MDCK1	2560	2560	5120	2560	2560	2560
243420	A/Netherlands/10488/2024			Sa.2a.1 (D.3)	2024-02-15	MDCK-MIX2/MDCK1	1280	1280	5120	2560	2560	2560
< relates to the lowest dilution of antiserum used							Vaccine	Vaccine				
1 hyperimmune sheep serum: ND =							SH 2023	NH 2023-24				
								SH 2024				
								NH 2024-25				

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-09. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-06-26)

Viruses	Other information	Genetic group	Collection Date	Genetic group	Passage	Haemagglutination inhibition titre				
						Post-infection ferret antiserum				
						A/Sydney 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 A/Vic/4897/22 MDCK	IVR-238 A/Wisconsin/ 67/2022 Egg	
						F46/22	F09/24	F05/23	F07/23	F17/23
						5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES										
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280	
A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	2560	2560	2560	1280	
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	1280	2560	2560	2560	
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	2560	2560	1280	
A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560	
TEST VIRUSES										
243444	A/Netherlands/10548/2023	5a.2a (C.1)	2023-09-21	MDCK-MIX2/MDCK1	5120	2560	2560	2560	1280	
243369	A/Ireland/00011772/2024	5a.2a (C.1)	2024-02-06	MDCK1	2560	1280	2560	2560	1280	
243646	A/Kyiv/1682/2024	5a.2a (C.1)	2024-02-17	MDCK1/MDCK1	1280	1280	2560	1280	640	
243545	A/Ukraine/2618/2024	5a.2a (C.1)	2024-03-05	MDCK2/MDCK1	1280	1280	2560	1280	1280	
243542	A/Ukraine/2768/2024	5a.2a (C.1)	2024-03-07	MDCK2/MDCK1	2560	1280	1280	1280	640	
243361	A/Ireland/00024709/2024	5a.2a (C.1.8)	2024-03-19	MDCK1	2560	2560	2560	2560	1280	
243371	A/Ireland/00013906/2024	5a.2a (C.1.9)	2024-02-03	MDCK1	1280	1280	1280	1280	640	
243370	A/Ireland/00011037/2024	5a.2a (C.1.9)	2024-02-06	MDCK1	1280	1280	1280	1280	640	
243285	A/Turkey/27016821/2024	5a.2a (C.1.9)	2024-02-21	MDCK2	2560	2560	1280	2560	1280	
243362	A/Ireland/00020083/2024	5a.2a (C.1.9)	2024-03-03	MDCK1	2560	2560	2560	2560	1280	
243536	A/Ukraine/2732/2024	5a.2a (C.1.9)	2024-03-13	MDCK2/MDCK1	2560	1280	1280	1280	1280	
243359	A/Ireland/00030721/2024	5a.2a (C.1.9)	2024-04-12	MDCK1	5120	2560	2560	2560	1280	
< relates to the lowest dilution of antiserum used						Vaccine	Vaccine			
¹ hyperimmune sheep serum; ND =						SH 2023	NH 2023-24 SH 2024 NH 2024-25			

 < 4-fold  4-fold  8-fold  > 8-fold  not recognised by the antiserum  ≥ 160 (no homologous titre)

Table H1-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-07-03)

Viruses	Other information	Passage	Ferret number	Haemagglutination inhibition titre						
				Post-infection ferret antiserum						
				A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/67/2022 MDCK		
				F46/22	F09/24	F05/23	F07/23	F17/23		
		Genetic group	Collection Date	Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)	
REFERENCE VIRUSES										
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	2560	
A/Lisboa/188/2023	K169Q, T216A, D506E	5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	2560	2560	1280	2560	
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	1280	2560	2560	1280	
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	2560	2560	1280	
A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560	
TEST VIRUSES										
243657 A/Kyiv/1677/2024		5a.2a (C.1)	2024-02-15	MDCK1/MDCK1	640	640	640	640	320	
243218 A/Poland/55/2024		5a.2a (C.1)	2024-03-01	MDCK1	2560	2560	2560	2560	1280	
243206 A/Poland/56/2024		5a.2a (C.1)	2024-03-05	MDCK1	2560	1280	2560	2560	1280	
243648 A/Kyiv/2342/2024		5a.2a (C.1)	2024-03-18	MDCK1/SIAT1	1280	1280	640	640	640	
243368 A/Ireland/00012902/2024		5a.2a (C.1.8)	2024-02-09	MDCK2	2560	2560	2560	2560	1280	
243365 A/Ireland/00016437/2024		5a.2a (C.1.8)	2024-02-21	MDCK2	2560	2560	2560	2560	1280	
243207 A/Poland/48/2024		5a.2a (C.1.8)	2024-03-05	MDCK1	1280	1280	1280	1280	640	
243544 A/Ukraine/2701/2024		5a.2a (C.1.8)	2024-03-05	MDCK1	2560	2560	5120	2560	2560	
243366 A/Ireland/00014925/2024		5a.2a (C.1.9)	2024-02-15	MDCK2	2560	2560	1280	1280	640	
243656 A/Kyiv/1857/2024		5a.2a (C.1.9)	2024-02-25	MDCK1/MDCK1	5120	2560	5120	5120	5120	
243221 A/Poland/29/2024		5a.2a (C.1.9)	2024-03-01	MDCK1	5120	2560	5120	5120	2560	
243217 A/Poland/57/2024		5a.2a (C.1.9)	2024-03-01	MDCK1	5120	5120	5120	2560	2560	
243502 A/Al_Multaqa/7241679/2024		5a.2a (C.1.9)	2024-03-04	MDCK1	2560	2560	2560	2560	1280	
243503 A/lbra/7241609/2024		5a.2a (C.1.9)	2024-03-04	MDCK1	2560	1280	1280	2560	1280	
243211 A/Poland/49/2024		5a.2a (C.1.9)	2024-03-04	MDCK1	2560	2560	2560	2560	1280	
243655 A/Kyiv/2217/2024		5a.2a (C.1.9)	2024-03-05	MDCK1/MDCK1	1280	1280	1280	1280	640	
243201 A/Poland/27/2024		5a.2a (C.1.9)	2024-03-07	MDCK1	1280	1280	1280	1280	1280	
243541 A/Ukraine/2709/2024		5a.2a (C.1.9)	2024-03-11	MDCK1	2560	2560	2560	2560	1280	
243540 A/Ukraine/2702/2024		5a.2a (C.1.9)	2024-03-12	MDCK1	2560	2560	1280	1280	1280	
243539 A/Ukraine/2705/2024		5a.2a (C.1.9)	2024-03-12	MDCK1	1280	1280	1280	1280	640	
243650 A/Kyiv/2387/2024		5a.2a (C.1.9)	2024-03-13	MDCK2/SIAT1	640	640	640	640	320	
243537 A/Ukraine/2728/2024		5a.2a (C.1.9)	2024-03-13	MDCK1	2560	2560	2560	2560	1280	
243649 A/Kyiv/2390/2024		5a.2a (C.1.9)	2024-03-14	MDCK2/SIAT1	2560	1280	2560	2560	1280	
243498 A/Firq/7241871/2024		5a.2a (C.1.9)	2024-03-20	MDCK1	1280	1280	1280	1280	640	
< relates to the lowest dilution of antiserum used					Vaccine	Vaccine				
1° hvoerimmune sheeo serum: ND =					SH 2023	NH 2023-24 SH 2024 NH 2024-25				

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-07-17)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
							Post-infection ferret antiserum				
							A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 A/Vic/4897/22 MDCK	IVR-238 A/Wisconsin/67/2022 Egg	A/Wisconsin/67/2022 MDCK
							F46/22	F09/24	F05/23	F07/23	F17/23
							5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES											
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
A/Lisboa/188/2023	K169Q, T216A, D506E			5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	1280	2560	2560	1280
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	2560	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560
TEST VIRUSES											
243796	A/BanskaBystrica/50/2024			5a.2a (C.1)	2024-01-12	MDCKx/MDCK1	1280	1280	1280	1280	640
243794	A/BanskaBystrica/84/2024			5a.2a (C.1)	2024-01-16	MDCKx/MDCK1	2560	2560	2560	2560	1280
243842	A/Switzerland/48148/2024			5a.2a (C.1)	2024-03-04	MDCK1/MDCK1	2560	1280	2560	2560	1280
243840	A/Switzerland/23716/2024			5a.2a (C.1)	2024-04-03	MDCK1	2560	2560	2560	2560	1280
243795	A/BanskaBystrica/51/2024			5a.2a (C.1.8)	2024-01-15	MDCKx/MDCK1	640	640	640	320	320
243792	A/SlovenskaLupca/127/2024			5a.2a (C.1.8)	2024-01-22	MDCKx/MDCK1	320	320	320	320	160
243791	A/BanskaBystrica/126/2024			5a.2a (C.1.8)	2024-01-23	MDCKx/MDCK1	1280	1280	2560	2560	1280
243789	A/Bratislava/79/2024			5a.2a (C.1.8)	2024-01-31	MDCKx/MDCK1	1280	1280	640	1280	640
243845	A/Switzerland/16031/2024			5a.2a (C.1.8)	2024-02-08	MDCK1	2560	2560	2560	2560	1280
243788	A/Bratislava/125/2024			5a.2a (C.1.8)	2024-02-15	MDCKx/MDCK1	2560	1280	2560	1280	640
243797	A/BanskaBystrica/52/2024			5a.2a (C.1.9)	2024-01-08	MDCKx/MDCK1	1280	1280	1280	1280	1280
243793	A/BanskaBystrica/85/2024			5a.2a (C.1.9)	2024-01-16	MDCKx/MDCK1	2560	1280	2560	2560	1280
243790	A/BanskaBystrica/105/2024			5a.2a (C.1.9)	2024-01-24	MDCKx/MDCK1	2560	1280	1280	1280	1280
243846	A/Switzerland/67627/2024			5a.2a (C.1.9)	2024-02-02	MDCK1	2560	2560	2560	2560	1280
243844	A/Switzerland/68478/2024			5a.2a (C.1.9)	2024-02-23	MDCK1	1280	1280	2560	1280	1280
243843	A/Switzerland/92480/2024			5a.2a (C.1.9)	2024-02-27	MDCK1	1280	640	1280	1280	640
243841	A/Switzerland/59420/2024			5a.2a (C.1.9)	2024-03-05	MDCK1	2560	1280	2560	1280	1280
< relates to the lowest dilution of antiserum used * hyperimmune sheep serum; ND =							Vaccine SH 2023			Vaccine NH 2023-24 SH 2024 NH 2024-25	

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Viruses	Other Information	Haemagglutination inhibition titre										
		Post-infection ferret antiserum										
		A/Sydney/18/2023	Al/Lisboa/487/2022	A/Victoria/487/2022	NIR-238 A/Wisconsin/67/2022	A/Wisconsin/67/2022	A/Wisconsin/67/2022					
		MDCK	MDCK	MDCK	Egg	Egg						
		Passage	F46/22	F08/24	F05/23	F07/23	F07/23					
Ferret number		Sa.2a (C.1-1)		Sa.2a (C.1-3)		Sa.2a.1 (D)		Sa.2a.1 (D)		Sa.2a.1 (D)		
Genetic group		Collection Date		Genetic group		Sa.2a (C.1-1)		Sa.2a (C.1-3)		Sa.2a.1 (D)		
REFERENCE VIRUSES												
A/Sydney/5/2021	K540, A186T, E224A, R259K, D368R, D54N, T216A	Sa.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	2560	1280	2560	
Al/Lisboa/188/2023	K1690, T216A, D506E	Sa.2a (C.1-9)	2021-11-22	SIAT1/MDCK2	2560	2560	2560	2560	2560	2560	640	
A/Victoria/487/2022	P1375, K1426, T216A, D260E, T277A	Sa.2a (D)	2022-10-02	SIAT1/MDCK3	2560	2560	2560	2560	2560	2560	640	
NIR-238 (A/Victoria/487/2022)	P1375, K1426, T216A, D506E, T277A (R142K, Q223R)	Sa.2a.1 (D)	2022-10-02	ES1/DE1	1280	1280	1280	1280	1280	2560	2560	
A/Wisconsin/67/2022		Sa.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560	2560	2560	
TEST VIRUSES												
242985	Al/Albania/30203/2024	Sa.2a (C.1)	2024-02-04	MDCK1	2560	2560	5120	2560	2560	1280	1280	
243505	Al/Albania/29986/12024	Sa.2a (C.1-8)	2024-01-04	MDCK1	2560	2560	2560	2560	2560	2560	1280	
243919	A/Serbia/136/2024	Sa.2a (C.1-8)	2024-01-10	C1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243917	A/Serbia/338/2024	Sa.2a (C.1-8)	2024-01-18	C1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
242986	Al/Albania/30203/2024	Sa.2a (C.1-8)	2024-02-04	MDCK1	2560	2560	2560	2560	2560	2560	1280	
242983	Al/Albania/30203/2024	Sa.2a (C.1-8)	2024-02-04	MDCK1	5120	5120	5120	5120	5120	5120	5120	
243887	Al/Mauritius/11147/14/2024	Sa.2a (C.1-8)	2024-04-19	MDCK1	2560	1280	2560	2560	2560	640	640	
242888	Al/Mauritius/11147/21/2024	Sa.2a (C.1-8)	2024-04-19	MDCK1	2560	2560	1280	1280	1280	1280	1280	
243868	Al/Mauritius/11353/47/2024	Sa.2a (C.1-8)	2024-05-28	MDCK1	2560	2560	2560	2560	2560	2560	1280	
243867	Al/Mauritius/11353/48/2024	Sa.2a (C.1-8)	2024-05-28	MDCK1	2560	2560	2560	2560	2560	2560	1280	
243901	Al/Albania/29987/12023	Sa.2a (C.1-9)	2023-12-19	MDCK1	5120	2560	2560	2560	2560	2560	1280	
243903	Al/Albania/29991/12024	Sa.2a (C.1-9)	2024-01-04	MDCK1	2560	1280	2560	2560	2560	2560	640	
243901	Al/Albania/30029/2024	Sa.2a (C.1-9)	2024-01-09	MDCK1	5120	2560	2560	2560	2560	2560	1280	
242999	Al/Albania/30049/2024	Sa.2a (C.1-10)	2024-01-10	MDCK1	5120	2560	2560	2560	5120	2560	2560	
242997	Al/Albania/30030/2024	Sa.2a (C.1-9)	2024-01-11	MDCK1	5120	2560	5120	5120	5120	2560	1280	
242992	Al/Albania/30047/2024	Sa.2a (C.1-10)	2024-01-16	MDCK1	2560	2560	2560	2560	2560	2560	1280	
243915	A/Serbia/16/2024	Sa.2a (C.1-9)	2024-02-01	C1/MDCK1	5120	5120	5120	5120	5120	5120	2560	
242984	Al/Albania/30203/2024	Sa.2a (C.1-9)	2024-02-04	MDCK1	2560	2560	2560	2560	2560	2560	1280	
242981	Al/Albania/30203/2024	Sa.2a (C.1-9)	2024-02-05	MDCK1	2560	2560	2560	2560	2560	2560	1280	
243913	A/Serbia/1244/2024	Sa.2a (C.1-9)	2024-02-12	C1/MDCK1	5120	2560	2560	2560	2560	2560	1280	
243981	Al/Mauritius/111459/2024	Sa.2a (C.1-9)	2024-03-25	MDCK1	2560	2560	2560	2560	2560	2560	1280	
244004	Sa/SouthAfrica/R20480/2024	Sa.2a (C.1-9)	2024-04-05	MDCK1/MDCK1	5120	5120	5120	5120	5120	5120	5120	
244002	Sa/SouthAfrica/R30030/2024	Sa.2a (C.1-9)	2024-04-24	MDCK1/MDCK1	1280	1280	1280	1280	1280	1280	640	
244001	Al/Mauritius/11356/2024	Sa.2a (C.1-9)	2024-04-29	MDCK1	2560	2560	2560	2560	2560	2560	1280	
243880	Al/Mauritius/11356/2024	Sa.2a (C.1-9)	2024-04-29	MDCK1	2560	2560	2560	2560	2560	2560	1280	
244000	Sa/SouthAfrica/R3035/24	Sa.2a (C.1-9)	2024-04-30	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243998	Sa/SouthAfrica/R3089/24	Sa.2a (C.1-9)	2024-05-13	MDCK1/MDCK1	5120	2560	2560	2560	2560	2560	1280	
243996	Sa/SouthAfrica/R3032/2024	Sa.2a (C.1-9)	2024-05-14	MDCK1/MDCK1	5120	5120	5120	5120	5120	5120	2560	
243994	Sa/SouthAfrica/R30987/24	Sa.2a (C.1-9)	2024-05-16	MDCK1/MDCK1	5120	5120	5120	5120	5120	2560	2560	
243992	Sa/SouthAfrica/R0413/124	Sa.2a (C.1-9)	2024-05-20	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243990	Sa/SouthAfrica/R0415/24	Sa.2a (C.1-9)	2024-05-20	MDCK1/MDCK1	5120	5120	5120	2560	2560	2560	1280	
243988	Sa/SouthAfrica/R0419/2024	Sa.2a (C.1-9)	2024-05-20	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243986	Sa/SouthAfrica/R0420/124	Sa.2a (C.1-9)	2024-05-21	MDCK1/MDCK1	5120	2560	2560	2560	2560	2560	1280	
243984	Sa/SouthAfrica/R04357/24	Sa.2a (C.1-9)	2024-05-22	MDCK1/MDCK1	1280	1280	1280	1280	1280	1280	640	
243982	Sa/SouthAfrica/R04380/24	Sa.2a (C.1-9)	2024-05-23	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243980	Sa/SouthAfrica/R04387/24	Sa.2a (C.1-9)	2024-05-23	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243978	Sa/SouthAfrica/R04387/24	Sa.2a (C.1-9)	2024-05-24	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243970	Al/Mauritius/1135316/2024	Sa.2a (C.1-9)	2024-05-25	MDCK1	1280	2560	2560	2560	2560	2560	1280	
243969	Al/Mauritius/1135313/2024	Sa.2a (C.1-9)	2024-05-27	MDCK1	640	640	640	640	640	640	640	
243976	Sa/SouthAfrica/R04552/24	Sa.2a (C.1-9)	2024-05-27	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243974	Sa/SouthAfrica/R04561/24	Sa.2a (C.1-9)	2024-05-31	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243972	Sa/SouthAfrica/R04859/24	Sa.2a (C.1-9)	2024-06-01	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560	1280	
243968	Sa/SouthAfrica/R04870/24	Sa.2a (C.1-9)	2024-06-04	MDCK2/MDCK1	5120	2560	2560	2560	2560	2560	1280	
243966	Sa/SouthAfrica/R04880/24	Sa.2a (C.1-9)	2024-06-05	MDCK1/MDCK1	5120	2560	5120	2560	5120	2560	1280	
243921	A/Serbia/17/2024	Sa.2a.1 (D.1)	2024-01-04	C1/MDCK1	2560	1280	2560	2560	2560	2560	2560	
243956	Al/Albania/300401/2024	Sa.2a.1 (D.1)	2024-01-14	MDCK1	1280	1280	2560	2560	1280	1280	1280	
243002	Al/Albania/299981/2024	Sa.2a.1 (D.3)	2024-01-08	MDCK1	640	640	2560	2560	1280	1280	1280	
< relates to the lowest dilution of antiserum used												
† Inocritimmune sheo serum: N= 0												
					Vaccine SH 2023		Vaccine NI 2023-24 SH 2024		Vaccine NI 2024-25			

[†] hyperimmune sheep serum; ND =

[†] hyperimmune sheep serum; ND =Vaccine
SH 2023Vaccine
NH 2023-24

NH 2023-24

SH 2024
NW 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-08-06)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
							Post-infection ferret antiserum				
							A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 A/Vic/4897/22 MDCK	IVR-238 A/Wisconsin/67/2022 Egg	A/Wisconsin/67/2022 MDCK
							F46/22	F09/24	F05/23	F07/23	F17/23
							5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES											
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
A/Lisboa/188/2023	K169Q, T216A, D506E			5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK2	2560	2560	2560	2560	1280
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (D)	2022-10-02	E3/D6/E1	1280	1280	1280	2560	640
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	5120	2560
TEST VIRUSES											
244008	A/SouthAfrica/R02344/24			5a.2a (C.1.8)	2024-03-26	MDCK1/MDCK1	2560	2560	2560	2560	1280
244119	A/Ambositra/01885/2024			5a.2a (C.1.8)	2024-06-18	MDCK1	1280	1280	1280	1280	640
244117	A/Toamasina/01984/2024			5a.2a (C.1.8)	2024-06-19	MDCK1	2560	2560	2560	2560	2560
244116	A/Antananarivo/01916/2024			5a.2a (C.1.8)	2024-06-20	MDCK1	2560	2560	2560	2560	1280
244115	A/Toamasina/01947/2024			5a.2a (C.1.8)	2024-06-20	MDCK1	2560	2560	2560	2560	1280
244113	A/Toamasina/01961/2024			5a.2a (C.1.8)	2024-06-23	MDCK1	2560	2560	1280	2560	640
244112	A/Moramanga/01994/2024			5a.2a (C.1.8)	2024-06-24	MDCK1	1280	1280	1280	1280	640
244109	A/Moramanga/01996/2024			5a.2a (C.1.8)	2024-06-28	MDCK1	2560	1280	2560	2560	1280
244104	A/Ihosey/02027/2024			5a.2a (C.1.8)	2024-07-03	MDCK1	2560	2560	2560	2560	1280
244101	A/Moramanga/02084/2024			5a.2a (C.1.8)	2024-07-04	MDCK1	2560	1280	1280	1280	640
244010	A/SouthAfrica/R01727/24			5a.2a (C.1.9)	2024-03-06	MDCK1/MDCK1	1280	1280	640	1280	640
244006	A/SouthAfrica/R02440/24			5a.2a (C.1.9)	2024-04-02	MDCK1/MDCK1	2560	2560	2560	2560	1280
244118	A/Antsirabe/01918/2024			5a.2a (C.1.9)	2024-06-19	MDCK1	2560	2560	2560	2560	1280
244114	A/Antananarivo/01900/2024			5a.2a (C.1.9)	2024-06-21	MDCK1	2560	2560	2560	2560	640
244111	A/Toliary/02038/2024			5a.2a (C.1.9)	2024-06-25	MDCK1	2560	2560	5120	2560	2560
244108	A/Antsirabe/02030/2024			5a.2a (C.1.9)	2024-07-01	MDCK1	2560	2560	2560	2560	1280
244107	A/Antananarivo/02019/2024			5a.2a (C.1.9)	2024-07-02	MDCK1	1280	1280	1280	1280	640
244106	A/Antsirabe/02032/2024			5a.2a (C.1.9)	2024-07-02	MDCK1	2560	2560	2560	2560	1280
244102	A/Antananarivo/50005/2024			5a.2a (C.1.9)	2024-07-03	MDCK1	5120	2560	2560	2560	1280
244100	A/Itasy/02064/2024			5a.2a (C.1.9)	2024-07-05	MDCK1	640	1280	640	640	320
244099	A/Antananarivo/02098/2024			5a.2a (C.1.9)	2024-07-08	MDCK1	2560	2560	2560	2560	1280
244034	A/Argentina/1285/2024			5a.2a.1 (D)	2024-06-07	MDCK1	2560	2560	5120	2560	2560
244155	A/Moscow/22/2024			no sequence	2024-02-21	SIAT1/MDCK1	1280	1280	1280	1280	640
244154	A/Moscow/28/2024			no sequence	2024-04-03	SIAT1/MDCK1	1280	1280	1280	1280	640
< relates to the lowest dilution of antiserum used							Vaccine		Vaccine		
¹ hvoerimmune sheeo serum: ND =							SH 2023		NH 2023-24 SH 2024 NH 2024-25		

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-15. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-08-13)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
							Post-infection ferret antiserum				
							A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/67/2022 MDCK
							F46/22	F09/24	F05/23	F07/23	F17/23
							5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES											
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
A/Lisboa/188/2023	K169Q, T216A, D506E			5a.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	2560	2560
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	2560	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (D)	2022-10-02	E3/D6/E1	2560	1280	2560	2560	640
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560
TEST VIRUSES											
244182 A/England/137/2023				5a.2a (C.1.8)	2023-12-19	MDCK1/MDCK1	2560	2560	2560	2560	1280
244202 A/Hungary/165/2024				5a.2a (C.1.8)	2024-01-31	MDCK1/MDCK1	2560	2560	2560	2560	1280
244179 A/England/28/2024				5a.2a (C.1.8)	2024-02-05	SIAT1/MDCK1	2560	2560	2560	2560	1280
244122 A/Antananarivo/01878/2024				5a.2a (C.1.8)	2024-06-17	MDCK1	2560	2560	2560	2560	1280
244130 A/Italy/01819/2024				5a.2a (C.1.8)	2024-06-17	MDCK1	2560	1280	1280	640	640
244127 A/Italy/01829/2024				5a.2a (C.1.8)	2024-06-17	MDCK1	2560	2560	2560	2560	1280
244120 A/Italy/01813/2024				5a.2a (C.1.8)	2024-06-18	MDCK1	2560	2560	2560	2560	1280
244110 A/Tomasina/01960/2024				5a.2a (C.1.8)	2024-06-27	MDCK2	1280	1280	2560	1280	640
244146 A/Moramanga/02145/2024				5a.2a (C.1.8)	2024-07-09	SIAT1	2560	2560	2560	2560	1280
244145 A/Tomasina/02156/2024				5a.2a (C.1.8)	2024-07-11	SIAT1	2560	2560	2560	2560	1280
244144 A/Antananarivo/02160/2024				5a.2a (C.1.8)	2024-07-12	SIAT1	2560	2560	2560	2560	1280
244203 A/Hungary/140/2024				5a.2a (C.1.9)	2024-01-22	MDCK1/MDCK1	2560	2560	2560	2560	1280
244206 A/Hungary/142/2024				5a.2a (C.1.9)	2024-01-29	MDCK1/SIAT1	5120	5120	5120	5120	2560
244201 A/Hungary/139/2024				5a.2a (C.1.9)	2024-02-01	MDCK1/MDCK1	2560	2560	2560	2560	1280
244178 A/England/54/2024				5a.2a (C.1.9)	2024-02-20	MDCK1/MDCK1	2560	2560	2560	2560	1280
244200 A/Hungary/243/2024				5a.2a (C.1.9)	2024-03-04	MDCK1/MDCK1	2560	2560	2560	2560	1280
244136 A/Moramanga/01426/2024				5a.2a (C.1.9)	2024-05-08	MDCK1	5120	5120	5120	2560	1280
244199 A/Hungary/277/2024				5a.2a (C.1.9)	2024-05-13	MDCK1/MDCK1	2560	2560	2560	2560	1280
244219 A/Catalonia/NSVH172312537/2024				5a.2a (C.1.9)	2024-05-15	SIAT2/MDCK1	5120	5120	5120	5120	2560
244135 A/Antananarivo/01609/2024				5a.2a (C.1.9)	2024-05-30	MDCK1	1280	1280	2560	1280	640
244147 A/Tomasina/01641/2024				5a.2a (C.1.9)	2024-06-03	SIAT1	5120	5120	5120	5120	1280
244133 A/Antananarivo/01775/2024				5a.2a (C.1.9)	2024-06-12	MDCK1	2560	5120	5120	2560	1280
244132 A/Moramanga/01870/2024				5a.2a (C.1.9)	2024-06-13	MDCK1	5120	5120	5120	5120	2560
244124 A/Antananarivo/01849/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	2560	2560	5120	2560	1280
244123 A/Antananarivo/01850/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	2560	2560	2560	2560	1280
244131 A/Italy/01809/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	2560	2560	2560	1280	640
244129 A/Italy/01823/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	2560	2560	2560	2560	1280
244128 A/Italy/01824/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	1280	2560	640	1280	640
244126 A/Italy/01841/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	2560	2560	5120	2560	1280
244125 A/Italy/01843/2024				5a.2a (C.1.9)	2024-06-17	MDCK2	2560	2560	2560	2560	1280
244121 A/Taolagnaro/02106/2024				5a.2a (C.1.9)	2024-06-17	MDCK1	2560	2560	2560	2560	1280
244105 A/Tsiranomandidy/02035/2024				5a.2a (C.1.9)	2024-07-02	MDCK1	2560	1280	1280	1280	640
244216 A/Catalonia/NSVH102379173/2024				5a.2a (C.1.9)	2024-07-05	SIAT2/MDCK1	2560	2560	2560	2560	1280
244221 A/Catalonia/NSVH102332755/2024				5a.2a.1 (D)	2024-05-01	SIAT2/MDCK1	2560	2560	5120	2560	2560
244214 A/Catalonia/NSVH102380079/2024				5a.2a.1 (D)	2024-07-07	SIAT2/MDCK1	2560	2560	5120	2560	2560
244180 A/England/53/2024				5a.2a.1 (D.1)	2024-01-30	SIAT1/MDCK1	1280	1280	2560	1280	1280
244181 A/England/31/2024				5a.2a.1 (D.2)	2024-01-19	MDCK1/MDCK1	1280	1280	5120	2560	2560
< relates to the lowest dilution of antiserum used							Vaccine				
¹ hvoerimmune sheeo serum: ND = Not Done							SH 2023				
							NH 2023-24				
							SH 2024				
							NH 2024-25				

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-16. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-08-28)

Haemagglutination inhibition titre									
Viruses	Other information	Post-infection ferret antisera							
		Passage		A/Sydney/621/2009		A/Victoria/1193/2009		A/Wisconsin/329/2009	
		MDCK	MDCK	MDCK	MDCK	Egg	MDCK	MDCK	MDCK
		Ferret number	F46/22	F08/24	F05/23	F07/23	F17/23		
Genetic group	Collection Date	Genetic group	Sa.2a (C.1)	Sa.2a (C.1.9)	Sa.2a.1 (D)	Sa.2a.1 (D)	Sa.2a.1 (C.1.1)		
REFERENCE VIRUSES									
A/Sydney/621/2009	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	Sa.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
A/Labna/189/2023	K169Q, T216A, D506E	Sa.2a (C.1.9)	2023-11-22	SIAT1/MDCK3	2560	2560	2560	2560	1280
A/India/4897/2022	P137S, K142R, T216A, D260E, T277A	Sa.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	2560
VIR-23a (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142Q, Q223R)	Sa.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	1280	2560	640
A/Wisconsin/87/2022		Sa.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	2560	5120	2560	2560
TEST VIRUSES									
244292 A/Belgium/183/2024		Sa.2a (C.1)	2024-01-23	MDCK1	2560	2560	2560	2560	1280
244447 A/Bucuresi/9567/4620/2024		Sa.2a (C.1)	2024-03-05	SIAT1/MDCK1	5120	2560	2560	2560	2560
244337 A/Belgium/3460/2024		Sa.2a (C.1)	2024-03-25	MDCK1	2560	2560	2560	2560	1280
244452 A/Bucuresi/57040/2024		Sa.2a (C.1.8)	2024-02-13	SIAT1/MDCK1	2560	2560	2560	2560	1280
244286 A/Belgium/337/2024		Sa.2a (C.1.8)	2024-03-31	MDCK1	1280	1280	1280	1280	640
244439 A/Denmark/2116/2024		Sa.2a (C.1.9)	2024-06-25	MDCK3/MDCK1	2560	1280	2560	2560	1280
244465 A/Ara/565183/2024		Sa.2a (C.1.9)	2024-01-31	SIAT1/MDCK1	5120	5120	5120	2560	2560
244464 A/Ara/558363/2024		Sa.2a (C.1.9)	2024-02-01	SIAT1/MDCK1	2560	2560	2560	2560	1280
244463 A/Ara/558365/2024		Sa.2a (C.1.9)	2024-02-01	SIAT1/MDCK1	2560	2560	2560	2560	1280
244462 A/Ara/558367/2024		Sa.2a (C.1.9)	2024-02-01	SIAT1/MDCK1	5120	5120	5120	2560	1280
244461 A/Ara/558370/2024		Sa.2a (C.1.9)	2024-02-01	SIAT1/MDCK1	5120	5120	5120	2560	1280
244460 A/Ara/558373/2024		Sa.2a (C.1.9)	2024-02-01	SIAT1/MDCK1	1280	1280	1280	1280	640
244459 A/Ara/558400/2024		Sa.2a (C.1.9)	2024-02-05	SIAT1/MDCK1	5120	5120	2560	2560	1280
244458 A/Ara/558483/2024		Sa.2a (C.1.9)	2024-02-05	SIAT1/MDCK1	2560	1280	1280	1280	640
244457 A/Alsa/156657/2024		Sa.2a (C.1.9)	2024-02-05	SIAT1/MDCK1	2560	2560	2560	2560	1280
244456 A/Ara/558484/2024		Sa.2a (C.1.9)	2024-02-06	SIAT1/MDCK1	2560	2560	2560	2560	1280
244455 A/Ara/558595/2024		Sa.2a (C.1.9)	2024-02-06	SIAT1/MDCK1	2560	2560	2560	2560	1280
244454 A/Ara/558340/2024		Sa.2a (C.1.9)	2024-02-08	SIAT1/MDCK1	2560	2560	2560	2560	1280
244453 A/Bucuresi/957060/2024		Sa.2a (C.1.9)	2024-02-08	SIAT1/MDCK1	2560	2560	2560	2560	1280
244451 A/Ara/558260/2024		Sa.2a (C.1.9)	2024-02-14	SIAT1/MDCK1	2560	2560	2560	2560	1280
244450 A/Ara/558261/2024		Sa.2a (C.1.9)	2024-02-14	SIAT1/MDCK1	2560	2560	2560	2560	1280
244449 A/Ara/567331/2024		Sa.2a (C.1.9)	2024-02-28	SIAT1/MDCK1	2560	2560	2560	2560	1280
244448 A/Ara/567133/2024		Sa.2a (C.1.9)	2024-02-28	SIAT1/MDCK1	5120	2560	2560	2560	1280
244289 A/Belgium/324/2024		Sa.2a (C.1.9)	2024-03-18	MDCK1	2560	2560	2560	2560	1280
242828 A/Belgium/3256/2024		Sa.2a (C.1.9)	2023-01-21	MDCK1	1280	1280	1280	1280	640
244267 A/Cameroon/3391/2024		Sa.2a (C.1.9)	2024-04-01	MDCK2	2560	2560	2560	2560	1280
24265 A/Cameroon/3400/2024		Sa.2a (C.1.9)	2024-04-01	MDCK1	2560	2560	2560	2560	1280
242426 A/Cameroon/3615/2024		Sa.2a (C.1.9)	2024-04-08	MDCK3	2560	2560	2560	2560	1280
243115 A/SaudiArabia/9960/2024		Sa.2a (C.1.9)	2024-05-05	MDCK1	2560	2560	2560	2560	1280
244438 A/Denmark/2114/2024		Sa.2a (C.1.9)	2024-05-14	MDCK3/MDCK1	5120	2560	2560	2560	1280
244437 A/Denmark/2166/2024		Sa.2a (C.1.9)	2024-05-15	MDCK3/MDCK1	2560	2560	2560	2560	1280
242263 A/Cameroon/9552/2024		Sa.2a (C.1.9)	2024-05-16	MDCK1	1280	1280	1280	1280	640
244422 A/Lyon/RELAB.182.16.58/2024		Sa.2a (C.1.9)	2024-05-21	MDCK3/MDCK1	2560	1280	2560	2560	1280
242462 A/Cameroon/978/2024		Sa.2a (C.1.9)	2024-06-04	MDCK1	1280	1280	1280	1280	640
244430 A/Cameroon/8377/2024		Sa.2a (C.1.9)	2024-06-06	MDCK1	2560	2560	2560	2560	1280
242420 A/Cameroon/9385/2024		Sa.2a (C.1.9)	2024-06-08	MDCK1	5120	5120	5120	2560	1280
242429 A/Cameroon/6545/2024		Sa.2a (C.1.9)	2024-06-12	MDCK1	2560	2560	5120	2560	2560
244420 A/Lyon/RELAB.24.16.73/2024		Sa.2a (C.1.9)	2024-06-15	MDCK2/MDCK1	1280	1280	1280	1280	320
243113 A/SaudiArabia/752/2024		Sa.2a (C.1.9)	2024-06-21	MDCK1	2560	2560	2560	2560	1280
244436 A/Denmark/2204/2024		Sa.2a (C.1.9)	2024-06-24	MDCK3/MDCK1	2560	2560	2560	2560	1280
243112 A/SaudiArabia/7680/2024		Sa.2a (C.1.9)	2024-06-24	MDCK1	2560	2560	2560	2560	1280
244419 A/Lyon/RELAB.24.16.58/2024		Sa.2a (C.1.9)	2024-06-27	MDCK3/MDCK1	2560	2560	2560	2560	1280
244446 A/Ara/57544/2024		Sa.2a (C.1.9)	2024-06-28	SIAT1/MDCK1	2560	2560	2560	2560	1280
243110 A/SaudiArabia/13107/2024		Sa.2a (C.1.9)	2024-06-30	MDCK1	2560	2560	5120	2560	1280
243310 A/SaudiArabia/428/2024		Sa.2a (C.1.9)	2024-07-03	MDCK1	2560	2560	2560	2560	1280
242457 A/Cameroon/7186/2024		Sa.2a (C.1.9)	2024-07-05	MDCK1	2560	2560	2560	2560	1280
244435 A/Denmark/2233/2024		Sa.2a (C.1.9)	2024-07-06	MDCK3/MDCK1	1280	1280	1280	2560	1280
244430 A/Belgium/3210/2024		Sa.2a.1 (D)	2024-03-18	MDCK1	1280	1280	5120	2560	1280
244421 A/Lyon/CHURU.24.182.58/2024		Sa.2a.1 (D)	2024-06-06	MDCK3/MDCK1	1280	1280	2560	1280	1280
< relates to the lowest dilution of antiserum used									
1 Inverimmune sheo serum: ND =									
				Vaccine	Vaccine				
				SH 2023	NH 2023-24				
					SH 2024				
					NH 2024-25				

Table H1-17. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-09-06)

Viruses		Other information	Haemagglutination inhibition titre							
			Post-infection ferret antiserum							
			A/Sydney/5/21 MDCK	A/Lisboa/188/2023 MDCK	A/Victoria/4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/67/2022 MDCK			
		Passage	Genetic group	Collection Date	Genetic group	F46/22 Sa.2a (C.1.9)	F09/24 Sa.2a (C.1)	F05/23 Sa.2a.1 (D)	F07/23 Sa.2a.1 (D)	F17/23 Sa.2a.1 (C.1.1)
REFERENCE VIRUSES										
	A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	Sa.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	2560	2560	2560	1280
	A/Lisboa/188/2023	K169Q, T216A, D506E	Sa.2a (C.1.9)	2021-11-22	SIAT1/MDCK3	2560	2560	2560	2560	1280
	A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	Sa.2a.1 (D)	2022-10-02	SIAT2/MDCK3	2560	2560	5120	2560	2560
	IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	Sa.2a.1 (D)	2022-10-02	E3/D6/E1	2560	2560	2560	2560	1280
	A/Wisconsin/67/2022		Sa.2a.1 (C.1.1)	2022-10-25	MDCK2	1280	1280	2560	1280	1280
TEST VIRUSES										
244508	A/Lisboa/189/2023		Sa.2a (C.1)	2023-11-30	SIAT1	5120	5120	5120	2560	1280
244507	A/Lisboa/313/2023		Sa.2a (C.1)	2023-12-30	SIAT1	1280	1280	2560	1280	1280
244674	A/Nord_Pas_de_Calais/04703/2024		Sa.2a (C.1)	2024-03-12	C1/MDCK1	1280	1280	1280	1280	640
244379	A/Spain/1957/2024		Sa.2a (C.1.8)	2024-02-13	MDCK1	1280	1280	1280	1280	640
244378	A/Spain/1958/2024		Sa.2a (C.1.8)	2024-02-14	MDCK1	2560	2560	2560	2560	2560
244377	A/Spain/1963/2024		Sa.2a (C.1.8)	2024-02-16	MDCK1	1280	1280	2560	1280	1280
244688	A/Bourgogne/04731/2024		Sa.2a (C.1.8)	2024-03-16	C1/MDCK1	2560	2560	2560	2560	1280
244293	A/Belgium/177/2024		Sa.2a (C.1.9)	2024-01-22	MDCK1	1280	2560	2560	2560	1280
244381	A/Spain/1886/2024		Sa.2a (C.1.9)	2024-02-10	MDCK1	2560	2560	2560	2560	1280
244600	A/HongKong/457/2024		Sa.2a (C.1.9)	2024-02-13	MDCK 1/MDCK1	2560	2560	2560	2560	1280
244598	A/HongKong/590/2024		Sa.2a (C.1.9)	2024-02-23	MDCK 1/MDCK1	2560	2560	2560	2560	1280
244672	A/Bourgogne/04729/2024		Sa.2a (C.1.9)	2024-03-12	C1/MDCK1	1280	1280	1280	1280	640
244670	A/Bretagne/04605/2024		Sa.2a (C.1.9)	2024-03-14	C1/MDCK1	5120	5120	5120	2560	2560
244666	A/Bretagne/04840/2024		Sa.2a (C.1.9)	2024-03-22	C1/MDCK1	2560	2560	2560	2560	1280
244664	A/Pays_de_Loire/04970/2024		Sa.2a (C.1.9)	2024-03-28	C1/MDCK1	1280	640	640	640	640
244660	A/Haute_Normandie/05151/2024		Sa.2a (C.1.9)	2024-04-04	C1/MDCK1	2560	2560	2560	2560	2560
244594	A/HongKong/1028/2024		Sa.2a (C.1.9)	2024-04-14	MDCK 2/MDCK1	2560	1280	1280	2560	1280
244278	A/Cameroon/4233/2024		Sa.2a (C.1.9)	2024-04-25	SIAT1	5120	5120	5120	2560	2560
244374	A/Spain/2542/2024		Sa.2a (C.1.9)	2024-05-10	MDCK1	2560	1280	1280	1280	640
244627	A/Croatia/2720/2024		Sa.2a (C.1.9)	2024-05-13	MDCK1	2560	2560	2560	2560	1280
244372	A/Spain/2546/2024		Sa.2a (C.1.9)	2024-05-31	MDCK1	2560	2560	2560	2560	1280
244592	A/HongKong/1567/2024		Sa.2a (C.1.9)	2024-06-02	MDCK 1/MDCK1	2560	2560	2560	2560	1280
244371	A/Spain/2544/2024		Sa.2a (C.1.9)	2024-06-03	MDCK1	1280	2560	2560	1280	640
244595	A/Lisboa/165/2024		Sa.2a (C.1.9)	2024-06-07	SIAT1	2560	2560	2560	1280	1280
244370	A/Spain/2566/2024		Sa.2a (C.1.9)	2024-06-12	MDCK1	2560	2560	2560	2560	1280
244590	A/HongKong/2070/2024		Sa.2a (C.1.9)	2024-07-15	MDCK 1/MDCK1	2560	1280	2560	2560	1280
244588	A/HongKong/2071/2024		Sa.2a (C.1.9)	2024-07-16	MDCK 1/MDCK1	2560	2560	2560	2560	1280
244586	A/HongKong/2098/2024		Sa.2a (C.1.9)	2024-07-18	MDCK 1/MDCK1	5120	2560	2560	2560	1280
244317	A/SaudiArabia/1509/2024		Sa.2a (C.1.9)	2024-07-21	SIAT3	2560	2560	2560	2560	2560
244584	A/HongKong/2149/2024		Sa.2a (C.1.9)	2024-07-22	MDCK 1/MDCK1	5120	5120	5120	5120	5120
244582	A/HongKong/2150/2024		Sa.2a (C.1.9)	2024-07-22	MDCK 1/MDCK1	2560	2560	2560	2560	1280
244375	A/Spain/2517/2024		Sa.2a.1 (D)	2024-05-05	MDCK1	1280	640	2560	1280	1280
244233	A/Catalonia/NSVH198299547/2024		Sa.2a.1 (D)	2024-05-08	SIAT2/SIAT1	1280	1280	2560	1280	1280
244373	A/Spain/2552/2024		Sa.2a.1 (D.1)	2024-05-30	MDCK1	1280	1280	2560	1280	1280
244662	A/Bretagne/05126/2024		Sa.2a.1 (D.2)	2024-03-29	MDCK1	2560	2560	5120	2560	2560
244380	A/Spain/1906/2024		Sa.2a.1 (D.3)	2024-02-13	MDCK1	1280	1280	2560	2560	2560
244376	A/Spain/2508/2024		Sa.2a.1 (D.4)	2024-04-17	MDCK1	1280	1280	2560	1280	1280
< relates to the lowest dilution of antiserum used 1 hyperimmune sheeo serum: ND = Not Done						Vaccine SH 2023	Vaccine NH 2023-24 SH 2024 NH 2024-25			

Table H1-18. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-09-11)

Viruses	Other information	Haemagglutination inhibition titre																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
		Passage	Post-infection ferret antiserum																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
			A/Sydney/ 5/21 MDCK	A/Lisboa/ 188/2023 MDCK	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
			F46/22	F09/24	F05/23	F07/23	F17/23																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
			Genetic group	Collection Date	Genetic group	5a.2a (C.1)	5a.2a (C.1.9)	5a.2a.1 (D)	5a.2a.1 (D)	5a.2a.1 (C.1.1)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Fold-reduction table: A/H1N1

Reference Virus	Clade	<4-fold difference		4-fold difference		>4-fold difference		Total
		Number	Percentage	Number	Percentage	Number	Percentage	
IVR-215 (A/Victoria/2570/2019) Egg	5a.2 (C)	38	100.0	0	0.0	0	0.0	38
A/Sydney/5/2021 Cell	5a.2a (C.1)	358	95.5	17	4.5	0	0.0	375
A/Sydney/5/2021 Egg	5a.2a (C.1)	38	100.0	0	0.0	0	0.0	38
A/Lisboa/188/2023 Cell	5a.2a.1 (C.1)	340	95.2	16	4.5	1	0.3	357
A/Wisconsin/67/2022 Cell	5a.2a.1 (C.1.1)	303	80.8	67	17.8	5	1.3	375
A/Victoria/4897/2022 Cell	5a.2a.1 (D)	329	87.7	42	11.2	4	1.0	375
IVR-238 (A/Victoria/4897/2022) Egg	5a.2a.1 (D)	365	97.3	9	2.4	1	0.3	375

34

A/H1N1: Egg isolates

Virus	Status	Genetic group (subclade) [Additional subst.]	Comment
A/Norway/31694/2022	NIB-133 (2-way pass)	5a.2a.1 (C.1.1)	Egg adaptation: (Q223R)
A/Catalonia/NSVH161512065/2022	NIB-134 (2-way pass)	5a.2a.1 (C.1.1)	Egg adaptation: (Q223R)

A/H1N1: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Sydney/5/2021	5a.2a (C.1)	MDCK3/MDCK4	F46/22
A/Victoria/4897/2022	5a.2a.1 (D)	SIAT2/MDCK3	F05/23
IVR-238 (A/Victoria/4897/2022)	5a.2a.1 (D)	E3/D6/E1	F07/23
A/Wisconsin/67/2022	5a.2a.1 (C.1.1)	MDCK2	F17/23
A/Lisboa/188/2023	5a.2a.1 (C.1)	SIAT1/MDCK2	F09/24

Summary: A/H1N1

Genetic analyses

6B.1A.5a.2a and 6B.1A.5a.2a.1 clade viruses both continued to circulate with differing relative proportions depending on region, with an overall predominance of 5a.2a viruses.

Globally, there is predominance of clade 5a.2a except in the Americas where there is a slight predominance of clade 5a.2a.1. In Europe, viruses from clade 5a.2a were detected with much higher frequency than clade 5a.2a.1.

Within the 5a.2a viruses, characterised by substitutions K54Q, A186T, E224A, R259K and K308R, several subclades were observed: subclade C.1 defined by substitution I418V which split into two further subclades: C.1.8, characterised by V47I with I96T in most sequences, and C.1.9, with substitution K169Q; the majority (>80%) of H1pdm viruses sequenced globally belong to subclade C.1.9, with viruses from C.1.8 as a minority subclade mostly observed in Europe. Minor subclades C.1.7 with substitution I533V and C.1.7.2 with T120A and K142R were circulating in Indonesia in low proportions.

Within the 5a.2a.1 viruses, characterised by substitutions P137S, K142R, D260E, T277A, E356D and N451H, there are two main groups of viruses: major subclade D (former C.1.1.1) with T216A represented by A/Victoria/4897/2022, which has split into four subclades: D.1 with R45K, D.2 with R113K and V427I, D.3 with T120A and I372V and D.4 with T120A. Of these, only subclade D circulates in significant proportions, albeit predominating only in the Americas. Subclade C.1.1 is a minor clade within 5a.2a.1 viruses with no additional substitutions represented by A/Wisconsin/67/2022, which has been detected in significant proportions only in Iceland.

Antigenic analyses

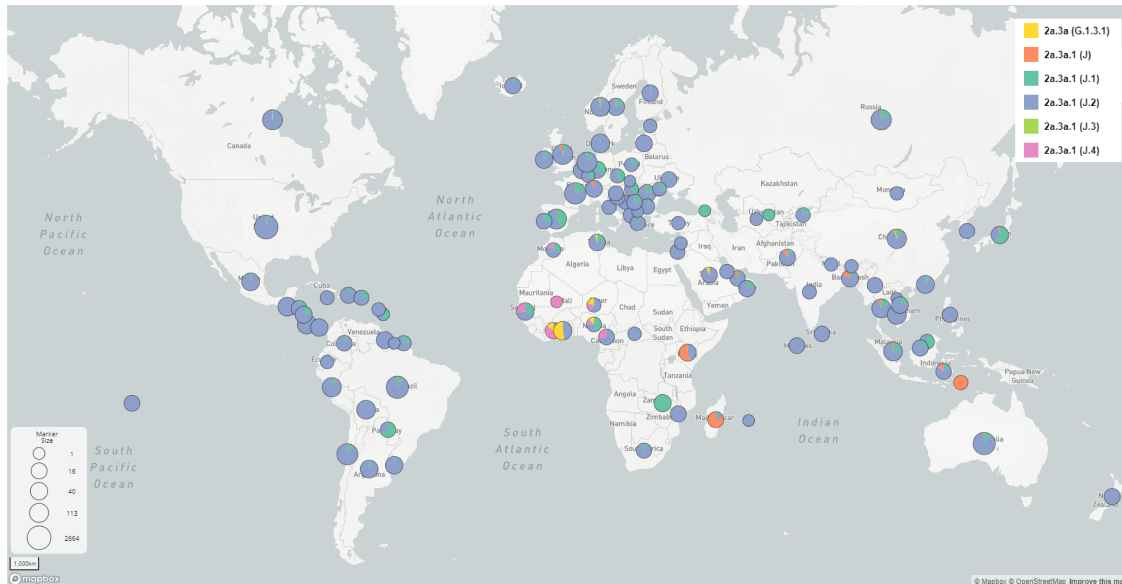
HI titres show that NH 2023-24, 2024-25 and SH 2024 strains, cell-based A/Wisconsin/67/2022 and egg-based A/Victoria/4897/2022, generally recognised both 5a.2a and 5a.2a.1 clade viruses well. Some four-fold drops were observed with the cell-based A/Wisconsin/67/2022 strain that were not seen with the egg-based A/Victoria/4897/2022.

Influenza A/H3N2

Genetic analyses

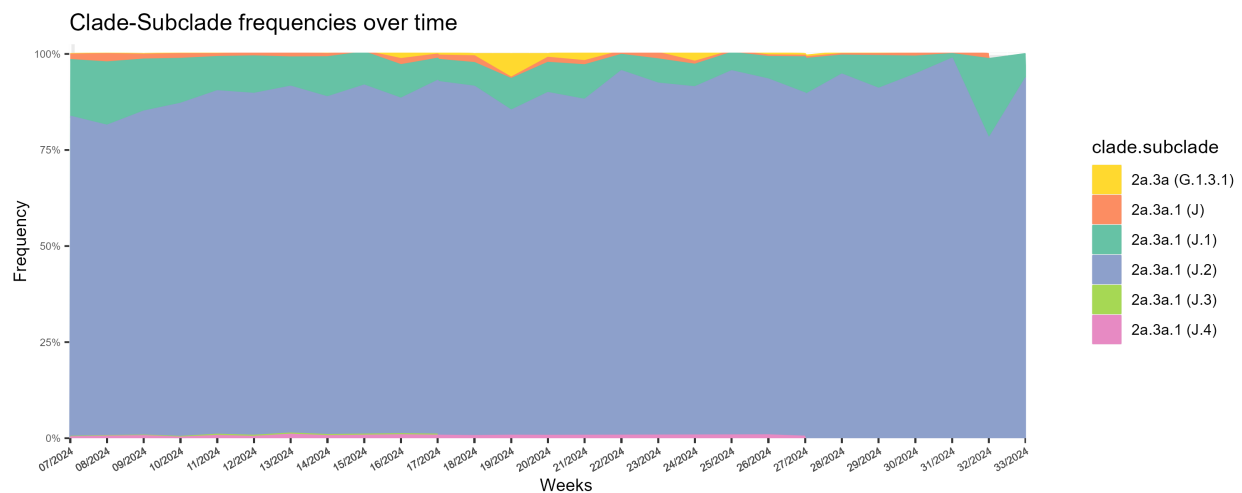
Global distribution of A/H3N2 subclades

Global geographical distribution of influenza A/H3N2 genetic clades (subclades) viruses with collection dates from 1st February 2024 to 31st August 2024, obtained with full HA sequences as deposited in GISAID, downloaded on 12th September 2024 and classified using [Nextclade](#). Map prepared with [Microreact](#). Geographic markers scaled to detection proportions.



Global time-dependent variation of A/H3N2 subclades

Global time-dependent variation in frequencies of genetic clades-subclades of A/H3N2 viruses collected from 1st February 2024 to 31st August 2024.



Maximum likelihood phylogenetic tree: A/H3N2

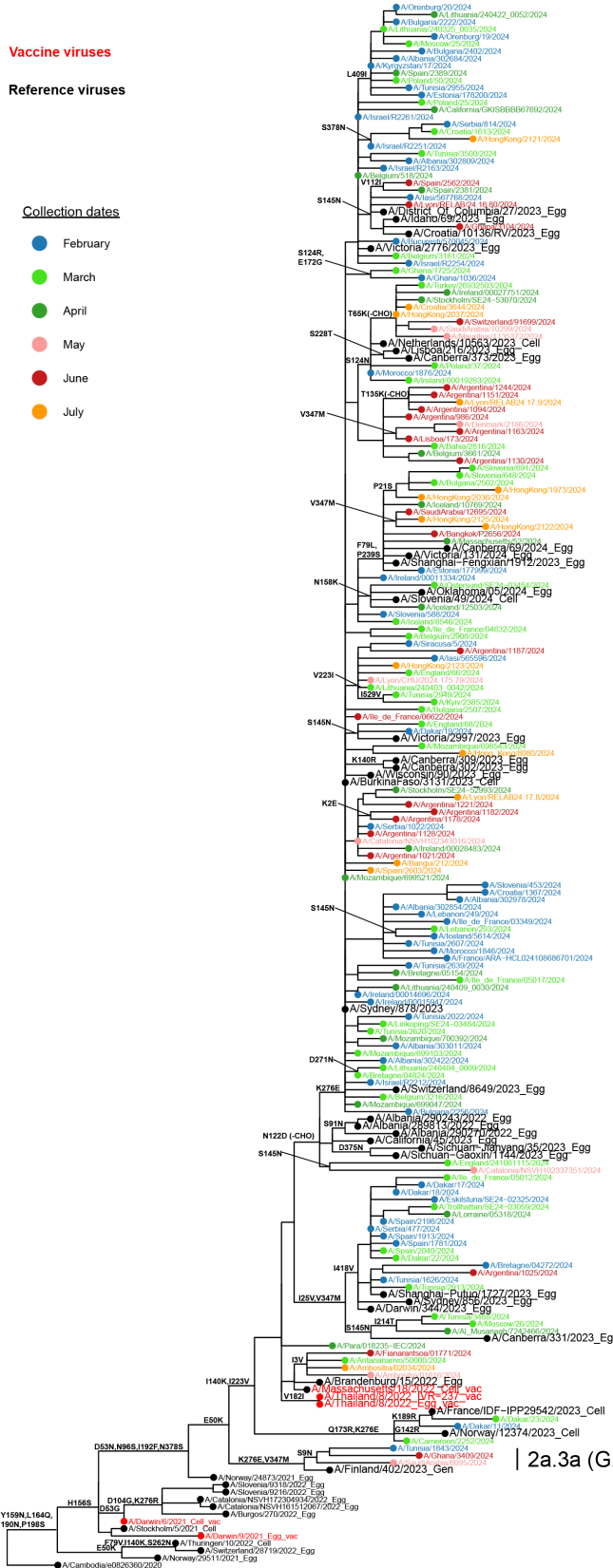
Maximum likelihood phylogenetic tree inferred using IQTree2 with HA sequence data obtained from GISAID, collection dates from 1st February 2024 to 31st August 2024, manually curated and then downsampled using Treemmer to retain a representative tree topology of 200 sequences, with emphasis on sequences generated at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

Vaccine viruses

Reference viruses

Collection dates

- February
- March
- April
- May
- June
- July



2a.3a.1 (J.2)

2a.3a.1 (J)

2a.3a.1 (J.1)

2a.3a.1 (J)

2a.3a.1 (J.4)

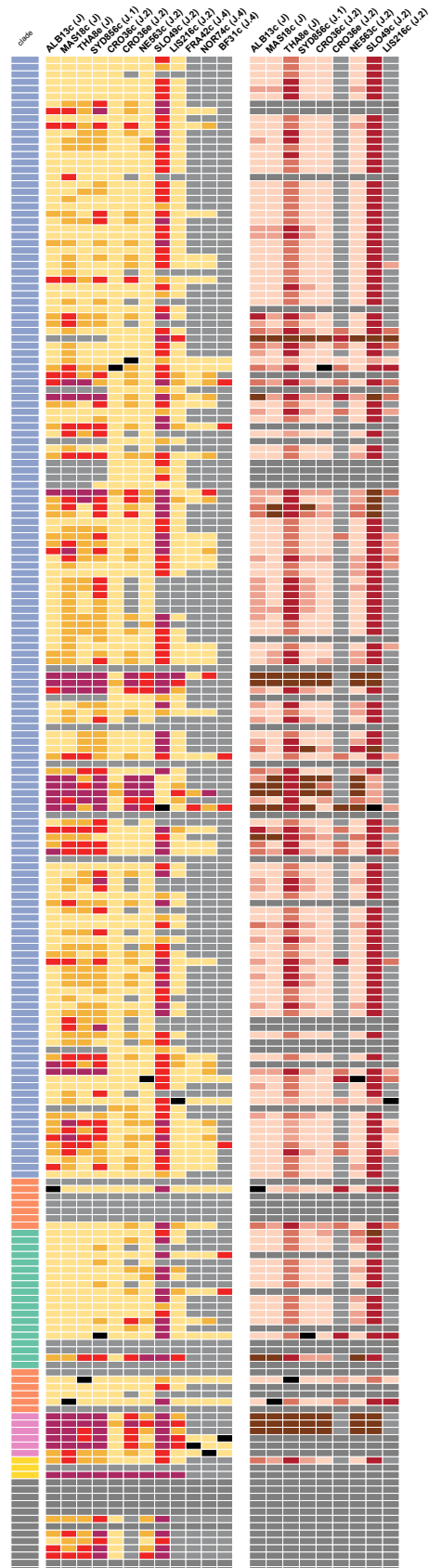
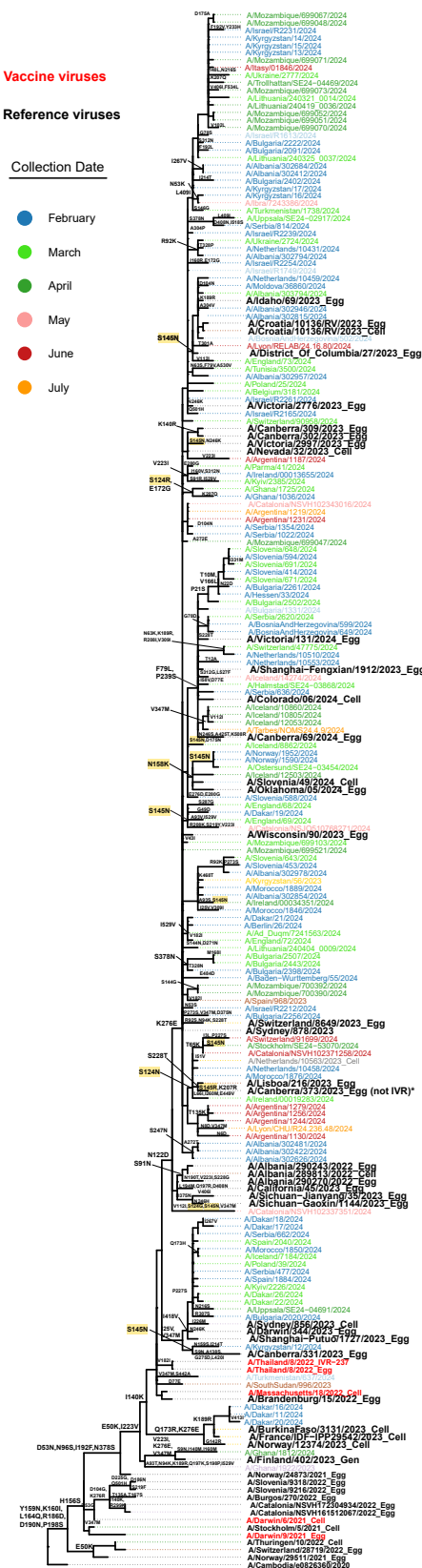
2a.3a (G.1.3.1)



Reference viruses

Collection Date

- February
- March
- April
- May
- June
- July



Clade-Subclade

-  2a.3a (G.1.3.1)
 2a.3a.1 (J)
 2a.3a.1 (J.1)
 2a.3a.1 (J.2)
 2a.3a.1 (J.4)

MN fold change

-  ≤2
 >2≤4
 >4≤8
 >8
 < (not recognised by antiserum)
 Homologous
 N/A

HI fold change

- < 4
 4
 8
 > 8
 Homologous
 N/A

Antigenic analyses: A/H3N2

Haemagglutination inhibition tables: A/H3N2

Table H3-01. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Osetamivir) 2024-03-15

Viruses	Other information	Haemagglutination inhibition titre															
		Post-infection ferret antiserum															
		A/Thuringen/10/2021	U/Switzerland/28719/2022	A/Stockholm/6/2021	A/Darwin/9/21	A/Catal/10/2022	A/Albania/1/10/2022	A/Massachusetts/10/2022	A/Thailand/10/2022	A/Sydney/10/2022	A/Croatia/10/2022	A/Vastara/SE23-14213/2023	A/Netherlands/10/2022	A/Thuringen/10/2021	U/Switzerland/28719/2022	A/Stockholm/6/2021	A/Darwin/9/21
		SIAT	SIAT	SIAT	Eq	SIAT	SIAT	SIAT	Eq	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	Eq
		Passage	F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F36/23	F34/23	F01/24	F06/24	F07/24	F08/24	F01/24	F06/24	F07/24
		Genetic group	Collection Date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)
REFERENCE VIRUSES																	
A/Thuringen/10/2021	S282N	2b	2022-04-01	P1/SIAT3	320	640	320	320	40	160	160	160	80	320	320	160	160
A/Switzerland/28719/2022	T135A	2b	2022-12-19	SIAT2	160	1280	160	160	40	160	160	160	80	320	320	160	160
A/Stockholm/6/2021		2a	2021-04-16	SIAT/SIAT5	80	320	320	640	160	160	160	320	160	320	320	320	320
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3E4	160	640	320	640	160	320	160	320	160	320	640	320	320
A/Catalonia/NSVH16151/2067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	160	320	320	640	320	160	320	160	320	320	320	320	320
A/Albania/28913/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK/SIAT1	160	320	160	320	80	640	320	640	840	320	320	640	640
A/Massachusetts/10/2022		2a.3a.1 (J)		SIAT3/SIAT1	160	640	320	640	160	1280	1280	1280	1280	640	640	640	640
A/Thailand/08/2022		2a.3a.1 (J)		E3E1	160	640	320	640	160	1280	320	1280	640	640	1280	640	640
A/Sydney/05/2023		2a.3a.1 (J)	2023-09-12	SIAT1/SIAT1	160	640	320	640	160	1280	640	1280	1280	640	640	640	640
A/Croatia/10136/RV/2023		2a.3a.1 (J)	2023-12-04	SIAT1	80	320	160	640	160	640	160	640	640	640	640	640	640
A/Vastara/SE23-14213/2023		2a.3a.1 (J)	2023-11-07	SIAT1/SIAT1	80	640	320	640	80	640	160	640	640	320	640	640	640
A/Netherlands/10663/2023		2a.3a.1 (J)	2023-11-10	CK-MIX2/SIAT1	320	640	320	1280	160	1280	320	1280	1280	640	640	1280	1280
TEST VIRUSES																	
A/Novelny/11294/2023		2a.3a (O, I, 3.1)	2023-11-02	SIAT1	80	320	160	320	80	160	160	320	320	320	320	320	320
A/Castilla-LaMancha/2303/2023		2a.3a (O, I, 3.1)	2023-11-22	SIAT2/SIAT1	80	320	160	640	80	320	160	640	320	320	320	640	640
A/Qatar/10-VI-23-0095120/2023		2a.3a.1 (J)	2023-09-16	SIAT1	160	1280	160	640	160	1280	640	1280	1280	320	640	640	640
A/Qatar/10-VI-23-0095639/2023		2a.3a.1 (J)	2023-09-17	SIAT1	80	320	160	320	80	640	320	1280	1280	320	640	640	640
A/Qatar/10-VI-23-0095639/2023		2a.3a.1 (J)	2023-09-18	SIAT1	160	320	160	640	160	1280	640	640	2560	640	640	640	640
A/Qatar/11-VI-23-0101037/2023		2a.3a.1 (J)	2023-10-08	SIAT1	160	640	160	640	160	1280	640	1280	1280	320	640	640	640
A/Spain/197/2023		2a.3a.1 (J)	2023-12-12	SIAT1	160	640	160	320	160	640	640	1280	1280	320	640	640	640
A/Spain/210/2023		2a.3a.1 (J)	2023-12-14	SIAT1	160	640	320	640	160	1280	640	1280	1280	640	640	640	640
A/Spain/499/2023		2a.3a.1 (J)	2023-12-14	SIAT1	160	640	320	640	160	1280	640	1280	1280	640	640	640	640
A/Spain/575/2023		2a.3a.1 (J)	2023-12-20	SIAT1	80	320	160	320	80	640	320	640	1280	320	320	320	320
A/Spain/909/2023		2a.3a.1 (J)	2023-12-24	SIAT1	160	320	160	320	80	1280	640	640	1280	320	320	640	640
A/Spain/940/2023		2a.3a.1 (J)	2023-12-27	SIAT1	160	640	320	640	160	1280	640	1280	1280	320	640	640	640
A/Spain/959/2023		2a.3a.1 (J)	2023-12-27	SIAT1	160	640	320	640	160	1280	640	1280	1280	320	640	640	640
A/Israel/R1564/2024		2a.3a.1 (J)	2024-01-23	P/SIAT1	160	320	160	320	80	640	320	640	640	320	640	320	320
A/Israel/R1571/2024		2a.3a.1 (J)	2024-01-24	P/SIAT1	160	640	320	640	160	1280	640	1280	1280	640	640	640	640
A/Israel/R1942/2024		2a.3a.1 (J)	2024-01-30	P/SIAT1	160	640	320	640	160	1280	640	1280	2560	640	1280	640	640
A/Israel/R2130/2024		2a.3a.1 (J)	2024-01-31	P/SIAT1	320	640	320	640	320	1280	640	2560	2560	640	1280	1280	1280
A/Qatar/10-VI-23-0092075/2023		2a.3a.1 (J)	2023-09-06	SIAT1	320	640	320	640	160	1280	320	1280	1280	640	640	640	640
A/Qatar/10-VI-23-0095639/2023		2a.3a.1 (J)	2023-09-27	SIAT1	160	640	320	640	160	1280	320	1280	1280	640	640	640	640
A/Qatar/10-VI-23-0101936/2023		2a.3a.1 (J)	2023-10-10	SIAT1	320	640	320	1280	160	1280	320	1280	1280	1280	1280	1280	1280
A/Qatar/10-VI-23-0102647/2023		2a.3a.1 (J)	2023-10-12	SIAT1	160	640	160	640	80	640	320	640	640	640	640	640	640
A/Spain/204/2023		2a.3a.1 (J)	2023-12-14	SIAT1	160	320	320	640	160	640	320	640	640	640	640	640	640
A/Spain/970/2023		2a.3a.1 (J)	2023-12-25	SIAT1	160	640	320	640	160	1280	320	1280	1280	640	640	640	640
A/Spain/968/2023		2a.3a.1 (J)	2023-12-27	SIAT1	80	320	160	320	80	320	160	320	320	320	320	320	320
A/Macedonia/289/2023		2a.3a.1 (J)	2023-12-28	SIAT1	320	640	320	1280	160	1280	640	1280	1280	1280	1280	1280	1280
A/Macedonia/5/2024		2a.3a.1 (J)	2024-01-04	SIAT1	160	640	320	640	160	640	320	640	1280	640	640	640	640
A/Macedonia/337/2024		2a.3a.1 (J)	2024-01-12	SIAT1	160	640	320	640	160	1280	320	1280	1280	640	640	640	640
A/Israel/G77/2024		2a.3a.1 (J)	2024-01-13	P/SIAT1	80	160	80	320	80	320	160	320	640	320	320	320	320
A/Israel/R1285/2024		2a.3a.1 (J)	2024-01-19	P/SIAT1	160	320	160	320	80	640	160	320	320	320	320	320	320
A/Israel/G733/2024		2a.3a.1 (J)	2024-01-21	P/SIAT1	160	640	320	640	160	1280	320	1280	1280	640	640	640	640
A/Israel/G584/2024		2a.3a.1 (J)	2024-01-22	P/SIAT1	160	320	160	320	80	320	160	320	320	320	320	320	320
A/Israel/R1578/2024		2a.3a.1 (J)	2024-01-23	P/SIAT1	160	320	160	320	80	320	160	320	320	320	320	320	320
A/Israel/R1544/2024		2a.3a.1 (J)	2024-01-24	P/SIAT1	160	640	160	640	160	1280	320	640	640	640	640	640	640
A/Israel/R1613/2024		2a.3a.1 (J)	2024-01-24	P/SIAT1	160	320	160	320	80	640	160	640	640	320	640	640	640
A/Israel/R1829/2024		2a.3a.1 (J)	2024-01-27	P/SIAT1	160	640	160	640	160	640	320	640	640	640	640	640	640
A/Israel/R1805/2024		2a.3a.1 (J)	2024-01-28	P/SIAT1	80	160	160	160	80	320	160	320	640	320	320	320	320
A/Israel/R1895/2024		2a.3a.1 (J)	2024-01-29	P/SIAT1	160	320	160	640	160	640	320	640	640	640	640	640	640
A/Israel/R1960/2024		2a.3a.1 (J)	2024-01-29	P/SIAT1	160	640	320	640	160	1280	320	1280	1280	640	1280	1280	1280
A/Israel/R2036/2024		2a.3a.1 (J)	2024-01-30	P/SIAT1	320	640	320	640	160	1280	320	1280	1280	640	1280	1280	1280
A/Israel/R2037/2024		2a.3a.1 (J)	2024-01-31	P/SIAT1	160	640	320	640	160	640	320	640	640	640	640	640	640
A/Israel/R2165/2024		2a.3a.1 (J)	2024-02-01	P/SIAT1	160	640	320	640	160	1280	320	1280	1280	640	640	640	640
A/Israel/R2231/2024		2a.3a.1 (J)	2024-02-01	P/SIAT1	320	1280	640	1280	320	640	640	1280	640	1280	1280	1280	1280
A/Israel/R2230/2024		2a.3a.1 (J)	2024-02-01	P/SIAT1	160	640	320	640	160	640	320	640	640	640	640	640	640
A/Israel/R2254/2024		2a.3a.1 (J)	2024-02-02	P/SIAT1	160	320	160	640	80	640	320	640	320	320	640	640	640
< relates to the lowest dilution of antiserum used																	
hyperimmune sheep serum; ND = Done																	

< relates to the lowest dilution of antiserum used
hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24

Vaccine
SH 2024
NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-03-22

Viruses		Other information		Haemagglutination inhibition titre													
				Post-infection ferret antiserum													
				A/Thuringen/10/22	A/Switzerland/28719/22	A/Stockholm/5/21	A/Darwin/9/21	A/Catal/2067/22	A/Albania/289813/2022	A/Massachusetts/18/2022	A/Thailand/08/2022	A/Sydney/056/2023	A/Croatia/10136/RV/2023	A/Netherlands/10563/2023			
				Passage	SIAT	SIAT	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	SIAT	SIAT	SIAT	
				Ferret number	F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F36/23	F34/23	F01/24	F06/24	F08/24		
				Genetic group	Collection Date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	
REFERENCE VIRUSES																	
A/Thuringen/10/2022		S262N	2b	2022-04-01	P1/SIAT3	320	640	320	160	40	80	160	160	40	160	160	
A/Switzerland/28719/2022		T135A	2b	2022-12-19	SIAT2	160	1280	160	160	40	80	80	160	80	160	80	
A/Stockholm/5/2021			2a	2021-04-16	SIAT0/SIAT5	80	320	320	640	160	160	160	320	160	320	320	
A/Darwin/9/2021		D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	640	160	320	160	320	160	320	320	
A/Catalonia/NSVH161512067/2022			2a.1b	2022-09-14	SIAT1/SIAT4	80	320	160	320	320	160	160	320	160	160	160	
A/Albania/289813/2022		N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1	80	320	80	320	80	640	320	640	640	640	160	
A/Massachusetts/18/2022			2a.3a.1 (J)	2022-12-13	SIAT3/SIAT1	160	640	160	640	160	1280	640	1280	1280	320	640	
A/Thailand/08/2022			2a.3a.1 (J)		E3/E1	160	640	320	640	160	640	320	640	640	320	640	
A/Sydney/056/2023			2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT1	160	640	320	640	160	1280	640	2560	1280	320	640	
A/Croatia/10136/RV/2023			2a.3a.1 (J.2)	2023-12-04	SIAT2	320	160	160	320	80	320	160	320	320	160	160	
A/Netherlands/10563/2023			2a.3a.1 (J.2)	2023-11-10	CK-MIX2/SIAT2	160	320	320	640	160	640	320	640	640	640	640	
TEST VIRUSES																	
242310	A/Cameroon/8474/2023		2a.3a.1 (J)	2023-09-03	SIAT2	80	160	80	160	80	640	640	640	1280	320	640	
240149	A/AI Musanaah/7234062/2023		2a.3a.1 (J)	2023-09-17	SIAT1	160	320	160	320	160	640	640	1280	1280	320	320	
241038	A/Kazakhstan/270/2023		2a.3a.1 (J)	2023-11-15	SIAT1	80	320	80	320	80	640	320	640	640	160	320	
242485	A/SouthSudan/986/2023		2a.3a.1 (J)	2023-12-07	SIAT2	80	320	80	160	80	640	320	640	640	160	320	
242068	A/Antananarivo/03320/2023		2a.3a.1 (J)	2023-12-14	SIAT1	80	160	80	160	80	320	160	320	320	160	160	
242067	A/Antananarivo/03308/2023		2a.3a.1 (J)	2023-12-15	SIAT1	80	320	160	320	80	640	320	640	640	160	320	
242065	A/Antananarivo/03371/2023		2a.3a.1 (J)	2023-12-20	SIAT1	80	320	80	320	80	320	320	640	640	160	320	
242064	A/Manandona/03462/2023		2a.3a.1 (J)	2023-12-28	SIAT1	80	160	80	160	80	320	320	640	640	80	160	
242063	A/Antananarivo/50000/2024		2a.3a.1 (J)	2024-01-03	SIAT1	80	160	80	160	80	320	320	640	640	160	320	
242500	A/Moldova/36850/2024		2a.3a.1 (J.1)	2024-02-05	SIAT1	80	160	80	160	80	320	160	320	320	160	160	
240152	A/Muscat/52326031/2023		2a.3a.1 (J.2)	2023-09-12	SIAT1	80	320	160	320	80	320	160	640	320	320	320	
240148	A/AI Musanaah/7234094/2023		2a.3a.1 (J.2)	2023-09-17	SIAT1	160	320	160	320	80	320	160	640	320	320	640	
240147	A/AI Musanaah/7234100/2023		2a.3a.1 (J.2)	2023-09-18	SIAT1	80	320	160	320	80	640	160	640	320	320	640	
240144	A/AI Musanaah/7234130/2023		2a.3a.1 (J.2)	2023-09-18	SIAT1	80	320	160	320	80	320	160	640	640	320	640	
240143	A/AI Musanaah/7233924/2023		2a.3a.1 (J.2)	2023-09-19	SIAT1	160	320	160	320	80	640	320	640	640	320	640	
240138	A/Muscat/7234348/2023		2a.3a.1 (J.2)	2023-09-24	SIAT1	160	640	160	640	160	640	320	640	640	640	640	
240140	A/Sohar/7234299/2023		2a.3a.1 (J.2)	2023-09-24	SIAT1	80	320	160	320	80	640	160	640	640	320	640	
240133	A/Salalah/52328322/2023		2a.3a.1 (J.2)	2023-09-25	SIAT1	160	320	160	320	80	640	160	640	320	320	640	
240132	A/Salalah/52328327/2023		2a.3a.1 (J.2)	2023-09-25	SIAT1	160	320	160	640	80	640	320	640	640	320	640	
240219	A/Qatar/15-VI-23-0098046/2023		2a.3a.1 (J.2)	2023-09-26	SIAT3	80	160	80	160	80	160	80	160	160	160	320	
241043	A/Kazakhstan/13/2023		2a.3a.1 (J.2)	2023-11-05	SIAT1	80	160	80	160	80	320	160	320	160	160	320	
241042	A/Kazakhstan/289/2023		2a.3a.1 (J.2)	2023-11-09	SIAT1	80	160	80	160	80	320	160	320	320	160	320	
241041	A/Kazakhstan/58/2023		2a.3a.1 (J.2)	2023-11-12	SIAT1	80	320	80	320	80	320	160	320	320	160	320	
241040	A/Kazakhstan/59/2023		2a.3a.1 (J.2)	2023-11-12	SIAT1	80	320	80	320	80	320	160	320	320	160	320	
241035	A/Kazakhstan/309/2023		2a.3a.1 (J.2)	2023-11-21	SIAT1	80	320	160	320	80	320	160	640	320	320	320	
241032	A/Kazakhstan/277/2023		2a.3a.1 (J.2)	2023-11-22	SIAT1	80	320	160	320	80	320	160	320	640	320	640	
241028	A/Kazakhstan/275/2023		2a.3a.1 (J.2)	2023-11-23	SIAT1	80	160	80	160	40	320	160	320	320	160	320	
241026	A/Kazakhstan/332/2023		2a.3a.1 (J.2)	2023-11-24	SIAT1	80	160	80	320	80	320	160	320	320	160	320	
241025	A/Kazakhstan/281/2023		2a.3a.1 (J.2)	2023-11-25	SIAT1	80	320	160	320	80	320	160	640	320	320	640	
241023	A/Kazakhstan/336/2023		2a.3a.1 (J.2)	2023-11-25	SIAT1	80	160	80	160	40	320	160	320	320	160	320	
242501	A/Moldova/36681/2024		2a.3a.1 (J.2)	2024-01-30	SIAT1	80	160	80	320	80	320	160	320	320	320	320	
242499	A/Moldova/36852/2024		2a.3a.1 (J.2)	2024-02-06	SIAT1	80	320	160	320	80	320	160	320	320	320	320	
242498	A/Moldova/36829/2024		2a.3a.1 (J.2)	2024-02-07	SIAT1	80	320	160	320	80	640	160	640	640	320	320	
242497	A/Moldova/36860/2024		2a.3a.1 (J.2)	2024-02-08	SIAT1	40	160	80	160	40	160	80	160	320	160	320	
242305	A/Cameroon/9092/2023		2a.3a.1 (J.4)	2023-09-28	SIAT1	80	160	80	320	80	320	160	320	160	160	320	
242303	A/Cameroon/9812/2023		2a.3a.1 (J.4)	2023-10-10	SIAT1	80	160	80	320	80	320	160	320	320	160	320	

< relates to the lowest dilution of antiserum used
hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24

Vaccine
SH 2024
NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-03. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-04-03

Haemagglutination inhibition titre																			
Viruses	Other Information	Post-infection ferret antiserum																	
		Passage	A/Thuringen /10/22 SIAT																
			A/Switzerland /28719/22 SIAT																
			A/Stockholm /5/21 SIAT																
			A/Darwin /9/21 Egg																
			A/Catal /NSVH-2067/22 SIAT																
A/Albania /289813/2022 SIAT																			
A/Massachusetts /18/2022 SIAT																			
A/Thailand /09/2022 Egg																			
A/Sydney /8/56/2023 SIAT																			
A/Croatia/ 10136/RV/2023 SIAT																			
A/Netherlands /10563/2023 SIAT																			
Genetic group		Collection Date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)				
REFERENCE VIRUSES																			
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT3	320	640	160	160	40	80	160	160	40	160	160				
A/Switzerland/28719/2022	T135A	2b	2022-12-19	SIAT2	160	1280	160	160	40	80	80	80	80	80	80				
A/Stockholm/5/2021		2a	2021-04-16	SIAT5/SIAT5	80	320	320	640	160	160	80	320	80	160	160				
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	1280	160	320	160	640	160	320	320				
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	80	160	160	320	320	160	80	320	80	80	160				
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1	80	160	80	160	80	640	320	640	640	160	320				
A/Massachusetts/18/2022		2a.3a.1 (J)	2022-12-13	SIAT3/SIAT1	160	640	160	640	160	1280	640	1280	1280	320	640				
A/Thailand/09/2022		2a.3a.1 (J)		E3/E1	160	640	320	640	160	1280	320	1280	640	640	640				
A/Sydney/8/56/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT1	160	640	320	640	160	1280	1280	1280	1280	1280	640				
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	2023-12-04	SIAT2	40	160	80	320	80	160	160	320	320	320	320				
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	CK-MDX2/SIAT2	160	640	320	640	160	640	320	640	640	640	640				
TEST VIRUSES																			
240587	A/Togo/1057/2023	2a.3a (G.1.3.1)	2023-09-03	SIAT1	80	160	80	160	80	320	160	640	640	160	160				
240585	A/Togo/1136/2023	2a.3a (G.1.3.1)	2023-09-08	SIAT1	80	160	160	160	80	320	320	640	320	160	320				
242082	A/Sainte Marie/03442/2023	2a.3a.1 (J)	2023-11-23	SIAT1	160	320	160	320	160	640	640	1280	640	160	320				
242084	A/Tollary/03036/2023	2a.3a.1 (J)	2023-11-23	SIAT1	160	320	160	320	160	1280	640	1280	640	320	320				
242081	A/Antananarivo/02981/2023	2a.3a.1 (J)	2023-11-24	SIAT1	160	320	160	320	160	1280	640	1280	1280	320	640				
242080	A/Toamasina/03143/2023	2a.3a.1 (J)	2023-11-28	SIAT1	160	320	160	320	160	640	320	640	640	320	640				
242079	A/Tollary/03152/2023	2a.3a.1 (J)	2023-11-30	SIAT1	80	320	160	320	80	640	640	640	640	160	320				
242077	A/Toamasina/03221/2023	2a.3a.1 (J)	2023-12-01	SIAT1	80	320	160	320	80	640	320	640	640	320	320				
242075	A/Antsaholity/03201/2023	2a.3a.1 (J)	2023-12-05	SIAT1	160	320	160	320	80	640	640	640	640	160	320				
242073	A/Maranandona/03216/2023	2a.3a.1 (J)	2023-12-06	SIAT1	160	320	160	320	160	640	320	640	640	160	320				
242072	A/Antananarivo/03253/2023	2a.3a.1 (J)	2023-12-08	SIAT1	80	320	80	320	80	640	320	640	640	160	320				
242069	A/Toamasina/03297/2023	2a.3a.1 (J)	2023-12-14	SIAT1	160	320	160	320	80	640	320	640	640	160	320				
242180	A/Kyrgyzstan/39/2023	2a.3a.1 (J.1)	2023-12-16	SIAT1	40	160	80	160	80	640	320	640	640	160	320				
242078	A/Toamasina/0319/2023	2a.3a.1 (J.2)	2023-12-01	SIAT1	80	320	160	320	160	640	640	1280	640	160	320				
242193	A/Kyrgyzstan/110/2023	2a.3a.1 (J.2)	2023-12-13	SIAT1	160	320	160	320	80	640	160	640	320	320	640				
242192	A/Kyrgyzstan/99/2023	2a.3a.1 (J.2)	2023-12-13	SIAT1	80	160	80	160	40	160	80	320	160	160	320				
242191	A/Kyrgyzstan/117/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	80	160	80	160	40	160	80	160	160	160	320				
242190	A/Kyrgyzstan/118/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	160	320	160	320	160	320	160	320	320	160	320				
242189	A/Kyrgyzstan/120/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	80	320	160	320	160	320	160	320	320	320	320				
242188	A/Kyrgyzstan/121/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	80	320	160	320	80	320	160	320	320	160	320				
242187	A/Kyrgyzstan/122/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	160	320	160	320	80	320	160	320	320	320	320				
242186	A/Kyrgyzstan/123/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	80	320	160	320	80	320	160	320	320	320	320				
242185	A/Kyrgyzstan/124/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	80	320	160	320	80	320	160	320	320	160	320				
242183	A/Kyrgyzstan/126/2023	2a.3a.1 (J.2)	2023-12-14	SIAT1	<40	160	80	320	80	320	80	320	320	320	160				
240562	A/Togo/1593/2023	2a.3a.1 (J.4)	2023-10-22	SIAT1	40	80	40	80	40	40	40	<40	80	40	80				
240569	A/Togo/1460/2023	2a.3a.1 (J.4)	2023-10-23	SIAT1	40	40	40	40	40	40	40	<40	80	40	80				
< relates to the lowest dilution of antiserum used																			
1 hyperimmune sheep serum; ND = Not Done																			

Table H3-04. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-04-19

Viruses	Other information	Haemagglutination inhibition titre																
		Passage	Post-infection ferret antiserum															
			A/Thuringen /10/22 SIAT	U/Switzerland /28/19/22 SIAT	A/Stockholm /9/21 SIAT	A/Darwin /9/21 Egg	A/Catal /NSVH-2067/22 SIAT	A/Albania /289813/2022 SIAT	A/Massachusetts /19/2022 SIAT	A/Thailand /08/2022 Egg	A/BYdney /856/2023 SIAT	A/Croatia /10136/RV/2023 SIAT	A/Netherlands /10663/2023 SIAT	A/Slovenia /49/2024 SIAT	A/Switzerland /8649/2023 Egg			
			F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F36/23	F34/23	F01/24	F06/24	F08/24	F11/24	F12/24			
Genetic group		Collection Date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J-1)	2a.3a.1 (J-2)	2a.3a.1 (J-2)	2a.3a.1 (J-2)	2a.3a.1 (J-2)	2a.3a.1 (J-2)		
REFERENCE VIRUSES																		
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT3	320	640	160	160	80	160	160	160	80	160	160	160	320	
A/Switzerland/28719/2022	T136A	2a	2022-12-19	SIAT2	160	1280	160	160	40	160	160	160	80	160	160	160	160	
A/Stockholm/9/2021		2a	2021-04-16	SIAT0/SIAT5	80	320	320	640	160	160	80	320	160	160	320	80	320	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	ESI2	160	320	320	640	160	320	160	320	160	320	320	320	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	80	320	160	320	320	320	160	320	160	160	160	160	640	
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1	80	320	160	320	80	640	320	640	640	320	320	160	320	
A/Massachusetts/19/2022		2a.3a.1 (J)		SIAT3/SIAT1	160	640	320	640	160	1280	640	1280	1280	320	640	320	640	
A/Thailand/08/2022		2a.3a.1 (J)		ESI1	160	640	320	640	160	1280	320	1280	640	640	640	640	640	
A/BYdney/856/2023		2a.3a.1 (J-1)	2023-09-12	SIAT1/SIAT1	160	640	320	640	160	1280	640	1280	1280	640	640	640	640	
A/Croatia/10136/RV/2023		2a.3a.1 (J-2)	2023-12-04	SIAT2	40	160	80	320	80	320	160	320	640	320	320	320	320	
A/Netherlands/10663/2023		2a.3a.1 (J-2)	2023-11-10	2K-MIX2/SIAT2	160	640	320	640	160	640	320	1280	1280	640	1280	640	1280	
A/Slovenia/49/2024	N188K	2a.3a.1 (J-2)	2024-01-08	MDCK1/SIAT2	80	320	40	160	40	80	80	320	40	320	160	2560	320	
A/Switzerland/8649/2023	N122D (-CHO), K276E	2a.3a.1 (J-2)	2023-11-24	ES1Am1A12	80	160	80	320	80	320	160	320	640	160	320	320	640	
TEST VIRUSES																		
242383 A/Kyrgyzstan/15/2023		2a.3a.1 (J)	2023-12-13	MDCK1/SIAT1	80	320	160	320	80	640	320	640	640	320	320	160	320	
242382 A/Kyrgyzstan/16/2023		2a.3a.1 (J)	2023-12-13	MDCK1/SIAT1	160	320	320	320	160	640	320	1280	1280	320	640	320	640	
242387 A/Kyrgyzstan/1/2023		2a.3a.1 (J-1)	2023-12-04	MDCK1/SIAT1	40	160	80	160	80	640	320	640	640	160	320	320	320	
242388 A/Kyrgyzstan/10/2023		2a.3a.1 (J-2)	2023-11-30	MDCK1/SIAT1	80	320	160	320	80	320	160	640	320	320	320	320	640	
242386 A/Kyrgyzstan/12/2023		2a.3a.1 (J-2)	2023-12-05	MDCK1/SIAT1	80	320	160	320	80	640	160	640	640	320	320	320	320	
242385 A/Kyrgyzstan/13/2023		2a.3a.1 (J-2)	2023-12-07	MDCK1/SIAT1	80	160	160	320	160	320	160	320	640	320	320	320	640	
242384 A/Kyrgyzstan/14/2023		2a.3a.1 (J-2)	2023-12-07	MDCK1/SIAT1	40	160	160	320	160	320	160	640	640	320	320	320	320	
242381 A/Kyrgyzstan/17/2023		2a.3a.1 (J-2)	2023-12-15	MDCK1/SIAT1	160	640	320	640	160	640	320	640	640	320	640	320	640	
242380 A/Kyrgyzstan/18/2023		2a.3a.1 (J-2)	2023-12-15	MDCK1/SIAT1	160	320	160	640	160	640	160	640	640	320	640	320	640	
242379 A/Kyrgyzstan/19/2023		2a.3a.1 (J-2)	2023-12-15	MDCK1/SIAT1	80	320	160	320	80	320	160	640	320	320	320	160	640	
242378 A/Kyrgyzstan/20/2023		2a.3a.1 (J-2)	2023-12-18	MDCK1/SIAT1	160	320	320	640	160	640	320	640	640	320	640	320	640	
242377 A/Kyrgyzstan/21/2023		2a.3a.1 (J-2)	2023-12-18	MDCK1/SIAT1	160	320	160	320	160	640	320	640	640	320	640	320	640	
242376 A/Kyrgyzstan/22/2023		2a.3a.1 (J-2)	2023-12-18	MDCK1/SIAT1	80	320	160	320	80	640	160	640	640	320	640	320	640	
242375 A/Kyrgyzstan/23/2023		2a.3a.1 (J-2)	2023-12-19	MDCK1/SIAT1	160	320	160	320	80	640	320	640	640	320	640	320	640	
242374 A/Kyrgyzstan/24/2023		2a.3a.1 (J-2)	2023-12-19	MDCK1/SIAT1	160	320	160	320	80	640	160	640	640	320	640	320	640	
242373 A/Kyrgyzstan/25/2023		2a.3a.1 (J-2)	2023-12-20	MDCK1/SIAT1	160	640	320	640	160	640	320	1280	640	640	640	320	640	
242372 A/Kyrgyzstan/26/2023		2a.3a.1 (J-2)	2023-12-28	MDCK1/SIAT1	160	320	160	320	80	640	320	640	640	320	640	320	640	
242371 A/Kyrgyzstan/27/2023		2a.3a.1 (J-2)	2023-12-28	MDCK1/SIAT1	160	320	160	320	80	640	160	640	640	320	640	320	640	
242370 A/Kyrgyzstan/28/2023		2a.3a.1 (J-2)	2023-12-28	MDCK1/SIAT1	160	320	160	320	80	640	160	640	640	320	640	320	640	
242369 A/Kyrgyzstan/29/2023		2a.3a.1 (J-2)	2023-12-28	MDCK1/SIAT1	80	160	160	320	160	320	160	640	640	320	640	320	320	
242756 A/Baden-Wuerttemberg/28/2024		2a.3a.1 (J-2)	2024-01-28	P1/SIAT1	80	320	160	320	80	320	160	320	320	320	320	320	640	
242755 A/Niedersachsen/33/2024		2a.3a.1 (J-2)	2024-01-30	P1/SIAT1	80	320	160	320	160	640	160	640	640	320	640	320	640	
242754 A/Baden-Wuerttemberg/55/2024		2a.3a.1 (J-2)	2024-02-08	P1/SIAT1	160	320	160	320	160	640	160	640	640	320	640	320	640	
242753 A/Hessen/33/2024		2a.3a.1 (J-2)	2024-02-16	P1/SIAT1	80	160	80	320	80	320	160	320	320	320	320	160	320	
242752 A/Berlin/26/2024		2a.3a.1 (J-2)	2024-02-26	P1/SIAT1	80	320	160	320	80	320	160	320	320	320	320	320	640	
< relates to the lowest dilution of antiserum used * hyperimmune sheep serum; ND = Not Done																		
Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24																		
Vaccine BH 2024 NH 2024-25																		

< relates to the lowest dilution of antiserum used
 1 hyperimmune sheep serum; ND = Not Done

Vaccine
 SH 2022
 NH 2022-23
 SH 2023
 NH 2023-24

Vaccine
 SH 2024
 NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-05. Antigenic analyses of Influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-04-26

Viruses	Other information	Haemagglutination inhibition titre														
		Post-infection ferret antiserum														
		A/Turkey/10/2022					A/Croatia/19136/RV/2023					A/Sydney/05/2023				
		Passage	Ferret number	F36/22	F29/23	F15/22	F38/21	F41/22	F21/23	F36/23	F34/23	F01/24	F06/24	F08/24	F08/24	
Genetic group	Collection Date	Genetic group	2b	2b	2a	2a	2a-1b	2a-3a-1 (J)	2a-3a-1 (J)	2a-3a-1 (J-1)	2a-3a-1 (J-2)	2a-3a-1 (J-2)	2a-3a-1 (J-2)			
REFERENCE VIRUSES																
A/Turkey/10/2022	S262N	2b	2022-04-01	P1/SIAT3	320	640	320	160	40	80	160	80	40	160	160	160
A/Sydney/05/2023	T135A	2a	2022-12-19	SIAT2	160	1280	320	160	40	160	160	160	80	160	160	160
A/Sydney/05/2021	D18N, D225S (egp)	2a	2021-04-16	SIAT0/SIAT5	160	640	640	640	160	160	160	160	160	160	160	160
A/Darwin/2021		2a	2021-04-17	E3E2	160	320	320	640	160	320	160	320	160	320	320	320
A/Catalonia/NDVH115/12047/2022		2a-1b	2022-08-14	SIAT1/SIAT4	80	320	160	320	320	160	160	80	320	80	160	160
A/Albania/288813/2022		2a-3a-1 (J)	2022-12-13	MOCK/SIAT1	80	320	160	320	80	640	320	320	640	160	320	320
A/Massachusetts/18/2022	H12D	2a-3a-1 (J)		SIAT3/SIAT1	160	640	160	640	160	1280	640	1280	1280	320	640	640
A/Thailand/08/2022		2a-3a-1 (J)		E3E1	160	640	320	640	160	1280	320	1280	640	640	640	640
A/Sydney/05/2023		2a-3a-1 (J-1)	2023-09-12	SIAT1/SIAT1	160	640	320	640	160	1280	1280	2560	2560	640	640	640
A/Croatia/19136/RV/2023		2a-3a-1 (J-2)	2023-12-04	SIAT2	40	160	80	320	80	320	80	320	320	320	320	320
A/Netherlands/10563/2023		2a-3a-1 (J-2)	2023-11-10	CK-MIX2/SIAT2	160	320	160	640	160	640	320	640	640	320	320	640
TEST VIRUSES																
242615	A/Serbia/41/2024	2a-3a-1 (J)	2024-01-12	SIAT1	160	320	160	320	80	640	320	1280	640	160	320	320
242616	A/Morocco/1317/2024	2a-3a-1 (J)	2024-01-19	SIAT1	80	320	160	160	80	640	320	640	640	160	320	320
242413	A/Israel/1836/2024	2a-3a-1 (J)	2024-01-28	SIAT1	80	160	80	160	80	640	320	640	640	160	320	320
242414	A/Israel/1880/2024	2a-3a-1 (J)	2024-01-28	SIAT1	40	160	80	160	40	640	320	640	640	160	320	320
242409	A/Israel/1938/2024	2a-3a-1 (J)	2024-01-29	SIAT1	80	320	160	320	80	640	320	640	640	160	320	320
242412	A/Israel/2322/2024	2a-3a-1 (J)	2024-01-30	SIAT1	80	320	80	160	80	640	320	640	640	160	320	320
242007	A/Serbia/423/2024	2a-3a-1 (J)	2024-01-31	SIAT1	80	320	160	320	80	1280	320	640	640	320	320	320
242005	A/Serbia/477/2024	2a-3a-1 (J)	2024-02-02	SIAT1	80	160	80	160	80	640	320	640	640	160	320	320
242003	A/Serbia/882/2024	2a-3a-1 (J)	2024-02-12	SIAT1	80	320	160	320	80	640	320	1280	640	160	320	320
242573	A/Morocco/1859/2024	2a-3a-1 (J)	2024-02-15	SIAT1	80	160	80	160	80	640	320	640	640	160	320	320
242712	A/Croatia/NDVH115/12047/2023	2a-3a-1 (J-2)	2023-01-17	SIAT1/SIAT1	160	320	160	320	80	320	160	640	640	320	320	320
242020	A/Serbia/16653/2023	2a-3a-1 (J-2)	2023-12-27	SIAT1	80	320	80	320	80	320	160	640	640	320	320	320
242619	A/Serbia/16670/2023	2a-3a-1 (J-2)	2023-12-29	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242618	A/Serbia/40/2024	2a-3a-1 (J-2)	2024-01-02	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242617	A/Serbia/93/2024	2a-3a-1 (J-2)	2024-01-09	SIAT1	80	320	80	160	40	320	160	320	320	160	320	320
242616	A/Serbia/70/2024	2a-3a-1 (J-2)	2024-01-10	SIAT1	40	160	80	160	80	320	160	320	320	160	320	320
242731	A/Norway/319/2024	2a-3a-1 (J-2)	2024-01-14	SIAT1	80	320	80	320	80	320	160	320	320	320	320	320
242581	A/Morocco/1200/2024	2a-3a-1 (J-2)	2024-01-16	SIAT1	40	160	80	160	40	320	80	320	160	160	320	320
242582	A/Slovenia/206/2024	2a-3a-1 (J-2)	2024-01-17	MOCK/SIAT1	160	320	160	320	80	640	160	320	320	320	320	320
242577	A/Morocco/1290/2024	2a-3a-1 (J-2)	2024-01-18	SIAT1	80	160	160	320	40	320	160	160	320	320	320	320
242576	A/Morocco/1291/2024	2a-3a-1 (J-2)	2024-01-18	SIAT1	40	160	80	160	40	320	80	320	160	160	320	320
242613	A/Serbia/248/2024	2a-3a-1 (J-2)	2024-01-19	SIAT1	80	320	160	320	80	320	160	320	320	320	320	320
242612	A/Serbia/233/2024	2a-3a-1 (J-2)	2024-01-19	SIAT1	160	640	320	640	160	640	320	640	640	160	320	320
242580	A/Slovenia/223/2024	2a-3a-1 (J-2)	2024-01-20	MOCK/SIAT1	80	320	80	320	80	320	160	320	320	320	320	320
242628	A/Slovenia/240/2024	2a-3a-1 (J-2)	2024-01-23	MOCK/SIAT1	40	160	80	160	40	320	80	320	160	160	320	320
242611	A/Slovenia/320/2024	2a-3a-1 (J-2)	2024-01-24	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242610	A/Serbia/324/2024	2a-3a-1 (J-2)	2024-01-25	SIAT1	80	320	160	320	80	640	160	320	320	320	320	320
242610	A/Serbia/1962/2024	2a-3a-1 (J-2)	2024-01-29	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242626	A/Slovenia/323/2024	2a-3a-1 (J-2)	2024-01-29	MOCK/SIAT1	80	320	160	320	80	320	160	320	320	320	320	320
242608	A/Israel/1991/2024	2a-3a-1 (J-2)	2024-01-30	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242607	A/Israel/1997/2024	2a-3a-1 (J-2)	2024-01-30	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242604	A/Israel/2042/2024	2a-3a-1 (J-2)	2024-01-30	SIAT1	80	320	80	320	80	640	160	320	320	320	320	320
242603	A/Israel/2205/2024	2a-3a-1 (J-2)	2024-01-30	SIAT1	160	320	160	320	80	640	160	640	640	320	320	320
242606	A/Serbia/400/2024	2a-3a-1 (J-2)	2024-01-30	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242624	A/Slovenia/370/2024	2a-3a-1 (J-2)	2024-01-30	MOCK/SIAT1	80	320	160	320	80	640	160	640	640	320	320	320
242602	A/Israel/12107/2024	2a-3a-1 (J-2)	2024-01-31	SIAT1	160	320	160	320	80	640	320	640	640	320	320	320
242606	A/Serbia/441/2024	2a-3a-1 (J-2)	2024-01-31	SIAT1	160	640	160	320	80	640	320	640	640	320	320	320
242601	A/Slovenia/163/2024	2a-3a-1 (J-2)	2024-02-01	SIAT1	160	320	160	640	160	640	320	640	640	320	320	320
242600	A/Israel/2179/2024	2a-3a-1 (J-2)	2024-02-01	SIAT1	80	160	80	160	40	320	80	320	320	160	320	320
242601	A/Israel/2210/2024	2a-3a-1 (J-2)	2024-02-01	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242600	A/Israel/2235/2024	2a-3a-1 (J-2)	2024-02-02	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242600	A/Israel/2259/2024	2a-3a-1 (J-2)	2024-02-02	SIAT1	80	160	80	160	40	320	80	320	160	160	320	320
242600	A/Israel/2261/2024	2a-3a-1 (J-2)	2024-02-02	SIAT1	80	320	80	160	80	320	160	320	160	160	320	320
242600	A/Slovenia/414/2024	2a-3a-1 (J-2)	2024-02-04	MOCK/SIAT1	80	320	80	320	80	640	160	640	640	320	320	320
242604	A/Serbia/336/2024	2a-3a-1 (J-2)	2024-02-09	SIAT1	80	320	160	320	80	640	160	640	640	320	320	320
242604	A/Slovenia/433/2024	2a-3a-1 (J-2)	2024-02-09	M/SIAT1	160	320	160	320	80	320	160	640	640	320	320	320
242574	A/Morocco/1889/2024	2a-3a-1 (J-2)	2024-02-12	SIAT1	160	320	160	320	80	320	160	320	320	320	320	320
242572	A/Morocco/1846/2024	2a-3a-1 (J-2)	2024-02-19	SIAT1	80	320	80	320	80	320	160	320	320	160	320	320
242570	A/Morocco/1876/2024	2a-3a-1 (J-2)	2024-02-21	SIAT1	80	320	160	320	80	640	160	640	640	320	320	320
242618	A/Slovenia/580/2024	2a-3a-1 (J-2)	2024-02-21	MOCK/SIAT1	80	320	160	320	80	320	160	320	320	320	320	320
242616	A/Slovenia/594/2024	2a-3a-1 (J-2)	2024-02-22	MOCK/SIAT1	80	320	80	320	80	320	160	320	320	160	320	320
242614	A/Slovenia/640/2024	2a-3a-1 (J-2)	2024-03-05	MOCK/SIAT1	80	320	80	320	40	320	160	320	320	320	320	320
242612	A/Slovenia/643/2024	2a-3a-1 (J-2)	2024-03-06	MOCK/SIAT1	80	320	80	320	80	320	160	320	320	160	320	320
242610	A/Slovenia/671/2024	2a-3a-1 (J-2)	2024-03-07	MOCK/SIAT1	160	320	160	640	80	640	160					
242608	A/Slovenia/691/2024	2a-3a-1 (J-2)	2024-03-12	MOCK/SIAT1	80	320	160	320	80	320	160	640	640	320	320	640

Viruses	Other information	Haemagglutination inhibition titre																
		Post-infection ferret antiserum																
		A/Thuringen /10/2022	S/Witzerland /287/2021	A/Stockholm /9/2021	A/Darwin /8/2021	A/Catal /2896/2021	A/Albania /2896/2021	A/Massachusetts /18/2021	A/Thailand /10/2021	A/Sydney /10/2021	A/Croatia /10136/RV/2023	A/Netherlands /10/2021						
		Passage	SIAT	SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	SIAT	SIAT	F01/24	F06/24	F08/24			
		Ferret number	F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F36/23	F34/23	F01/24	F06/24	F08/24					
		Genetic group	Collection Date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)			
REFERENCE VIRUSES																		
A/Thuringen/10/2022	S262N		P1/SIAT3	320	640	160	160	40	80	160	80	40	320	160				
A/Switzerland/287/10/2022	T135A		SIAT2	640	160	160	160	80	160	80	160	40	80	160				
A/Stockholm/5/2021		2a	SIAT0/SIAT5	80	320	320	640	160	160	80	320	80	320	160				
A/Darwin/9/2021	D186N, D225G (egg)	2a	E3/E2	160	320	320	640	160	320	160	320	160	320	320				
Catolonia/NSVH16115/01/2067/2022		2a.1b	SIAT1/SIAT4	80	320	160	640	320	320	320	160	320	160	320				
A/Albania/289813/2022	N122D	2a.3a.1 (J)	MDC/K1/SIAT1	160	320	160	320	80	1280	320	640	1280	320	320				
A/Massachusetts/18/2022		2a.3a.1 (J)	SIAT3/SIAT1	160	640	320	640	160	1280	640	1280	1280	320	640				
A/Thailand/08/2022		2a.3a.1 (J)	E3/E1	160	640	320	640	160	1280	320	1280	640	640	640				
A/Sydney/16/2022		2a.3a.1 (J.1)	SIAT1/SIAT1	160	320	160	640	1280	640	1280	1280	320	320	320				
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	SIAT2	40	160	80	320	160	320	160	320	320	320	320				
A/Netherlands/10563/2023		2a.3a.1 (J.2)	CK-MK2/SIAT2	160	320	160	320	160	640	320	640	640	320	640				
TEST VIRUSES																		
242230 A/Extremadura/240458/2023		2a.3a.1 (J.1)	SIAT1	80	320	80	320	80	640	320	640	1280	320	320				
242334 A/France/OF-RELAB-PPQZC589/2024		2a.3a.1 (J.1)	SIAT1	40	80	80	160	80	320	320	640	640	160	160				
242232 A/Extremadura/240460/2023		2a.3a.1 (J.2)	SIAT1	80	320	160	320	80	640	320	640	640	320	640				
242231 A/Extremadura/240459/2023		2a.3a.1 (J.2)	SIAT1	80	320	80	320	80	640	160	320	640	320	640				
242241 A/Serbia/1664/2023		2a.3a.1 (J.2)	SIAT2	160	320	160	640	80	640	320	640	640	320	640				
242406 Altraud/R2007/2024		2a.3a.1 (J.2)	SIAT2	80	320	160	320	160	640	640	640	640	320	640				
242637 A/BurkinaFaso/182/2023		2a.3a.1 (J.4)	SIAT1	<40	40	80	80	40	40	<40	160	<40	80	80				
242636 A/BurkinaFaso/203/2023		2a.3a.1 (J.4)	SIAT1	<40	<40	<40	40	40	<40	<40	40	<40	40	40				
242635 A/BurkinaFaso/205/2023		2a.3a.1 (J.4)	SIAT1	<40	<40	40	40	40	<40	<40	40	<40	40	40				
242634 A/BurkinaFaso/206/2023		2a.3a.1 (J.4)	SIAT1	<40	<40	80	160	80	40	40	160	40	160	160				
242631 A/BurkinaFaso/3140/2023		2a.3a.1 (J.4)	SIAT1	4														

47

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H3-08. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-05-22

Viruses	Other information	Haemagglutination inhibition titre												
		Passage	Post-infection ferret antiserum											
			A/Albania /289813/2022 SIAT	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg	A/Sydney /856/2023 SIAT	A/Croatia/ 10136/RV/2023 SIAT	A/Croatia/ 10136/RV/2023 Egg	A/Netherlands /10563/2023 SIAT	A/Slovenia /49/2024 SIAT	A/Lisboa/ 216/2023 Egg			
			Ferret number	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24		
			Genetic group	Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES	N122D		2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1	640	320	640	640	160	320	320	160	320
A/Albania/289813/2022			2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	160	320	320	160	320
A/Massachusetts/18/2022			2a.3a.1 (J)		E3/E1	640	640	640	640	320	640	320	320	640
A/Thailand/08/2022			2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	640	320	640	640	160	320	320	160	320
A/Sydney/856/2023			2a.3a.1 (J.2)	2023-12-04	SIAT2	160	80	160	320	160	320	320	160	320
A/Croatia/10136/RV/2023			2a.3a.1 (J.2)		E2 (Am1A1)	640	160	320	640	320	640	320	320	640
A/Croatia/10136/RV/2023			2a.3a.1 (J.2)	2023-11-10	CK-MIX2/SIAT2	640	160	640	640	320	640	640	320	640
A/Netherlands/10563/2023			2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT2	40	40	160	40	160	40	80	2560	320
A/Slovenia/49/2024	N158K		2a.3a.1 (J.2)		E2 (Am1A1)	320	80	320	320	320	320	320	320	320
A/Lisboa/216/2023			2a.3a.1 (J.2)											
TEST VIRUSES			2a.3a (G.1.3.1)	2023-09-12	MDCK2	<40	<40	<40	<40	<40	<40	<40	<40	<40
A/Ghana/1922/2023			2a.3a.1 (J)	2023-10-03	SIAT1	640	320	640	640	160	160	320	160	320
242038 A/Antsirabe/02539/2023			2a.3a.1 (J)	2023-10-25	SIAT1	640	320	640	640	160	160	160	80	320
242109 A/Antananarivo/02681/2023			2a.3a.1 (J)	2023-10-31	SIAT1	640	320	640	320	160	160	320	80	320
242106 A/Antananarivo/02702/2023			2a.3a.1 (J)	2023-10-31	SIAT1	1280	320	1280	640	160	320	320	80	320
242105 A/Antsirabe/02735/2023			2a.3a.1 (J)	2023-11-06	SIAT1	1280	320	640	640	160	320	320	160	320
242103 A/Antananarivo/02757/2023			2a.3a.1 (J)	2023-11-14	SIAT1	640	320	640	320	80	160	160	80	160
242097 A/Antananarivo/02860/2023			2a.3a.1 (J)	2023-11-14	SIAT1	320	160	320	320	80	80	80	40	160
242095 A/Antananarivo/02916/2023			2a.3a.1 (J)	2023-11-15	SIAT1	320	160	320	320	80	80	160	80	160
242094 A/Antananarivo/04306/2023			2a.3a.1 (J)	2023-11-17	SIAT1	320	160	320	320	80	80	160	80	160
242093 A/Toamasina/02944/2023			2a.3a.1 (J)	2023-11-17	SIAT1	320	160	320	320	80	80	160	80	160
242092 A/Toamasina/02991/2023			2a.3a.1 (J)	2023-11-18	SIAT1	640	320	640	640	80	160	160	80	160
242196 A/Kyrgyzstan/106/2023			2a.3a.1 (J)	2023-12-13	SIAT1	1280	640	1280	1280	160	320	320	160	320
242219 A/Kyrgyzstan/60/2023			2a.3a.1 (J.1)	2023-11-30	SIAT1	320	160	320	320	40	160	160	40	80
242225 A/Kyrgyzstan/8/2023			2a.3a.1 (J.2)	2023-11-28	SIAT1	160	80	160	80	80	80	160	80	160
242223 A/Kyrgyzstan/21/2023			2a.3a.1 (J.2)	2023-11-30	SIAT1	160	80	160	160	160	160	160	160	160
242221 A/Kyrgyzstan/56/2023			2a.3a.1 (J.2)	2023-11-30	SIAT1	160	80	160	160	160	160	320	160	320
242220 A/Kyrgyzstan/58/2023			2a.3a.1 (J.2)	2023-11-30	SIAT1	320	160	320	320	320	320	320	160	320
242217 A/Kyrgyzstan/9/2023			2a.3a.1 (J.2)	2023-12-01	SIAT1	320	80	160	160	160	160	160	160	320
242216 A/Kyrgyzstan/318/2023			2a.3a.1 (J.2)	2023-12-04	SIAT1	80	40	80	40	80	80	80	40	80
241949 A/Estonia/176992/2023			2a.3a.1 (J.2)	2023-12-05	SIAT2	80	40	80	80	80	160	160	80	160
242209 A/Kyrgyzstan/91/2023			2a.3a.1 (J.2)	2023-12-11	SIAT1	80	40	80	80	80	160	160	80	160
242204 A/Kyrgyzstan/98/2023			2a.3a.1 (J.2)	2023-12-11	SIAT1	320	160	320	160	160	320	320	160	320
242200 A/Kyrgyzstan/101/2023			2a.3a.1 (J.2)	2023-12-13	SIAT1	320	160	320	320	160	320	320	160	320
242195 A/Kyrgyzstan/108/2023			2a.3a.1 (J.2)	2023-12-13	SIAT1	320	160	640	320	320	320	320	320	640
240635 A/Dakar/40/2023			2a.3a.1 (J.4)	2023-09-11	MDCK1/MDCK1	<40	<40	40	<40	40	40	40	<40	40
240625 A/Dakar/39/2023			2a.3a.1 (J.4)	2023-10-04	MDCK1/MDCK2	<40	<40	40	<40	40	40	40	<40	40
241617 A/France/IDF-IPP29542/2023			2a.3a.1 (J.4)	2023-11-30	MDCK1/SIAT1	<40	<40	40	<40	40	80	80	<40	40
< relates to the lowest dilution of antiserum used						Vaccine SH 2024 NH 2024-25								
* hyperimmune sheep serum; ND = Not Done														

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H3-09. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-05-30

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre									
					Post-infection ferret antiserum									
					A/Albania/289813/2022 SIAT	A/Massachusetts/18/2022 SIAT	A/Thailand/08/2022 Egg	A/Sydney/856/2003 SIAT	A/Croatia/10136/RV/2023 SIAT	A/Croatia/10136/RV/2023 Egg	A/Netherlands/10563/2023 SIAT	A/Slovenia/149/2024 SIAT	A/Lisboa/216/2023 Egg	
					Passage	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24
					2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	
REFERENCE VIRUSES														
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1	1280	640	1280	1280	320	640	640	320	640	
A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	320	640	640	320	640	
A/Thailand/08/2022		2a.3a.1 (J)		ES/E1	1280	320	1280	640	640	640	640	640	640	
A/Sydney/856/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	640	320	640	1280	320	640	320	320	640	
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	2023-12-04	SIAT3/SIAT1	320	160	320	320	320	320	320	320	160	640
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)		E2 (Am1A1)	640	160	640	1280	320	1280	640	320	640	
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	CK-MDX2/SIAT2	1280	320	640	640	640	640	640	640	640	1280
A/Slovenia/149/2024	N158K	2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT2	40	40	320	80	320	80	320	80	2560	320
A/Lisboa/216/2023		2a.3a.1 (J.2)		E2 (Am1A1)	320	160	320	640	160	320	320	160	640	
TEST VIRUSES														
243154 A/Turkmenistan/637/2024		2a.3a.1 (J)	2024-01-23	SIAT1	640	320	640	640	320	640	640	320	640	
240628 A/Dakar/41/2023		2a.3a.1 (J.1)	2023-09-25	MDCK2/MDCK1	1280	640	1280	1280	320	640	640	320	640	
242091 A/Sainte Marie/03340/2023		2a.3a.1 (J.1)	2023-11-19	SIAT3	1280	640	1280	1280	640	640	640	320	640	
241671 A/Belgium/112/2023		2a.3a.1 (J.1)	2023-12-01	MDCK1/MDCK1	1280	640	1280	1280	640	640	640	320	640	
242732 A/Norway/47/2024		2a.3a.1 (J.2)	2024-01-02	SIAT1	80	40	320	40	160	80	160	2560	320	
243155 A/Turkmenistan/69/2024		2a.3a.1 (J.2)	2024-01-20	SIAT1	640	320	640	1280	640	640	640	320	640	
242432 A/Israel/R1287/2024		2a.3a.1 (J.2)	2024-01-21	SIAT1	640	320	320	320	320	640	320	160	640	
242431 A/Israel/R1511/2024		2a.3a.1 (J.2)	2024-01-23	SIAT1	640	160	640	640	320	640	640	320	640	
242430 A/Israel/R1550/2024		2a.3a.1 (J.2)	2024-01-24	SIAT1	640	160	640	640	320	640	640	320	640	
242428 A/Israel/R1896/2024		2a.3a.1 (J.2)	2024-01-24	SIAT1	640	320	640	640	320	640	640	320	640	
242427 A/Israel/R1617/2024		2a.3a.1 (J.2)	2024-01-25	SIAT1	640	160	640	640	320	640	640	320	640	
242426 A/Israel/R1642/2024		2a.3a.1 (J.2)	2024-01-25	SIAT1	640	320	640	640	320	640	320	320	640	
242424 A/Israel/R1690/2024		2a.3a.1 (J.2)	2024-01-25	SIAT1	640	320	640	1280	640	640	640	320	640	
242423 A/Israel/R1721/2024		2a.3a.1 (J.2)	2024-01-26	SIAT1	640	160	640	640	320	640	640	320	640	
242422 A/Israel/R1723/2024		2a.3a.1 (J.2)	2024-01-26	SIAT1	640	160	640	640	320	640	640	320	640	
242421 A/Israel/R1739/2024		2a.3a.1 (J.2)	2024-01-26	SIAT1	1280	320	640	1280	640	640	640	320	1280	
242420 A/Israel/R1749/2024		2a.3a.1 (J.2)	2024-01-26	SIAT1	1280	320	640	640	320	640	640	320	640	
242419 A/Israel/R1753/2024		2a.3a.1 (J.2)	2024-01-26	SIAT1	1280	320	640	640	320	640	640	320	640	
242418 A/Israel/R1792/2024		2a.3a.1 (J.2)	2024-01-28	SIAT1	1280	320	640	1280	320	640	640	320	1280	
242417 A/Israel/R1810/2024		2a.3a.1 (J.2)	2024-01-28	SIAT1	1280	320	640	1280	320	640	640	640	1280	
242416 A/Israel/R1817/2024		2a.3a.1 (J.2)	2024-01-28	SIAT1	320	160	320	320	160	320	320	160	320	
242414 A/Israel/R1823/2024		2a.3a.1 (J.2)	2024-01-28	SIAT1	640	320	640	640	320	640	640	320	640	
243153 A/Turkmenistan/131/2024		2a.3a.1 (J.2)	2024-01-31	SIAT1	640	320	640	640	320	640	640	320	640	
243152 A/Turkmenistan/142/2024		2a.3a.1 (J.2)	2024-01-31	SIAT1	1280	320	640	1280	640	640	640	320	1280	
243021 A/Albania/302626/2024		2a.3a.1 (J.2)	2024-02-13	SIAT1	1280	640	1280	1280	640	1280	1280	640	1280	
243020 A/Albania/302636/2024		2a.3a.1 (J.2)	2024-02-13	SIAT1	640	320	640	640	320	640	320	320	640	
243019 A/Albania/302660/2024		2a.3a.1 (J.2)	2024-02-13	SIAT1	1280	640	640	640	640	640	640	640	1280	
243018 A/Albania/302684/2024		2a.3a.1 (J.2)	2024-02-14	SIAT1	640	640	640	1280	640	640	640	640	640	
243017 A/Albania/302794/2024		2a.3a.1 (J.2)	2024-02-15	SIAT1	1280	320	640	1280	640	640	640	640	1280	
243016 A/Albania/302765/2024		2a.3a.1 (J.2)	2024-02-16	SIAT1	1280	320	640	1280	640	640	640	640	320	
243015 A/Albania/302809/2024		2a.3a.1 (J.2)	2024-02-17	SIAT1	1280	640	640	1280	640	1280	640	640	1280	
243014 A/Albania/302815/2024		2a.3a.1 (J.2)	2024-02-18	SIAT1	640	160	640	640	320	640	640	320	640	
243013 A/Albania/302840/2024		2a.3a.1 (J.2)	2024-02-19	SIAT1	1280	320	1280	1280	640	1280	1280	640	1280	
243012 A/Albania/302854/2024		2a.3a.1 (J.2)	2024-02-19	SIAT1	1280	320	1280	1280	640	640	1280	640	1280	
243011 A/Albania/302845/2024		2a.3a.1 (J.2)	2024-02-20	SIAT1	640	160	640	1280	640	640	640	640	640	
243010 A/Albania/302957/2024		2a.3a.1 (J.2)	2024-02-20	SIAT1	1280	320	1280	1280	640	640	1280	640	1280	
243008 A/Albania/302978/2024		2a.3a.1 (J.2)	2024-02-22	SIAT1	1280	320	1280	1280	640	1280	1280	640	1280	
243007 A/Albania/303794/2024		2a.3a.1 (J.2)	2024-03-14	SIAT1	640	160	640	640	640	640	640	320	640	
243151 A/Turkmenistan/1738/2024		2a.3a.1 (J.2)	2024-03-25	SIAT1	640	320	640	640	320	640	640	320	640	
243044 A/Ghana/1725/2024		2a.3a.1 (J.2)	2024-03-27	SIAT1	1280	320	1280	1280	640	640	640	320	640	

< relates to the lowest dilution of antiserum used
 1 hyperimmune sheep serum; ND = Not Done

Vaccine
 SH 2024
 NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-06-07

Haemagglutination inhibition titre															
Viruses	Other information	Passage	Post-infection ferret antiserum												
			A/Albania /2896/13/2022 SIAT	A/Massachusetts /19/2002 SIAT	A/Thailand /08/2002 Egg	A/Sydney /856/2002 SIAT	A/Croatia /10136/RV/2023 SIAT	A/Croatia /10136/RV/2023 SIAT	A/Netherlands /10563/2023 SIAT	A/Slovenia /49/2002 SIAT	A/Isbova /216/2023 Egg				
			F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24				
			Genetic group	Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES															
A/Albania/2896/13/2022	N122D		2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1	640	640	640	640	320	640	320	160	640	
A/Massachusetts/18/2002			2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	320	640	640	160	640	
A/Thailand/08/2002			2a.3a.1 (J)		E3/E1	320	320	1280	640	640	640	640	320	640	
A/Sydney/856/2023			2a.3a.1 (J.1)	2023-08-12	SIAT1/SIAT2	1280	640	1280	1280	320	320	320	320	160	640
A/Croatia/10136/RV/2023			2a.3a.1 (J.2)	2023-12-04	SIAT2	160	80	320	320	320	640	640	320	160	640
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)		E3 (Am1A2)	640	160	640	640	640	640	640	640	320	640	
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	CK-MIX2/SIAT2	640	320	640	640	640	640	1280	320	640	640	
A/Slovenia/49/2002	N158K		2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT2	40	40	160	40	160	40	80	2560	320	640
A/Isbova/216/2023			2a.3a.1 (J.2)		E2 (Am1A1)	320	160	320	640	160	320	80	160	320	320
TEST VIRUSES															
2a.3a.1 (G.1.3.1)			2a.3a.1 (G.1.3.1)	2024-11-10	SIAT1	160	80	320	320	160	160	320	160	320	320
A/China/1812/2004			2a.3a.1 (G.1.3.1)	2024-04-04	SIAT2	160	80	320	320	320	320	640	320	640	320
A/Poland/39/2024			2a.3a.1 (J.1)	2024-03-20	SIAT1	1280	640	1280	1280	160	320	320	160	320	320
2a.3a.1 (J.2)			2a.3a.1 (J.2)	2024-02-05	MDCK1	320	160	320	160	160	160	320	160	320	320
A/Albania/303011/2024			2a.3a.1 (J.2)	2024-02-21	MDCK1	320	80	320	160	160	320	320	160	320	320
A/China/1036/2024			2a.3a.1 (J.2)	2024-02-21	SIAT2	320	160	320	320	320	320	640	320	640	320
A/Albania/303097/2024			2a.3a.1 (J.2)	2024-02-22	MDCK1	320	160	320	160	320	320	640	160	320	320
A/Albania/303141/2024			2a.3a.1 (J.2)	2024-02-24	MDCK1	160	80	160	160	320	320	320	160	320	320
A/Lithuania/240404_0009/2024			2a.3a.1 (J.2)	2024-03-01	SIAT1	320	160	320	320	160	320	320	160	320	320
A/Poland/50/2024			2a.3a.1 (J.2)	2024-03-05	SIAT1	640	320	640	320	320	320	640	320	640	320
A/Lithuania/240321_0014/2024			2a.3a.1 (J.2)	2024-03-15	SIAT1	320	160	320	160	320	320	320	160	320	320
A/Lebanon/293/2024			2a.3a.1 (J.2)	2024-03-18	SIAT1	640	320	320	320	320	320	640	320	640	320
A/Lithuania/240322_0054/2024			2a.3a.1 (J.2)	2024-03-20	SIAT1	320	160	320	320	160	320	320	160	320	320
A/Lithuania/240322_0019/2024			2a.3a.1 (J.2)	2024-03-21	SIAT1	320	160	320	320	160	320	320	160	320	320
A/Lithuania/240325_0035/2024			2a.3a.1 (J.2)	2024-03-25	SIAT1	160	80	160	160	160	160	160	160	160	160
A/Lithuania/240325_0036/2024			2a.3a.1 (J.2)	2024-03-22	SIAT1	320	160	320	160	160	320	320	160	320	320
A/Lithuania/240325_0037/2024			2a.3a.1 (J.2)	2024-03-22	SIAT1	640	160	640	640	320	320	640	320	640	320
A/Lithuania/240327_0011/2024			2a.3a.1 (J.2)	2024-03-25	SIAT1	320	160	320	320	160	320	320	160	320	320
A/Lithuania/240403_0042/2024			2a.3a.1 (J.2)	2024-03-25	SIAT1	160	80	160	160	160	160	320	160	320	320
A/Lithuania/240328_0119/2024			2a.3a.1 (J.2)	2024-03-26	SIAT1	320	160	320	160	160	320	320	160	320	320
A/Lithuania/240411_0064/2024			2a.3a.1 (J.2)	2024-04-04	SIAT1	160	160	320	160	160	160	320	160	320	320
A/Lithuania/240409_0030/2024			2a.3a.1 (J.2)	2024-04-08	SIAT1	640	320	640	320	320	640	640	320	640	320
A/Lithuania/240410_0010/2024			2a.3a.1 (J.2)	2024-04-09	SIAT1	640	160	640	640	320	640	640	320	640	320
A/Lithuania/240412_0066/2024			2a.3a.1 (J.2)	2024-04-09	SIAT1	320	160	320	320	160	320	320	160	320	320
A/Lithuania/240419_0036/2024			2a.3a.1 (J.2)	2024-04-13	SIAT1	320	160	320	320	320	320	320	320	320	320
A/Lithuania/240422_0052/2024			2a.3a.1 (J.2)	2024-04-16	SIAT2	320	160	320	160	320	320	320	320	320	320
A/Lithuania/240425_0049/2024			2a.3a.1 (J.2)	2024-04-21	SIAT1	320	160	320	320	320	320	640	320	640	320
A/BurkinaFaso/3371/2023			2a.3a.1 (J.4)	2023-10-03	SIAT1	<40	<40	80	<40	80	80	80	<40	80	80
A/BurkinaFaso/3406/2023			2a.3a.1 (J.4)	2023-10-06	SIAT1	<40	<40	80	<40	80	80	80	<40	80	80
2a.3a.1 (J.4)			2a.3a.1 (J.4)	2023-10-09	SIAT1	<40	<40	40	<40	40	80	80	<40	40	40
A/BurkinaFaso/3433/2023			2a.3a.1 (J.4)	2023-10-10	SIAT1	<40	<40	40	<40	80	80	80	<40	40	40
A/BurkinaFaso/3441/2023			2a.3a.1 (J.4)	2023-10-11	SIAT1	<40	<40	40	<40	80	80	80	<40	40	40
A/BurkinaFaso/3442/2023			2a.3a.1 (J.4)	2023-10-11	SIAT1	<40	<40	80	<40	80	160	160	80	160	160
A/BurkinaFaso/3457/2023			2a.3a.1 (J.4)	2023-10-13	SIAT1	<40	<40	40	<40	40	80	80	<40	40	40
A/BurkinaFaso/153/2023			2a.3a.1 (J.4)	2023-11-11	SIAT1	<40	<40	40	<40	40	80	80	<40	40	40
A/BurkinaFaso/3131/2023			2a.3a.1 (J.4)	2023-11-20	SIAT1	<40	<40	80	<40	80	160	160	40	80	80
< relates to the lowest dilution of antiserum used															
hyperimmune sheep serum; ND = Not Done															
Vaccine SH 2024															
NH 2024-25															

Table H3-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-06-14

Viruses	Other information	Haemagglutination inhibition titre												
		Passage	Post-infection ferret antiserum											
			A/Albania/289813/2022	A/Massachusetts/18/2022	A/Thailand/08/2022	A/Sydney/856/2023	A/Croatia/10136/RV/2023	A/Croatia/10136/RV/2023	A/Netherlands/10563/2023	A/Slovenia/149/2024	A/Lisboa/216/2023			
			SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	SIAT	Egg			
			F21/23	F38/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24			
Genetic group		Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	
REFERENCE VIRUSES														
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT2	640	640	1280	1280	320	640	320	160	640	
A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT1	1280	320	1280	1280	160	320	640	160	320	
A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	1280	320	1280	640	640	640	640	320	640	
A/Sydney/856/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	1280	640	1280	1280	320	320	320	160	320	
A/Croatia/10136/RV/2023	N158K	2a.3a.1 (J.2)	2023-12-04	SIAT3	160	80	320	320	160	320	320	160	320	
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)		E3 (Am1A12)	640	320	640	640	640	640	640	320	640	
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	CK-MIX2/SIAT2	640	160	640	320	320	320	640	320	640	
A/Slovenia/49/2024		2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT2	40	40	320	40	160	80	80	2560	320	
A/Lisboa/216/2023		2a.3a.1 (J.2)		E3 (Am1A12)	320	160	320	640	160	320	320	160	640	
TEST VIRUSES														
243276 A/Iceland/7184/2024		2a.3a.1 (J.1)	2024-03-03	SIAT1	640	320	640	640	80	160	160	80	160	
243279 A/Iceland/5614/2024		2a.3a.1 (J.2)	2024-02-19	SIAT1	160	80	160	160	160	320	320	160	320	
243278 A/Iceland/5655/2024		2a.3a.1 (J.2)	2024-02-20	SIAT1	160	80	160	80	160	320	320	80	320	
243277 A/Iceland/6456/2024		2a.3a.1 (J.2)	2024-02-26	SIAT1	320	160	320	160	160	160	320	160	320	
243275 A/Iceland/8546/2024		2a.3a.1 (J.2)	2024-03-13	SIAT1	160	80	160	160	160	320	320	160	160	
243225 A/Poland/25/2024		2a.3a.1 (J.2)	2024-03-13	SIAT2	320	160	640	320	320	320	640	160	320	
243291 A/Turkey/26932503/2024		2a.3a.1 (J.2)	2024-03-14	SIAT1	160	80	160	80	160	160	160	160	320	
243274 A/Iceland/8622/2024		2a.3a.1 (J.2)	2024-03-16	SIAT1	160	80	160	160	160	320	320	160	320	
243289 A/Turkey/27154072/2024		2a.3a.1 (J.2)	2024-03-21	SIAT1	160	80	160	160	160	160	320	160	160	
243273 A/Iceland/9695/2024		2a.3a.1 (J.2)	2024-03-24	SIAT1	320	160	320	160	160	320	320	160	320	
243288 A/Turkey/27174042/2024		2a.3a.1 (J.2)	2024-03-27	SIAT1	160	80	160	160	160	160	320	160	320	
243272 A/Iceland/10182/2024		2a.3a.1 (J.2)	2024-03-29	SIAT1	320	160	320	160	320	320	320	160	320	
243271 A/Iceland/10404/2024		2a.3a.1 (J.2)	2024-04-02	SIAT1	320	160	320	320	320	320	320	160	640	
243270 A/Iceland/10426/2024		2a.3a.1 (J.2)	2024-04-02	SIAT1	320	160	320	320	320	320	320	160	320	
243269 A/Iceland/10769/2024		2a.3a.1 (J.2)	2024-04-05	SIAT1	320	160	320	320	320	320	320	160	320	
243268 A/Iceland/10805/2024		2a.3a.1 (J.2)	2024-04-05	SIAT1	320	160	320	320	320	320	320	160	320	
243267 A/Iceland/10860/2024		2a.3a.1 (J.2)	2024-04-06	SIAT1	320	160	320	320	320	320	320	160	320	
243266 A/Iceland/10877/2024		2a.3a.1 (J.2)	2024-04-06	SIAT1	320	160	160	160	160	160	320	160	320	
243265 A/Iceland/10961/2024		2a.3a.1 (J.2)	2024-04-08	SIAT1	320	160	320	320	320	320	320	160	320	
243310 A/Mozambique/699071/2024		2a.3a.1 (J.2)	2024-04-10	SIAT1	320	80	320	160	160	160	320	160	320	
243309 A/Mozambique/699073/2024		2a.3a.1 (J.2)	2024-04-10	SIAT1	320	160	320	320	160	160	320	160	320	
243264 A/Iceland/11668/2024		2a.3a.1 (J.2)	2024-04-12	SIAT1	320	160	320	160	320	320	320	160	320	
243308 A/Mozambique/699621/2024		2a.3a.1 (J.2)	2024-04-13	SIAT1	320	160	320	160	160	160	320	160	320	
243263 A/Iceland/12053/2024		2a.3a.1 (J.2)	2024-04-17	SIAT1	320	160	320	320	320	320	320	160	320	
243307 A/Mozambique/700390/2024		2a.3a.1 (J.2)	2024-04-19	SIAT1	320	80	320	320	160	160	320	160	320	
243306 A/Mozambique/700392/2024		2a.3a.1 (J.2)	2024-04-19	SIAT1	320	160	320	320	160	320	320	160	320	
243262 A/Iceland/12503/2024		2a.3a.1 (J.2)	2024-04-20	SIAT1	40	40	80	40	160	80	80	2560	160	
243261 A/Iceland/12516/2024		2a.3a.1 (J.2)	2024-04-21	SIAT1	320	160	320	320	320	320	320	320	320	
243305 A/Mozambique/700571/2024		2a.3a.1 (J.2)	2024-04-23	SIAT1	160	160	320	160	160	320	160	160	320	
243260 A/Iceland/14274/2024		2a.3a.1 (J.2)	2024-05-07	SIAT1	320	160	320	320	320	320	320	160	320	

< relates to the lowest dilution of antiserum used
† hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2024
NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-06-26

Haemagglutination inhibition titre													
Viruses	Other information	Passage	Ferret number	Post-infection ferret antiserum									
				A/Albania/289813/2022 SIAT	A/Massachusetts/18/2022 SIAT	A/Thailand/08/2022 Egg	A/Sydney/856/2023 SIAT	A/Croatia/10136/RV/2023 SIAT	A/Croatia/10136/RV/2023 Egg	A/Netherlands/10563/2023 SIAT	A/Slovenia/49/2024 SIAT	A/Lisboa/216/2023 Egg	
				F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	
				Genetic group	Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES													
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT2	1280	640	1280	1280	320	320	640	160	640
A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	320	640	640	160	640
A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	1280	320	640	640	320	640	640	320	640
A/Sydney/856/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	1280	640	1280	1280	320	320	320	160	320
A/Croatia/10136/RV/2023	N158K	2a.3a.1 (J.2)	2023-12-04	SIAT3	160	160	160	320	320	320	320	320	320
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)		E3 (Am1A12)	640	320	640	640	640	1280	640	640	640
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	MDCK-MIX2/SIAT2	640	320	640	640	320	640	640	320	640
A/Slovenia/49/2024		2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT2	80	80	160	40	160	80	80	2560	320
A/Lisboa/216/2023		2a.3a.1 (J.2)		E3 (Am1A12)	320	160	640	640	320	320	640	320	640
TEST VIRUSES													
243652 A/Kyiv/2226/2024		2a.3a.1 (J.1)	2024-03-10	MDCK1/SIAT1	1280	640	1280	1280	320	640	640	160	640
243662 A/Norway/423/2024		2a.3a.1 (J.2)	2024-01-12	SIAT1	40	40	160	<40	160	40	80	2560	320
243661 A/Norway/805/2024		2a.3a.1 (J.2)	2024-01-25	SIAT1	40	40	160	<40	160	40	80	2560	160
243388 A/Ireland/00011334/2024		2a.3a.1 (J.2)	2024-02-02	SIAT1	640	160	320	320	320	320	640	320	640
243386 A/Ireland/00024704/2024		2a.3a.1 (J.2)	2024-02-09	SIAT1	320	160	640	320	320	320	640	320	640
243385 A/Ireland/00013633/2024		2a.3a.1 (J.2)	2024-02-12	SIAT1	640	320	640	320	320	320	640	320	640
243384 A/Ireland/00013655/2024		2a.3a.1 (J.2)	2024-02-12	SIAT1	320	80	320	160	320	320	320	160	320
243660 A/Norway/1590/2024		2a.3a.1 (J.2)	2024-02-15	SIAT1	<40	<40	80	<40	80	40	80	1280	160
243382 A/Ireland/00014686/2024		2a.3a.1 (J.2)	2024-02-16	SIAT1	320	80	320	160	160	160	320	160	320
243659 A/Norway/1592/2024		2a.3a.1 (J.2)	2024-02-16	SIAT1	<40	<40	80	<40	160	80	40	1280	160
243381 A/Ireland/00016985/2024		2a.3a.1 (J.2)	2024-02-22	SIAT1	40	40	160	<40	160	40	80	2560	320
243658 A/Norway/1952/2024		2a.3a.1 (J.2)	2024-02-23	SIAT1	<40	40	160	<40	160	80	40	2560	320
243380 A/Ireland/00017646/2024		2a.3a.1 (J.2)	2024-02-27	SIAT1	640	320	640	320	320	320	640	320	640
243379 A/Ireland/00019283/2024		2a.3a.1 (J.2)	2024-03-01	SIAT1	320	160	320	320	320	320	320	320	640
243507 A/Ad_Duqm/7241563/2024		2a.3a.1 (J.2)	2024-03-03	SIAT1	320	160	320	320	320	320	640	320	320
243378 A/Ireland/00019853/2024		2a.3a.1 (J.2)	2024-03-04	SIAT1	320	80	320	640	320	320	320	160	320
243651 A/Kyiv/2385/2024		2a.3a.1 (J.2)	2024-03-12	MDCK2/SIAT1	320	160	320	160	320	160	320	160	320
243547 A/Ukraine/2777/2024		2a.3a.1 (J.2)	2024-03-18	MDCK1/SIAT1	640	320	640	640	320	640	640	320	640
243647 A/Kyiv/2462/2024		2a.3a.1 (J.2)	2024-03-21	MDCK1/SIAT1	320	160	320	160	160	320	320	160	320
243506 A/Firq/7242349/2024		2a.3a.1 (J.2)	2024-04-15	SIAT1	320	160	320	320	320	320	320	160	320
243504 A/Ibra/7243386/2024		2a.3a.1 (J.2)	2024-05-14	SIAT1	320	160	320	320	320	320	320	160	320
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done					Vaccine SH 2024 NH 2024-25								

 < 4-fold  4-fold  8-fold  > 8-fold  not recognised by the antiserum  ≥ 160 (no homologous titre)

Table H3-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Osetamivir) 2024-07-05

Haemagglutination inhibition titre																
Viruses	Other information	Post-infection ferret antiserum														
		Passage	A/Albania /28913/2022 SIAT	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg	A/Sydney /856/2023 SIAT	A/Croatia/ 10136/RV/2023 SIAT	A/Croatia/ 10136/RV/2023 Egg	A/Netherlands /10663/2023 SIAT	A/Slovenia /49/2024 SIAT	A/Lisboa/ 216/2023 Egg	A/France/DF- IPP29542/2023 Cell	A/Norway /12374/2023 Cell	A/Norway /12374/2023 Cell		
		Ferret number	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	F22/24	F21/24	F10/24		
		Genetic group	Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.4)	2a.3a.1 (J.4)	2a.3a.1 (J.4)
REFERENCE VIRUSES																
A/Albania/28913/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCX1/SIAT2	1280	640	1280	1280	160	320	320	160	640	320	160	160
A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	160	320	320	160	640	320	160	160
A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	1280	320	640	640	320	640	640	320	640	320	320	320
A/Sydney/856/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	2560	640	1280	1280	160	320	320	160	320	320	160	160
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	2023-12-04	SIAT3	160	80	160	320	320	640	160	320	320	320	160	160
A/Croatia/10136/RV/2023	N158K S124N K169R	2a.3a.1 (J.2)		E3 (Am1A2)	1280	320	1280	1280	640	1280	640	1280	1280	640	640	640
A/Netherlands/10663/2023		2a.3a.1 (J.2)	2023-11-10	MDCX1/SIAT2	1280	640	640	640	640	640	640	1280	640	320	640	640
A/Slovenia/49/2024		2a.3a.1 (J.2)	2024-01-08	MDCX1/SIAT2	40	40	160	40	160	80	1280	160	40	40	40	40
A/Lisboa/216/2023		2a.3a.1 (J.2)	2023-12-15	E3 (Am1A2)	320	160	640	640	320	320	640	320	640	320	160	320
A/France/DF-IPP29542/2023		2a.3a.1 (J.4)	2023-11-30	MDCX1/SIAT1	40	<40	160	<40	80	160	160	80	80	160	80	80
A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT3	640	160	320	320	320	320	640	320	640	320	320	160
TEST VIRUSES																
243750	A/Eskilstuna/SE24-02325/2024	2a.3a.1 (J.1)	2024-02-15	SIAT1/SIAT1	640	320	640	640	80	160	160	80	160	160	80	80
243748	A/Eskilstuna/SE24-03387/2024	2a.3a.1 (J.1)	2024-03-01	SIAT1/SIAT1	320	160	320	320	80	80	160	80	160	80	40	40
243744	A/Trollhattan/SE24-03089/2024	2a.3a.1 (J.1)	2024-03-03	SIAT1/SIAT1	640	320	640	1280	160	160	320	160	320	160	80	80
243738	A/Trollhattan/SE24-03639/2024	2a.3a.1 (J.1)	2024-03-19	SIAT1/SIAT1	640	320	640	640	160	160	320	80	160	160	80	80
243732	A/Uppsala/SE24-04691/2024	2a.3a.1 (J.1)	2024-04-14	SIAT1/SIAT1	640	320	640	320	160	160	160	80	320	160	80	80
243034	A/Albania/29847/2022	2a.3a.1 (J.2)	2023-12-19	SIAT1	320	80	320	160	160	160	320	160	320	160	80	80
243033	A/Albania/299565/2023	2a.3a.1 (J.2)	2023-12-22	SIAT1	320	160	320	320	160	320	320	160	320	160	80	80
243032	A/Albania/300922/2024	2a.3a.1 (J.2)	2024-01-23	SIAT1	320	160	320	160	160	320	320	160	320	320	160	160
243031	A/Albania/301074/2024	2a.3a.1 (J.2)	2024-01-24	SIAT1	640	320	640	320	320	320	640	320	640	320	320	320
243030	A/Albania/301155/2024	2a.3a.1 (J.2)	2024-01-24	SIAT1	160	80	160	80	160	160	160	160	160	160	80	80
243029	A/Albania/301159/2024	2a.3a.1 (J.2)	2024-01-25	SIAT1	80	80	160	160	160	320	320	160	320	160	80	80
243028	A/Albania/301277/2024	2a.3a.1 (J.2)	2024-01-25	SIAT1	160	80	160	160	160	160	320	160	320	160	80	80
243668	A/BosniaAndHerzegovina/483/2024	2a.3a.1 (J.2)	2024-01-30	SIAT1	320	160	320	320	160	320	320	160	320	320	160	160
243667	A/BosniaAndHerzegovina/502/2024	2a.3a.1 (J.2)	2024-01-30	SIAT1	160	80	160	160	160	320	320	160	160	160	80	80
243027	A/Albania/302203/2024	2a.3a.1 (J.2)	2024-02-06	SIAT1	160	80	160	160	160	160	320	160	320	160	80	80
243666	A/BosniaAndHerzegovina/599/2024	2a.3a.1 (J.2)	2024-02-06	SIAT1	320	160	320	320	320	320	320	160	320	320	160	160
243026	A/Albania/302195/2024	2a.3a.1 (J.2)	2024-02-07	SIAT1	320	160	320	160	320	320	640	320	320	320	160	160
243665	A/BosniaAndHerzegovina/649/2024	2a.3a.1 (J.2)	2024-02-08	SIAT1	320	160	320	160	160	320	320	160	320	320	160	160
243025	A/Albania/302510/2024	2a.3a.1 (J.2)	2024-02-09	SIAT1	160	160	160	160	160	160	320	160	320	320	160	80
243024	A/Albania/302412/2024	2a.3a.1 (J.2)	2024-02-10	SIAT1	320	160	320	160	160	320	320	160	320	320	160	160
243023	A/Albania/302422/2024	2a.3a.1 (J.2)	2024-02-10	SIAT1	160	160	320	160	320	320	320	160	320	320	160	160
243022	A/Albania/302481/2024	2a.3a.1 (J.2)	2024-02-12	SIAT1	640	160	320	320	320	320	640	160	320	320	160	160
243602	A/Tunisia/2955/2024	2a.3a.1 (J.2)	2024-02-19	SIAT1	320	160	320	320	160	320	320	160	320	320	160	160
243746	A/Uppsala/SE24-02917/2024	2a.3a.1 (J.2)	2024-03-03	SIAT1/SIAT1	640	160	320	320	320	320	640	160	320	320	160	160
243742	A/Ostersund/SE24-03454/2024	2a.3a.1 (J.2)	2024-03-04	SIAT1/SIAT1	<40	<40	40	<40	80	<40	<40	640	80	40	<40	<40
243663	A/BosniaAndHerzegovina/1149/2024	2a.3a.1 (J.2)	2024-03-07	SIAT1	320	160	160	160	160	160	320	160	320	320	160	160
243740	A/Linköping/SE24-03484/2024	2a.3a.1 (J.2)	2024-03-12	SIAT1/SIAT1	320	80	160	160	160	160	320	160	320	160	80	80
243680	A/Slovenia/695/2024	2a.3a.1 (J.2)	2024-03-14	MDCX1/SIAT1	640	160	640	320	320	320	320	320	640	160	160	160
243595	A/Tunisia/3500/2024	2a.3a.1 (J.2)	2024-03-14	SIAT1	320	160	320	160	160	320	640	160	320	320	160	160
243736	A/Halmstad/SE24-03688/2024	2a.3a.1 (J.2)	2024-03-21	SIAT1/SIAT1	640	160	640	320	320	320	640	320	640	160	160	160
243546	A/Ukraine/2724/2024	2a.3a.1 (J.2)	2024-03-25	SIAT3	160	160	160	160	160	160	320	320	320	320	160	80
243374	A/Ireland/90028483/2024	2a.3a.1 (J.2)	2024-04-04	SIAT3	320	160	320	320	160	320	320	160	320	320	160	160
243734	A/Trollhattan/SE24-04469/2024	2a.3a.1 (J.2)	2024-04-09	SIAT1/SIAT1	160	80	160	160	160	160	320	160	320	160	80	80
243730	A/Stockholm/SE24-52993/2024	2a.3a.1 (J.2)	2024-04-15	SIAT1/SIAT1	160	80	160	160	160	160	320	160	160	160	80	80
243728	A/Stockholm/SE24-53070/2024	2a.3a.1 (J.2)	2024-04-22	SIAT1/SIAT1	80	80	160	160	160	160	320	160	160	160	80	40
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done																
Vaccine SH 2024 NH 2024-25																
< 4-fold 4-fold 8-fold > 8-fold not recognised by the antiserum ≥ 160 (no homologous titre)																

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Viruses	Other information	Haemagglutination inhibition titre													
		Post-infection ferrit antiserum													
		A/Albania /2998/13/2022	A/Massachusetts /16/2022	A/Thailand /08/2022	A/Sydney /856/2023	A/Croatia /10136/RV/2023	A/Croatia /10136/RV/2023	A/Netherlands /10563/2023	A/Slovenia /46/2023	A/Libya /216/2023	A/France/IDF- PP2954/2023	A/Norway /12374/2023			
		Passage	SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	SIAT	Egg	Cell	Cell		
Ferret number		F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	F22/24	F21/24			
Genetic group		Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.4)	2a.3a.1 (J.4)	
REFERENCE VIRUSES		N122D													
A/Albania/2998/13/2022		2a.3a.1 (J)	MDCK1/SIAT2	1280	1280	1280	1280	320	320	640	160	640	320	320	
A/Massachusetts/16/2022		2a.3a.1 (J)	SIAT3/SIAT1	1280	1280	1280	320	320	320	640	160	640	320	320	
A/Thailand/08/2022		2a.3a.1 (J)	E3E1	1280	640	1280	640	640	640	640	640	640	320	320	
A/Sydney/856/2023		2a.3a.1 (J.1)	SIAT1/SIAT2	2560	640	1280	1280	320	320	320	160	1280	640	320	
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	SIAT3	320	160	320	640	320	320	320	160	1280	320	160	
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	E3 (Am1A2)	1280	1280	1280	320	320	320	640	1280	640	320	320	
A/Netherlands/10563/2023		2a.3a.1 (J.2)	MDCK-MIX2/SIAT2	1280	320	640	640	640	640	1280	320	640	320	320	
A/Slovenia/46/2024		2a.3a.1 (J.2)	MDCKx3/SIAT3	80	80	320	40	320	80	160	2560	640	40	80	
A/Libya/216/2023		2a.3a.1 (J.2)	E3 (Am1A2)	1280	320	640	640	320	640	640	320	640	320	320	
A/France/IDF-PP2954/2023		2a.3a.1 (J.4)	MDCK1/SIAT1	80	40	320	40	160	160	160	40	160	320	160	
A/Norway/12374/2023		2a.3a.1 (J.4)	SIAT3	640	320	640	640	320	640	640	320	320	320	320	
TEST VIRUSES															
A/Serbia/8424/2023		2a.3a.1 (J.2)	C1/SIAT1	640	320	640	640	320	640	640	320	640	320	320	
A/Serbia/100/2024		2a.3a.1 (J.2)	C1/SIAT1	640	320	640	640	320	640	640	320	640	320	320	
A/Serbia/1010/2024		2a.3a.1 (J.1)	C1/SIAT1	640	320	640	640	320	640	640	320	640	320	320	
A/Serbia/362/2024		2a.3a.1 (J.2)	C1/SIAT1	640	320	640	640	640	640	640	320	640	320	320	
A/Serbia/468/2024		2a.3a.1 (J.2)	C1/SIAT1	1280	320	1280	1280	640	640	640	640	640	320	320	
A/Serbia/13/2024		2a.3a.1 (J.2)	C1/SIAT1	1280	320	640	640	640	640	640	320	640	320	320	
A/Serbia/814/2024		2a.3a.1 (J.2)	C1/SIAT1	640	320	640	640	320	320	640	320	640	320	160	
A/Serbia/1022/2024		2a.3a.1 (J.2)	C1/SIAT1	1280	320	1280	1280	640	640	640	320	640	320	320	
A/SouthAfrica/R08623/24		2a.3a.1 (J.2)	MDCK1/SIAT1	640	320	640	640	320	640	640	320	640	320	320	
A/Serbia/154/2024		2a.3a.1 (J.1)	C1/SIAT1	640	320	640	320	320	640	640	320	640	320	160	
A/Serbia/7620/2024		2a.3a.1 (J.2)	C1/SIAT1	640	320	640	640	320	640	640	320	640	320	320	
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done															
Vaccine SH 2024 SH 2024-25															

Table H3-17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-08-02

Viruses		Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre													
								Post-infection ferret antiserum													
								A/Albania/289813/2022	A/Massachusetts/18/2022	A/Thailand/08/2022	A/Sydney/05/2023	A/Croatia/10136/RV/2023	A/Croatia/10663/2023	A/Slovenia/49/2024	A/Lisboa/216/2023	U France/DF-12374/2023	A/Norway/12374/2023				
								SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	Egg	Cell	Cell				
								F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	F22/24	F21/24			
								2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.4)	2a.3a.1 (J.4)		
REFERENCE VIRUSES		N122D			2a.3a.1 (J)	2022-12-13	IDCK1/SIAT2	1280	640	1280	1280	320	320	640	160	640	640	320			
A/Albania/289813/2022				2a.3a.1 (J)		SIAT3/SIAT1	2560	1280	2560	320	320	320	640	320	640	640	320				
A/Massachusetts/18/2022				2a.3a.1 (J)		ESIE1	1280	320	1280	640	320	640	640	320	640	320	160				
A/Sydney/05/2023				2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	1280	640	1280	1280	160	320	320	160	640	320	160				
A/Croatia/10136/RV/2023				2a.3a.1 (J.2)	2023-12-04	SIAT3	320	80	320	320	320	320	320	320	320	320	640	160			
A/Croatia/10136/RV/2023	N158K			2a.3a.1 (J.2)		E3 (Am1A2)	1280	320	640	640	640	640	640	640	640	640	640	320			
A/Netherlands/10663/2023				2a.3a.1 (J.2)	2023-11-10	C-MIX2/SIAT3	640	320	320	320	320	320	320	640	320	640	320	320			
A/Slovenia/49/2024				2a.3a.1 (J.2)	2024-01-08	IDCKx/SIAT3	40	40	160	40	160	40	160	80	2560	320	<40	40			
A/Lisboa/216/2023		S124N			2a.3a.1 (J.2)	2023-12-15	E3 (Am1A2)	320	160	320	320	160	320	320	160	320	320	160			
A/France/IDF-IP29542/2023		K189R			2a.3a.1 (J.4)	2023-11-30	IDCK1/SIAT1	40	<40	80	40	160	80	160	<40	80	160	80			
A/Norway/12374/2023				2a.3a.1 (J.4)	2023-11-28	SIAT3	320	160	640	320	320	320	640	320	640	320	640	320	320		
TEST VIRUSES																					
244030	A/Ghana/2545/2024			2a.3a (G.1.3.1)	2024-05-02	SIAT1	640	160	640	320	320	320	320	160	640	160	160				
244028	A/Ghana/3195/2024			2a.3a (G.1.3.1)	2024-05-04	SIAT1	160	80	160	80	160	160	160	160	160	160	80				
244027	A/Ghana/818/2024			2a.3a (G.1.3.1)	2024-05-05	SIAT1	80	40	160	160	160	160	160	320	160	320	160	80			
244024	A/Ghana/3409/2024			2a.3a (G.1.3.1)	2024-05-18	SIAT1	80	40	160	160	160	160	160	320	160	160	80				
244070	A/Argentina/1025/2024			2a.3a.1 (J.1)	2024-06-04	SIAT1	320	160	320	320	80	80	80	40	160	80	40				
244038	A/Argentina/1211/2024			2a.3a.1 (J.1)	2024-06-25	SIAT1	640	320	1280	640	320	160	320	80	320	160	160				
244031	A/Ghana/1824/2024			2a.3a.1 (J.2)	2024-04-03	SIAT1	640	160	320	320	320	320	320	320	640	320	160				
244056	A/Argentina/1290/2024			2a.3a.1 (J.2)	2024-06-06	SIAT1	320	80	160	80	80	160	160	160	320	160	80				
244026	A/Ghana/3104/2024			2a.3a.1 (J.2)	2024-06-06	SIAT1	640	160	640	640	640	640	640	640	640	640	160				
244064	A/Argentina/1065/2024			2a.3a.1 (J.2)	2024-06-07	SIAT1	320	160	320	160	320	320	320	160	320	320	160				
244060	A/Argentina/1071/2024			2a.3a.1 (J.2)	2024-06-09	SIAT1	320	160	160	160	160	160	320	160	320	320	160				
244056	A/Argentina/1215/2024			2a.3a.1 (J.2)	2024-06-15	SIAT1	320	160	320	160	160	160	320	160	320	160	80				
244055	A/Argentina/986/2024			2a.3a.1 (J.2)	2024-06-17	SIAT1	160	80	160	160	160	160	320	160	160	160	160				
244054	A/Argentina/1077/2024			2a.3a.1 (J.2)	2024-06-18	SIAT1	160	80	160	80	160	160	320	80	160	320	80				
244053	A/Argentina/1084/2024			2a.3a.1 (J.2)	2024-06-18	SIAT1	320	80	320	320	160	160	320	160	320	160	80				
244052	A/Argentina/1130/2024			2a.3a.1 (J.2)	2024-06-18	SIAT1	320	160	320	320	160	320	320	160	320	320	160				
244050	A/Argentina/1178/2024			2a.3a.1 (J.2)	2024-06-18	SIAT1	320	160	320	320	160	320	320	160	320	160	160				
244049	A/Argentina/1128/2024			2a.3a.1 (J.2)	2024-06-19	SIAT1	320	160	320	320	160	320	320	160	320	320	160				
244048	A/Argentina/1163/2024			2a.3a.1 (J.2)	2024-06-19	SIAT1	320	160	320	320	160	320	320	160	320	320	160				
244044	A/Argentina/1162/2024			2a.3a.1 (J.2)	2024-06-21	SIAT1	320	160	320	320	160	320	320	160	320	320	160				
244043	A/Argentina/1221/2024			2a.3a.1 (J.2)	2024-06-21	SIAT1	160	80	320	160	160	160	320	160	320	160	80				
244042	A/Argentina/1244/2024			2a.3a.1 (J.2)	2024-06-21	SIAT1	160	80	160	160	160	160	160	160	320	160	80				
244045	A/Argentina/996/2024			2a.3a.1 (J.2)	2024-06-21	SIAT1	320	160	320	320	320	320	320	160	320	320	160				
244041	A/Argentina/1167/2024			2a.3a.1 (J.2)	2024-06-24	SIAT1	80	40	80	80	80	80	160	80	160	80	40				
244040	A/Argentina/1231/2024			2a.3a.1 (J.2)	2024-06-24	SIAT1	160	80	320	160	160	160	320	160	320	160	80				
244039	A/Argentina/1256/2024			2a.3a.1 (J.2)	2024-06-24	SIAT1	320	160	320	160	160	320	320	320	320	160	160				
244036	A/Argentina/1276/2024			2a.3a.1 (J.2)	2024-06-28	SIAT1	320	80	160	160	160	160	160	160	320	160	80				
244035	A/Argentina/1219/2024			2a.3a.1 (J.2)	2024-07-05	SIAT1	320	160	320	320	160	320	320	160	320	160	160				
< relates to the lowest dilution of antiserum used																					
1 hyperimmune sheep serum; ND = Not Done																					
										Vaccine SH 2024											
										NH 2024-25											

< relates to the lowest dilution of antiserum used
 hyperimmune sheep serum; ND = Not Done

Vaccine
 SH 2024
 NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H3-18. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-08-09

Viruses	Other Information	Haemagglutination inhibition titre																			
		Post-infection ferret antiserum																			
		A/Albania/289813/2022		A/Massachusetts/18/2022		A/Thailand/08/2022		A/Sydney/856/2023		A/Croatia/10136/RV/2023		A/Netherlands/10563/2023		A/Slovenia/49/2024		A/Lisboa/216/2023		U France/IDF-IPP/29542/2023		A/Norway/12374/2023	
		Passage	SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	SIAT	SIAT	SIAT	SIAT	Egg	Cell	Cell	Cell	
		Ferret number	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	F22/24	F21/24								
Genetic group		Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.4)	2a.3a.1 (J.4)						
REFERENCE VIRUSES																					
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	IDCK3/SIAT2	1280	640	1280	1280	160	320	320	80	640	320	160						
A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	160	160	320	320	160	320	160						
A/Thailand/08/2022		2a.3a.1 (J)		ES/E1	1280	320	1280	640	320	640	640	320	640	320	160						
A/Sydney/856/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	1280	640	1280	1280	160	320	320	160	320	320	160						
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)	2023-12-04	SIAT3	160	80	160	160	160	320	160	320	160	320	160						
A/Croatia/10136/RV/2023	N158K	2a.3a.1 (J.2)		E3 (Am1A2)	640	160	640	640	320	640	320	320	640	320	160						
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	C-MIX2/SIAT3	320	320	320	320	320	320	640	320	640	320	160						
A/Slovenia/49/2024		2a.3a.1 (J.2)	2024-01-08	IDCK3/SIAT3	40	80	160	40	320	160	160	2560	320	80	160						
A/Lisboa/216/2023		2a.3a.1 (J.2)	2023-12-15	E3 (Am1A2)	320	160	640	640	320	320	640	320	640	320	160						
A/France/IDF-IPP/29542/2023		K189R	2a.3a.1 (J.4)	2023-11-30	IDCK1/SIAT2	40	<40	80	<40	80	80	160	<40	40	160	80					
A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	80	320	160	160	160	320	320	320	320	160						
TEST VIRUSES																					
244142	A/Antsirabe/01553/2024	2a.3a.1 (J)	2024-05-24	SIAT1	640	320	320	320	80	160	160	80	160	80	40						
244148	A/Ambositra/01616/2024	2a.3a.1 (J)	2024-05-30	SIAT1	640	320	640	640	80	80	160	80	160	160	80						
244141	A/Fianarantsoa/01771/2024	2a.3a.1 (J)	2024-06-11	SIAT1	640	320	640	320	80	80	160	40	160	160	40						
244138	A/Ambositra/01969/2024	2a.3a.1 (J)	2024-06-25	SIAT1	640	160	320	160	320	640	640	320	320	320	160						
244137	A/Ambositra/02034/2024	2a.3a.1 (J)	2024-07-01	SIAT1	320	160	320	160	160	160	320	160	320	320	160						
244157	A/Moscow/26/2024	2a.3a.1 (J.1)	2024-03-18	SIAT2/SIAT1	640	320	640	640	80	160	160	40	80	160	40						
244164	A/Moscow/3/2024	2a.3a.1 (J.2)	2024-01-19	SIAT2/SIAT1	160	80	160	80	80	160	160	80	160	160	80						
244163	A/Orenburg/20/2024	2a.3a.1 (J.2)	2024-02-06	IDCK3/SIAT1	160	80	320	160	160	160	320	160	320	160	80						
244162	A/Orenburg/19/2024	2a.3a.1 (J.2)	2024-02-07	IDCK3/SIAT1	320	160	320	160	160	320	320	160	320	160	160						
244161	A/Orenburg/21/2024	2a.3a.1 (J.2)	2024-02-07	IDCK3/SIAT1	160	80	160	160	160	160	320	160	320	160	80						
244160	A/Orenburg/24/2024	2a.3a.1 (J.2)	2024-03-04	IDCK3/SIAT1	160	80	160	80	160	160	160	80	320	160	80						
244159	A/Moscow/24/2024	2a.3a.1 (J.2)	2024-03-18	SIAT1/SIAT1	160	80	160	80	160	160	320	160	320	160	80						
244158	A/Moscow/25/2024	2a.3a.1 (J.2)	2024-03-18	SIAT1/SIAT1	160	80	160	160	160	160	320	160	320	160	80						
244156	A/Moscow/27/2024	2a.3a.1 (J.2)	2024-04-01	SIAT1/SIAT1	160	80	160	80	80	160	160	80	160	160	80						
244071	A/Argentina/1114/2024	2a.3a.1 (J.2)	2024-05-01	SIAT2	320	160	320	160	160	320	320	320	320	320	160						
244069	A/Argentina/1021/2024	2a.3a.1 (J.2)	2024-05-05	SIAT2	320	160	320	160	320	640	320	320	320	640	320						
244068	A/Argentina/1151/2024	2a.3a.1 (J.2)	2024-06-06	SIAT2	160	80	320	160	160	160	320	160	320	160	80						
244062	A/Argentina/1167/2024	2a.3a.1 (J.2)	2024-06-08	SIAT2	80	40	80	80	160	160	320	160	160	160	80						
244061	A/Argentina/1173/2024	2a.3a.1 (J.2)	2024-06-08	SIAT2	640	320	640	640	160	320	320	160	320	160	80						
244059	A/Argentina/1094/2024	2a.3a.1 (J.2)	2024-06-09	SIAT2	640	320	640	320	320	320	640	320	640	640	320						
244057	A/Argentina/1080/2024	2a.3a.1 (J.2)	2024-06-14	SIAT2	320	160	320	320	160	320	320	320	320	160	160						
244140	A/Itasy/01844/2024	2a.3a.1 (J.2)	2024-06-17	SIAT1	160	80	160	80	160	160	320	160	320	160	160						
244139	A/Itasy/01846/2024	2a.3a.1 (J.2)	2024-06-17	SIAT1	160	80	320	80	160	160	320	160	320	160	80						
244051	A/Argentina/1166/2024	2a.3a.1 (J.2)	2024-06-18	SIAT3	320	160	320	320	160	320	640	320	640	320	160						
244047	A/Argentina/1225/2024	2a.3a.1 (J.2)	2024-06-19	SIAT2	160	80	320	160	160	160	320	160	320	320	160						
< relates to the lowest dilution of antiserum used																					
1 hyperimmune sheep serum; ND = Not Done																					
Vaccine SH 2024																					
NH 2024-25																					

Table H3-19. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-08-13

Viruses		Other information	Haemagglutination inhibition titre																
			Post-infection ferret antiserum																
			A/Albania/289813/2022	A/Massachusetts/18/2022	A/Thailand/08/2022	A/Sydney/05/2023	A/Croatia/10136/RV/2023	A/Croatia/10136/RV/2023	A/Netherlands/10563/2023	A/Slovenia/49/2024	A/Lisboa/216/2023	VFrance/IDF-PP29542/2023	A/Norway/12374/2023						
			Passage	SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	SIAT	Egg	Cell	Cell					
		Ferret number	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	F22/24	F21/24						
		Genetic group	Collection Date	Genetic group	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.4)	2a.3a.1 (J.4)					
REFERENCE VIRUSES																			
A/Albania/289813/2022	N122D	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT2	1280	640	1280	1280	160	320	640	160	320	640					
A/Massachusetts/18/2022		2a.3a.1 (J)		SIAT3/SIAT1	1280	640	1280	1280	320	160	320	160	320	160					
A/Thailand/08/2022		2a.3a.1 (J)		E3/E1	1280	320	1280	640	640	640	640	640	640	320					
A/Sydney/05/2023		2a.3a.1 (J.1)	2023-09-12	SIAT1/SIAT2	1280	640	1280	1280	160	320	320	160	320	160					
A/Croatia/10136/RV/2023	N158K	2a.3a.1 (J.2)	2023-12-04	SIAT3	160	80	160	160	320	320	160	320	320	80					
A/Croatia/10136/RV/2023		2a.3a.1 (J.2)		E3 (Am1A12)	1280	320	640	640	640	640	640	640	640	320					
A/Netherlands/10563/2023		2a.3a.1 (J.2)	2023-11-10	JCK-MIX2/SIAT3	320	160	320	160	160	320	320	320	640	320	160				
A/Slovenia/49/2024		2a.3a.1 (J.2)	2024-01-08	MDCKx/SIAT3	40	40	160	<40	160	40	80	2560	160	<40	40				
A/Lisboa/216/2023	S124N	2a.3a.1 (J.2)	2023-12-15	E3 (Am1A12)	320	160	640	640	320	320	640	320	640	320	160				
A/France/IDF-PP29542/2023	K189R	2a.3a.1 (J.4)	2023-11-30	MDCK1/SIAT2	40	<40	40	<40	80	80	80	<40	40	80	80				
A/Norway/12374/2023		2a.3a.1 (J.4)	2023-11-28	SIAT4	160	80	320	160	160	160	320	320	160	320	160				
TEST VIRUSES																			
244234	A/Catalonia/NSVH102337351/2024	2a.3a.1 (J)	2024-05-07	SIAT2/SIAT1	640	320	640	640	80	160	160	160	80	160	80				
244196	A/England/189/2023	2a.3a.1 (J.1)	2023-12-28	SIAT1/SIAT1	640	320	640	320	80	160	160	160	80	160	80				
244195	A/England/19/2024	2a.3a.1 (J.2)	2024-01-09	MDCK1/SIAT1	80	40	80	40	80	80	160	80	160	80	40				
244194	A/England/64/2024	2a.3a.1 (J.2)	2024-01-30	MDCK1/SIAT1	160	80	320	160	160	160	320	160	320	160	80				
244193	A/England/61/2024	2a.3a.1 (J.2)	2024-02-01	MDCK1/SIAT1	80	<40	80	40	80	80	160	80	160	80	40				
244192	A/England/62/2024	2a.3a.1 (J.2)	2024-02-02	MDCK1/SIAT1	320	160	320	320	80	80	160	40	160	80	40				
244205	A/Hungary/201/2024	2a.3a.1 (J.2)	2024-02-12	MDCK1/SIAT1	320	80	320	160	160	160	320	160	320	80	80				
244191	A/England/70/2024	2a.3a.1 (J.2)	2024-02-21	SIAT1/SIAT1	320	160	320	160	160	160	320	160	320	320	160				
244190	A/England/65/2024	2a.3a.1 (J.2)	2024-03-05	MDCK1/SIAT1	80	80	80	80	160	160	320	160	160	160	80				
244189	A/England/66/2024	2a.3a.1 (J.2)	2024-03-08	MDCK1/SIAT1	320	80	160	80	320	160	320	160	320	320	160				
244188	A/England/67/2024	2a.3a.1 (J.2)	2024-03-12	MDCK1/SIAT1	320	80	160	160	160	320	320	160	320	160	160				
244187	A/England/69/2024	2a.3a.1 (J.2)	2024-03-13	MDCK1/SIAT1	320	80	160	160	160	320	320	160	320	160	80				
244186	A/England/71/2024	2a.3a.1 (J.2)	2024-03-14	SIAT1/SIAT1	160	40	160	80	80	80	160	80	160	40	40				
244185	A/England/68/2024	2a.3a.1 (J.2)	2024-03-15	MDCK1/SIAT1	160	80	160	160	160	320	320	80	160	160	80				
244184	A/England/73/2024	2a.3a.1 (J.2)	2024-03-18	MDCK1/SIAT1	80	40	80	80	80	160	160	80	160	80	40				
244183	A/England/72/2024	2a.3a.1 (J.2)	2024-03-21	SIAT1/SIAT1	160	80	320	160	160	160	320	160	320	160	160				
244204	A/Hungary/264/2024	2a.3a.1 (J.2)	2024-03-26	MDCK1/SIAT1	80	40	80	40	80	80	160	80	160	80	40				
244230	A/Catalonia/NSVH102343016/2024	2a.3a.1 (J.2)	2024-05-15	SIAT2/SIAT1	320	160	320	160	160	320	320	160	320	160	160				
244228	A/Catalonia/NSJOS10768271/2024	2a.3a.1 (J.2)	2024-05-23	SIAT2/SIAT1	80	80	160	160	160	320	320	160	320	320	80				
244226	A/Catalonia/NSVH102367246/2024	2a.3a.1 (J.2)	2024-06-19	SIAT2/SIAT1	160	80	160	160	160	160	320	160	320	320	160				
244224	A/Catalonia/NSVH102371258/2024	2a.3a.1 (J.2)	2024-06-25	SIAT2/SIAT1	80	40	80	40	80	80	80	80	160	80	40				
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done					Vaccine SH 2024 NH 2024-25														

< relates to the lowest dilution of antiserum used
 † hyperimmune sheep serum; ND = Not Done

Vaccine
 SH 2024
 NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-20. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-08-29

Viruses	Other information	Haemagglutination inhibition titre																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
		Pre-infection serum antibodies												Post-infection serum antibodies																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
		A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202			A/Adana/0581/19202		

¹ hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2024
NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H3-21. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-09-03

[illegible]

¹ hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2024
NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H3-22. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-09-10

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre												
					Post-infection ferret antiserum												
					A/Albania /289813/2022	A/Massachusetts /19/2022	A/Thailand /08/2022	A/Sydney /856/2023	A/Croatia /136/RV/2023	A/Croatia /10563/2023	A/Slovenia /49/2024	A/Lisboa /P29542/2023	A/France/IDF- /12374/2023	A/Norway /BurkinaFaso /1313/2023			
					Passage SIAT	SIAT	Egg	SIAT	SIAT	Egg	SIAT	Egg	Cell	Cell	Cell		
Ferret number	F21/23	F36/23	F34/23	F01/24	F06/24	F16/24	F08/24	F11/24	F15/24	F22/24	F21/24	F32/24					

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not Done

Vaccine

SH 2024

NH 2024-25

< 4-fold

4-fold

8-fold

> 8-fold

<

not recognised by the antiserum

>= 160

(no homologous titre)

Table 23. (Additional) Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-09-10

Viruses	Other information	Haemagglutination inhibition titre																				
		Post-infection ferret antiserum																				
		A/Croatia/10136/RV/2023 SIAT	A/Croatia/10136/RV/2023 Egg	A/Netherlands/10963/2023 SIAT	A/Lisboa/216/2023 Egg	A/Slovenia/49/2024 umbria/27/2023 SIAT	A/District Of Umbria/27/2023 mbia/27/2023 SIAT	A/Colorado/06/2024 SIAT	A/Nevada/132/2023 SIAT	A/Sydney/07/8/2023 SIAT	A/Idaho/69/2023 SIAT	A/Oklahoma/05/2024 EGG										
		Passage	Ferret number	F06/24	F16/24	F08/24	F15/24	F11/24	F24-022	F24-037	F24-035	F24-012	A9825	F24-033	F24-044							
		Genetic group	Collection Date	Genetic group	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)		
REFERENCE VIRUSES																						
	S145N	A/Croatia/10136/RV/2023	2023-12-04	SIAT3	320	320	320	160	160	160	160	320	320	320	40	40	40	40	40	40		
	S145N	A/Croatia/10136/RV/2023		E3 (Am1A12)	640	640	640	640	320	640	640	640	640	640	80	80	80	80	80	80		
	S124N	A/Netherlands/10963/2023	2023-11-10	CA-MIX2/SIAT3	160	320	320	320	160	160	160	320	640	320	40	80	80	80	80	80		
	S124N	A/Lisboa/216/2023	2023-12-15	E3 (Am1A12)	320	320	320	320	160	160	320	640	640	640	40	80	80	80	80	80		
	N158K	A/Slovenia/49/2024	2024-01-08	IDCKx/SIAT3	160	40	80	160	1280	80	80	80	80	80	<40	640	640	640	640	640		
		A/District Of Columbia/27/2023		SIAT2	320	320	320	320	160	320	320	640	640	640	80	40	40	40	40	40		
		A/District Of Columbia/27/2023		SpPZE2	320	320	320	640	320	320	320	320	320	640	640	80	40	40	40	40		
		A/Idaho/69/2023		SPPE1	160	320	160	40	80	160	160	80	160	320	80	160	80	<40	1280	1280		
		A/Oklahoma/05/2024		SIAT1	160	40	80	320	2560	80	160	80	160	80	80	160	<40	1280	1280	1280		
		A/Colorado/06/2024		SIAT1	160	160	320	320	160	160	160	320	640	640	640	80	80	80	80	80		
		A/Nevada/32/2023		SIAT2	640	320	640	640	640	640	640	640	1280	1280	1280	640	1280	1280	1280	1280		
		A/Shanghai-Fengxian/1912/2023		E2/E10	320	640	640	640	320	640	1280	1280	1280	1280	1280	1280	160	160	160	160		
		A/Sydney/07/8/2023		SIAT1	640	640	640	640	320	320	320	640	1280	1280	640	1280	1280	1280	1280	1280		
	S145N	A/Victoria/2997/2023		SIAT2	160	320	320	160	160	160	160	320	640	640	320	80	40	40	40	40		
		A/Canberra/302/2023		E2	640	640	640	640	320	640	640	640	1280	1280	1280	1280	80	320	320	320		
		A/Canberra/309/2023		E2	160	320	320	320	160	320	320	320	640	640	320	40	80	80	80	80		
		A/Victoria/2776/2023		E2	640	640	640	1280	320	320	320	640	640	1280	1280	40	320	320	320	320		
		IVR-252 A/Canberra/302/2023		E2/D10	160	160	160	320	160	320	320	640	640	320	320	1280	40	160	160	160		
		IVR-253 A/Canberra/302/2023		EX/D7	160	160	160	320	320	160	160	160	320	640	640	320	40	40	40	40		
		IVR-254 A/Canberra/373/2023		E4/D7	80	160	160	160	160	160	160	160	320	320	320	320	<40	<40	<40	<40		
		IVR-258 A/Victoria/2997/2023		E4/D6	320	640	640	640	160	160	160	160	320	320	320	640	320	<40	80	80		
		A/Stockholm/SE24-53070/2024	2024-04-22	SIAT1/SIAT1	160	320	320	320	160	160	160	160	320	640	640	320	80	40	40	40		
243728		A/BosniaAndHerzegovina/502/2024	2024-01-30	SIAT1	160	160	160	320	160	160	160	160	320	640	640	320	80	40	40	40		
244224		A/Catalonia/NSVH102371258/2024	2024-06-25	SIAT2/SIAT1	160	320	320	320	160	160	160	160	320	640	640	320	80	40	40	40		
244234		A/Catalonia/NSVH10237351/2024	2024-05-07	SIAT2/SIAT1	160	320	320	320	160	160	160	160	320	640	640	320	80	40	40	40		
		A/Idaho/69/2023		S2	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND		
		A/Oklahoma/05/2024		SpPZE1	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND		

< relates to the lowest dilution of antiserum used
ND = Not Done

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum

Microneutralisation tables: A/H3N2

Table MN-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-06-20

Viruses				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand 8/2022 Egg F34/23 2a.3a.1 (J)	A/Albania 289813/2022 SIAT F21/23 2a.3a.1 (J)	A/Sydney 856/2023 SIAT F01/24 2a.3a.1 (J.1)	A/Croatia 10136/RV/2023 SIAT F06/24 2a.3a.1 (J.2)	A/Netherlands 10563/2023 SIAT F08/24 2a.3a.1 (J.3)	A/Slovenia 49/2024 SIAT F11/24 2a.3a.1 (J.3)
Genetic group	Collection Date	Passage history Ferret number Passage History								
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ³	218	405	431	252	223	506	95
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁶	590	2112	1511	1058	958	2710	694
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ³	273	629	535	427	279	381	102
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ³	252	437	440	288	248	517	102
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ³	97	182	120	206	273	402	135
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ³	242	472	466	342	363	833	178
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ³	<	195	<	40	171	58	3908
TEST VIRUSES										
A/Israel/R2254/2022	2a.3a.1 (J.2)	2024-02-02	Px/SIAT1	191	485	452	144	250	501	211
A/Israel/R2239/2022	2a.3a.1 (J.2)	2024-02-01	Px/SIAT1	256	505	480	256	348	461	123
A/Israel/R2231/2022	2a.3a.1 (J.2)	2024-02-01	Px/SIAT1	257	380	360	223	356	496	104
A/Israel/R2165/2022	2a.3a.1 (J.2)	2024-02-01	Px/SIAT1	238	437	400	236	316	566	169
A/Israel/R2261/2024	2a.3a.1 (J.2)	2024-02-02	SIAT1	331	711	576	377	389	940	189
A/Moldova/36860/2024	2a.3a.1 (J.2)	2024-02-08	SIAT1	114	270	152	307	341	594	147
A/Berlin/26/2024	2a.3a.1 (J.2)	2024-02-26	P1/SIAT2	252	512	504	298	386	568	136
A/Hessen/33/2024	2a.3a.1 (J.2)	2024-02-16	P1/SIAT2	152	293	297	168	267	576	104
A/Baden-Wuerttemberg/55/2024	2a.3a.1 (J.2)	2024-02-08	P1/SIAT2	164	391	320	230	283	473	128

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>4-8-fold

>8-fold

not recognised by the antiserum

Table MN-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-06-27

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	A/Albania 289813/2022 SIAT F21/23	A/Sydney 856/2023 SIAT F01/24	A/Croatia 10136/RV/2023 SIAT F06/24	A/Netherlands 10563/2023 SIAT F08/24	A/Slovenia 49/2024 SIAT F11/24
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	218	404.5	431	252	222	506	95
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁵	590	2112	1511	1058	958	2710	694
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	273	629	535	427	279	381	102
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	252	437	440	288	248	517	102
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	97	182	120	206	273	402	135
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	242	472	466	342	363	833	178
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	195	<	40	171	58	3908
TEST VIRUSES										
A/Morocco/1850/2024	2a.3a.1 (J.1)	2024-02-15	SIAT1	248	510	550	294	277	525	106
A/Morocco/1876/2024	2a.3a.1 (J.2)	2024-02-21	SIAT1	243	470	437	267	416	735	204
A/Morocco/1846/2024	2a.3a.1 (J.2)	2024-02-19	SIAT1	247	516	451	320	437	545	137
A/Morocco/1889/2024	2a.3a.1 (J.2)	2024-02-12	SIAT2	226	419	452	231	303	797	135
A/Slovenia/691/2024	2a.3a.1 (J.2)	2024-03-12	MDCKF1/SIAT1	153	232	217	131	232	489	144
A/Slovenia/671/2024	2a.3a.1 (J.2)	2024-03-07	MDCKF1/SIAT1	106	151	154	114	156	480	80
A/Slovenia/643/2024	2a.3a.1 (J.2)	2024-03-06	MDCKF1/SIAT1	127	220	223	129	169	371	76
A/Slovenia/648/2024	2a.3a.1 (J.2)	2024-03-05	MDCKF1/SIAT1	117	163	175	122	202	420	79
A/Slovenia/594/2024	2a.3a.1 (J.2)	2024-02-22	MDCKF1/SIAT1	113	165	148	112	160	354	74

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Table MN-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-07-04

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	A/Albania 289813/2022 SIAT F21/23	A/Sydney 856/2023 SIAT F01/24	A/Croatia 10136/RV/2023 SIAT F06/24	A/Netherlands 10563/2023 SIAT F08/24	A/Slovenia 49/2024 SIAT F11/24
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁶	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Serbia/662/2024	2a.3a.1 (J.1)	2024-02-12	SIAT2	133	222	232	133	120	363	64
A/Serbia/477/2024	2a.3a.1 (J.1)	2024-02-02	SIAT2	184	264	270	158	149	320	71
A/Serbia/636/2024	2a.3a.1 (J.2)	2024-02-09	SIAT1	100	135	137	100	129	356	64
A/Slovenia/588/2024	2a.3a.1 (J.2)	2024-02-21	MDCKF1/SIAT1	147	238	236	122	165	512	89
A/Slovenia/453/2024	2a.3a.1 (J.2)	2024-02-09	M/SIAT1	123	171	154	97	145	402	73
A/Slovenia/414/2024	2a.3a.1 (J.2)	2024-02-04	MDCKF1/SIAT1	94	132	123	82	129	366	64
A/Bulgaria/2507/2024	2a.3a.1 (J.2)	2024-03-18	SIAT2/SIAT1	105	129	136	101	141	289	77
A/Bulgaria/2502/2024	2a.3a.1 (J.2)	2024-03-18	SIAT1/SIAT1	126	220	260	136	234	450	77
A/Bulgaria/2443/2024	2a.3a.1 (J.2)	2024-03-14	SIAT2/SIAT1	95	137	192	103	155	315	86
A/Bulgaria/2402/2024	2a.3a.1 (J.2)	2024-02-29	SIAT2/SIAT1	155	217	193	139	196	513	97
A/Bulgaria/2398/2024	2a.3a.1 (J.2)	2024-02-29	SIAT2/SIAT1	203	276	276	133	171	614	110
A/Bulgaria/2261/2024	2a.3a.1 (J.2)	2024-02-14	SIAT2/SIAT1	107	158	166	107	181	456	70
A/Bulgaria/2222/2024	2a.3a.1 (J.2)	2024-02-13	SIAT2/SIAT1	204	265	265	151	255	373	103
A/Bulgaria/2256/2024	2a.3a.1 (J.2)	2024-02-12	SIAT2/SIAT1	117	208	192	109	195	391	85
A/Bulgaria/2091/2024	2a.3a.1 (J.2)	2024-02-08	SIAT2/SIAT1	155	257	256	227	280	243	94

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 not recognised by the antiserum

Table MN-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-07-11

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand 8/2022 Egg F34/23 2a.3a.1 (J)	A/Albania 289813/2022 SIAT F21/23 2a.3a.1 (J)	A/Sydney 856/2023 SIAT F01/24 2a.3a.1 (J.1)	A/Croatia 10136/RV/2023 SIAT F06/24 2a.3a.1 (J.2)	A/Netherlands 10563/2023 SIAT F08/24 2a.3a.1 (J.2)	A/Slovenia 49/2024 SIAT F11/24 2a.3a.1 (J.2)
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History							
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁶	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Ghana/1812/2024	2a.3a (G.1.3.1)	2024-04-04	SIAT2	61	142	83	98	120	353	53
A/Turkmenistan/1738/2024	2a.3a.1 (J.2)	2024-03-25	SIAT1	173	275	254	176	275	564	96
A/Albania/302978/2024	2a.3a.1 (J.2)	2024-02-22	SIAT1	201	299	303	209	274	349	132
A/Albania/302854/2024	2a.3a.1 (J.2)	2024-02-19	SIAT1	143	217	213	122	202	453	78
A/Albania/302815/2024	2a.3a.1 (J.2)	2024-02-18	SIAT1	116	146	90	142	222	435	108
A/Albania/302794/2024	2a.3a.1 (J.2)	2024-02-15	SIAT1	209	295	330	209	283	457	103
A/Albania/302684/2024	2a.3a.1 (J.2)	2024-02-14	SIAT1	190	285	293	156	253	434	101
A/Albania/302626/2024	2a.3a.1 (J.2)	2024-02-13	SIAT1	133	242	212	141	237	436	74
A/Ghana/1725/2024	2a.3a.1 (J.2)	2024-03-27	MDCK1	150	227	202	121	205	414	95
A/Ghana/1036/2024	2a.3a.1 (J.2)	2024-02-21	SIAT2	231	261	257	146	251	530	143
A/Lithuania/240419 0036/2024	2a.3a.1 (J.2)	2024-04-13	SIAT1	145	208	217	131	176	683	117
A/Lithuania/240325 0037/2024	2a.3a.1 (J.2)	2024-03-22	SIAT1	122	196	194	128	172	290	72
A/Lithuania/240321 0014/2024	2a.3a.1 (J.2)	2024-03-15	SIAT1	101	160	156	123	153	354	75
A/Lithuania/240404 0009/2024	2a.3a.1 (J.2)	2024-03-01	SIAT1	120	178	181	118	151	386	69

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not recognised by the antiserum

Table MN-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-07-18

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	A/Albania 289813/2022 SIAT F21/23	A/Sydney 856/2023 SIAT F01/24	A/Croatia 10136/RV/2023 SIAT F06/24	A/Netherlands 10563/2023 SIAT F08/24	A/Slovenia 49/2024 SIAT F11/24
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁵	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Poland/39/2024	2a.3a.1 (J.1)	2024-03-20	SIAT1	153	310	257	154	135	272	71
A/Iceland/7184/2024	2a.3a.1 (J.1)	2024-03-03	SIAT1	158	298	311	204	114	184	58
A/Poland/25/2024	2a.3a.1 (J.2)	2024-03-13	SIAT2	155	254	254	151	210	400	119
A/Iceland/14274/2024	2a.3a.1 (J.2)	2024-05-07	SIAT1	93	158	142	115	114	195	64
A/Iceland/12503/2024	2a.3a.1 (J.2)	2024-04-20	SIAT1	133	64	150	171	65	99	880
A/Iceland/12053/2024	2a.3a.1 (J.2)	2024-04-17	SIAT1	111	70	58	<	56	45	<
A/Iceland/10860/2024	2a.3a.1 (J.2)	2024-04-06	SIAT1	139	224	210	130	127	274	75
A/Iceland/10805/2024	2a.3a.1 (J.2)	2024-04-05	SIAT1	95	158	149	110	96	160	57
A/Iceland/8862/2024	2a.3a.1 (J.2)	2024-03-16	SIAT1	75	123	80	126	129	235	76
A/Mozambique/700392/2024	2a.3a.1 (J.2)	2024-04-19	SIAT1	92	156	145	80	120	232	68
A/Mozambique/700390/2024	2a.3a.1 (J.2)	2024-04-19	SIAT1	139	238	204	121	133	271	71

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 not recognised by the antiserum

Table MN-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-07-25

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand 8/2022 Egg F34/23 2a.3a.1 (J)	A/Albania 289813/2022 SIAT F21/23 2a.3a.1 (J)	A/Sydney 856/2023 SIAT F04/23 2a.3a.1 (J.1)	A/Croatia 10136/RV/2023 SIAT F06/24 2a.3a.1 (J.2)	A/Netherlands 10563/2023 SIAT F08/24 2a.3a.1 (J.2)	A/Slovenia 49/2024 SIAT F11/24 2a.3a.1 (J.2)
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History							
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁵	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Mozambique/699521/2024	2a.3a.1 (J.2)	2024-04-13	SIAT1	121	208	226	132	127	246	70
A/Mozambique/699073/2024	2a.3a.1 (J.2)	2024-04-10	SIAT1	109	168	188	123	175	361	76
A/Mozambique/699070/2024	2a.3a.1 (J.2)	2024-04-10	MDCK1	121	201	206	145	142	338	63
A/Mozambique/699067/2024	2a.3a.1 (J.2)	2024-04-10	MDCK1	149	230	246	137	176	435	113
A/Mozambique/699052/2024	2a.3a.1 (J.2)	2024-04-09	MDCK1	134	179	200	125	170	473	80
A/Mozambique/699051/2024	2a.3a.1 (J.2)	2024-04-09	MDCK1	175	240	260	149	240	369	108
A/Mozambique/699048/2024	2a.3a.1 (J.2)	2024-04-04	MDCK1	101	200	226	137	147	287	105
A/Mozambique/699047/2024	2a.3a.1 (J.2)	2024-04-04	MDCK1	108	228	215	138	160	183	73
A/Mozambique/699103/2024	2a.3a.1 (J.2)	2024-03-29	MDCK1	133	190	215	132	266	398	83
A/Kyrgyzstan/17/2024	2a.3a.1 (J.2)	2024-02-26	MDCK1/MDCK1	75	123	125	71	120	263	62
A/Kyrgyzstan/16/2024	2a.3a.1 (J.2)	2024-02-25	MDCK1/MDCK1	80	136	150	96	131	301	69
A/Kyrgyzstan/15/2024	2a.3a.1 (J.2)	2024-02-22	MDCK1/MDCK1	78	169	158	116	136	235	67
A/Kyrgyzstan/14/2024	2a.3a.1 (J.2)	2024-02-22	MDCK1/MDCK1	86	153	185	120	144	288	73
A/Kyrgyzstan/13/2024	2a.3a.1 (J.2)	2024-02-22	MDCK1/MDCK1	111	178	183	121	140	297	72
A/Kyrgyzstan/12/2024	2a.3a.1 (J.1)	2024-02-21	MDCK1/MDCK1	<	63	<	63	71	<	70

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Table MN-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-08-01

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 19/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	A/Albania 289813/2022 SIAT F21/23	A/Sydney 856/2023 SIAT F01/24	A/Croatia 10136/2023 SIAT F06/24	A/Netherlands 10563/2023 SIAT F08/24	A/Slovenia 49/2024 SIAT F11/24
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁶	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Dakar/22/2024	2a.3a.1 (J.1)	2024-03-25	MDCK1/SIAT1	151	253	306	204	170	220	55
A/Dakar/26/2024	2a.3a.1 (J.1)	2024-03-13	MDCK1/SIAT1	175	272	316	182	211	387	50
A/Dakar/18/2024	2a.3a.1 (J.1)	2024-02-28	MDCK1/SIAT1	137	260	259	154	76	108	<
A/Kyiv/2226/2024	2a.3a.1 (J.1)	2024-03-10	MDCK1/SIAT1	131	248	250	141	119	228	58
A/Netherlands/10553/2024	2a.3a.1 (J.2)	2024-02-22	MDCK-MIX1/MDCK1	88	150	147	94	142	363	67
A/Netherlands/10510/2024	2a.3a.1 (J.2)	2024-02-20	MDCK-MIX2/MDCK1	<	<	<	<	<	<	<
A/Netherlands/10459/2024	2a.3a.1 (J.2)	2024-02-13	MDCK-MIX2/MDCK1	49	72	40	70	115	250	61
A/Netherlands/10458/2024	2a.3a.1 (J.2)	2024-02-12	MDCK-MIX2/MDCK1	133	208	166	114	141	306	72
A/Netherlands/10431/2024	2a.3a.1 (J.2)	2024-02-08	MDCK-MIX2/MDCK1	107	156	177	94	134	331	63
A/Dakar/21/2024	2a.3a.1 (J.2)	2024-02-26	MDCK1/SIAT1	99	150	171	86	165	398	63
A/Dakar/19/2024	2a.3a.1 (J.2)	2024-02-24	MDCK1/SIAT1	<	49	<	45	60	120	44
A/Kyiv/2385/2024	2a.3a.1 (J.2)	2024-03-12	MDCK1/SIAT1	<	68	64	40	67	112	<
A/Ireland/00019283/2024	2a.3a.1 (J.2)	2024-03-01	SIAT1	111	193	166	103	130	264	74
A/Ireland/00013655/2024	2a.3a.1 (J.2)	2024-02-12	SIAT1	<	53	53	<	54	117	<
A/Dakar/11/2024	2a.3a.1 (J.4)	2024-02-02	MDCK1/SIAT1	<	<	<	<	<	<	<

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 not recognised by the antiserum

Table MN-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-08-08

Viruses				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand 8/2022 Egg F34/23 2a.3a.1 (J)	A/Albania 289813/2022 SIAT F21/23 2a.3a.1 (J)	A/Sydney 856/2023 SIAT F01/24 2a.3a.1 (J.1)	A/Croatia 10136/RV/2023 SIAT F06/24 2a.3a.1 (J.2)	A/Netherlands 10563/2023 SIAT F08/24 2a.3a.1 (J.2)	A/Slovenia 49/2024 SIAT F11/24 2a.3a.1 (J.2)
Genetic group	Collection Date	Passage history Ferret number Passage History								
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁵	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Uppsala/SE24-04691/2024	2a.3a.1 (J.1)	2024-04-14	SIAT1/SIAT1	140	249	260	148	91	144	55
A/Ibira/7243386/2024	2a.3a.1 (J.2)	2024-05-14	SIAT1	148	300	311	248	184	307	117
A/Ad Duqm /7241563/2024	2a.3a.1 (J.2)	2024-03-03	SIAT1	136	265	280	224	172	257	93
A/Ukraine/2777/2024	2a.3a.1 (J.2)	2024-03-18	MDCK1/SIAT1	230	366	307	198	264	480	155
A/Norway/1952/2024	2a.3a.1 (J.2)	2024-02-23	SIAT1	<	56	133	<	<	<	1085
A/Norway/1590/2024	2a.3a.1 (J.2)	2024-02-15	SIAT1	<	57	<	<	40	<	1020
A/Albania/302481/2024	2a.3a.1 (J.2)	2024-02-12	SIAT1	139	256	174	120	220	311	72
A/Albania/302422/2024	2a.3a.1 (J.2)	2024-02-10	SIAT1	160	264	302	230	240	231	80
A/Albania/302412/2024	2a.3a.1 (J.2)	2024-02-10	SIAT1	217	388	389	219	217	298	89
A/Ukraine/2724/2024	2a.3a.1 (J.2)	2024-03-25	SIAT1	146	281	240	135	158	291	104
A/Trollhattan/SE24-04469/2024	2a.3a.1 (J.2)	2024-04-09	SIAT1/SIAT1	148	294	265	195	200	237	79
A/Halmstad/SE24-03868/2024	2a.3a.1 (J.2)	2024-03-21	SIAT1/SIAT1	102	193	192	131	126	197	60
A/Ostersund/SE24-03454/2024	2a.3a.1 (J.2)	2024-03-04	SIAT1/SIAT1	<	40	<	<	<	<	793
A/Uppsala/SE24-02917/2024	2a.3a.1 (J.2)	2024-03-03	SIAT1/SIAT1	191	304	315	205	186	305	103

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not recognised by the antiserum

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 not recognised by the antiserum

Table MN-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-08-12

				Neutralisation Titre						
				Post-infection ferret antisera						
				A/Massachusetts 18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand 8/2022 Egg F34/23 2a.3a.1 (J)	A/Albania 289813/2022 SIAT F21/23 2a.3a.1 (J)	A/Sydney 856/2023 SIAT F01/24 2a.3a.1 (J.1)	A/Croatia 10136/RV/2023 SIAT F06/24 2a.3a.1 (J.2)	A/Netherlands 10563/2023 SIAT F08/24 2a.3a.1 (J.2)	A/Slovenia 49/2024 SIAT F11/24 2a.3a.1 (J.2)
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History							
REFERENCE VIRUSES										
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁵	471	1489	1078	555	648	2274	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	2594
TEST VIRUSES										
A/Dakar/17/2024	2a.3a.1 (J.1)	2024-02-22	MDCK1/SIAT1	146	257	262	144	102	190	51
A/Ireland/00034351/2024	2a.3a.1 (J.2)	2024-04-27	SIAT1	50	96	70	101	102	158	64
A/Tunisia/3500/2024	2a.3a.1 (J.2)	2024-03-14	SIAT1	124	236	232	118	144	367	77
A/BosniaAndHerzegovina/649/2024	2a.3a.1 (J.2)	2024-02-08	SIAT1	75	166	202	111	78	152	68
A/BosniaAndHerzegovina/599/2024	2a.3a.1 (J.2)	2024-02-06	SIAT1	77	182	188	135	93	160	64
A/Parma/41/2024	2a.3a.1 (J.2)	2024-03-05	SIAT2/SIAT1	101	173	157	109	149	133	<
A/Switzerland/91699/2024	2a.3a.1 (J.2)	2024-06-14	SIAT1	139	244	283	146	227	282	63
A/Switzerland/90958/2024	2a.3a.1 (J.2)	2024-03-18	MDCK1/SIAT1	105	215	194	107	114	165	70
A/Switzerland/47775/2024	2a.3a.1 (J.2)	2024-03-04	MDCK1/SIAT1	<	<	<	<	<	<	<
A/Dakar/20/2024	2a.3a.1 (J.4)	2024-02-22	MDCK1/SIAT1	<	<	<	<	<	<	<
A/Dakar/16/2024	2a.3a.1 (J.4)	2024-02-21	MDCK1/SIAT1	<	<	<	<	<	<	<
				<=2-fold	>2<4-fold	>4<8-fold	>8-fold	not recognised by the antiserum		

<=2-fold >2<4-fold >4<8-fold >8-fold not recognised by the antiserum

Table MN-10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-08-22

				Neutralisation Titre							
				Post-infection ferret antisera							
				A/Massachusetts 18/2022 SIAT F36/23 2a.3a.1 (J)	A/Thailand 8/2022 Egg F34/23 2a.3a.1 (J)	A/Albania 289813/2022 SIAT F21/23 2a.3a.1 (J)	A/Sydney 856/2023 SIAT F01/24 2a.3a.1 (J.1)	A/Croatia 16136/RV/2023 SIAT F06/24 2a.3a.1 (J.2)	A/Netherlands 10563/2023 SIAT F08/24 2a.3a.1 (J.2)	A/Lisboa 216/2023 Egg F15/24 2a.3a.1 (J.2)	A/Slovenia 49/2024 SIAT F11/24 2a.3a.1 (J.2)
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History								
REFERENCE VIRUSES											
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ⁻³	168	270	279	155	150	443	75	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁻⁶	471	1489	1078	555	648	2274	450	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ⁻³	197	382	355	225	182	311	79	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ⁻³	198	310	337	216	181	340	77	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ⁻³	60	127	74	128	181	302	84	84
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ⁻³	186	250	317	179	240	492	110	110
A/Lisboa/216/2023	2a.3a.1 (J.2)	2023-12-15	E3/E1 10 ⁻³	218	416	240	550	243	742	846	217
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ⁻³	<	131	<	<	117	<	232	2594
TEST VIRUSES											
A/Serbia/2620/2024	2a.3a.1 (J.2)	2024-03-27	C1/SIAT1	140	243	264	139	152	341	296	85
A/Serbia/1354/2024	2a.3a.1 (J.2)	2024-02-15	C1/SIAT1	92	130	158	80	76	192	198	68
A/Serbia/1022/2024	2a.3a.1 (J.2)	2024-02-06	C1/SIAT1	137	240	295	154	135	199	217	69
A/Serbia/814/2024	2a.3a.1 (J.2)	2024-02-01	C1/SIAT1	158	249	277	125	156	411	312	105
A/Argentina/1219/2024	2a.3a.1 (J.2)	2024-07-05	SIAT1	151	238	242	119	178	302	240	82
A/Argentina/1279/2024	2a.3a.1 (J.2)	2024-06-28	SIAT1	255	464	488	242	265	476	436	213
A/Argentina/1256/2024	2a.3a.1 (J.2)	2024-06-24	SIAT1	208	443	395	210	212	390	390	270
A/Argentina/1231/2024	2a.3a.1 (J.2)	2024-06-24	SIAT1	122	221	199	110	131	304	251	76
A/Argentina/1187/2024	2a.3a.1 (J.2)	2024-06-24	SIAT1	56	80	92	58	80	169	140	<
A/Argentina/1244/2024	2a.3a.1 (J.2)	2024-06-21	SIAT1	196	480	422	253	248	453	569	234
				<=2-fold	>2<4-fold	>4<8-fold	>8-fold	not recognised by the antiserum			

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 >8-fold
 not recognised by the antiserum

Table MN-11. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-09-09

				Neutralisation Titre								
				Post-infection ferret antisera								
				A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	A/Albania 289813/2022 SIAT F21/23	A/Sydney 856/2023 SIAT F01/24	A/Croatia 10136/RV/2023 SIAT F06/24	A/Croatia 10136/RV/2023 Egg F16/24	A/Netherlands 10563/2023 SIAT F08/24	A/Lisboa 216/2023 Egg F15/24	A/Slovenia 49/2024 SIAT F11/24
Viruses	Genetic group	Collection Date	Passage history Ferret number Passage History	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
REFERENCE VIRUSES												
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ³	168	270	279	155	150	152	443	75	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁴	471	1489	1078	555	648	528	2274	450	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ³	197	382	355	225	182	118	311	79	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ³	198	310	337	216	181	125	340	77	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ³	60	127	74	128	181	160	302	84	84
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	E3 (Am1A/2) 10 ³	243	522	236	355	458	1165	905	912	752
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ³	186	250	317	179	240	139	492	110	110
A/Lisboa/216/2023	2a.3a.1 (J.2)	2023-12-15	E3/E1 10 ³	218	416	240	550	243	404	742	846	217
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ³	<	131	<	<	117	<	<	232	2594
TEST VIRUSES												
A/Iadoho/69/2023	2a.3a.1 (J.2)	2024-02-27	SIAT2	<	<	<	<	<	47	<	<	<
A/Albania/303794/2024	2a.3a.1 (J.2)	2024-03-14	SIAT1	58	75	59	69	101	149	293	139	65
A/Albania/302957/2024	2a.3a.1 (J.2)	2024-02-20	SIAT1	91	127	132	71	121	154	320	252	61
A/Albania/302946/2024	2a.3a.1 (J.2)	2024-02-20	SIAT1	64	107	70	101	133	209	302	154	75
A/Catalonia/NSVH102371258/2024	2a.3a.1 (J.2)	2024-06-25	SIAT2/SIAT1	78	131	125	72	125	146	320	246	51
A/Catalonia/NSJOS10768271/2024	2a.3a.1 (J.2)	2024-05-23	SIAT2/SIAT1	50	78	51	71	127	181	259	123	59
A/Catalonia/NSVH102343016/2024	2a.3a.1 (J.2)	2024-05-15	SIAT2/SIAT1	160	267	263	120	190	204	304	309	76
A/Catalonia/NSVH102337351/2024	2a.3a.1 (J)	2024-05-07	SIAT2/SIAT1	66	95	68	98	84	165	247	120	58

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 >4<8-fold
 >8-fold
 not recognised by the antiserum

Table MN-12. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-09-12

Viruses				Neutralisation Titre								
				Post-infection ferret antisera								
				A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	A/Albania 289813/2022 SIAT F21/23	A/Sydney 856/2023 SIAT F01/24	A/Croatia 10136/RV/2023 SIAT F06/24	A/Croatia 10136/RV/2023 Egg F16/24	A/Netherlands 10563/2023 SIAT F08/24	A/Lisboa 216/2023 Egg F15/24	A/Slovenia 49/2024 SIAT F11/24
				2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J)	2a.3a.1 (J.1)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)	2a.3a.1 (J.2)
Passage history				Passage History								
Ferret number				Ferret number								
Collection Date				Collection Date								
Genetic group				Genetic group								
REFERENCE VIRUSES												
A/Massachusetts/18/2022	2a.3a.1 (J)	2022-06-04	SIAT3/SIAT1 10 ³	168	270	279	155	150	152	443	75	75
A/Thailand/8/2022	2a.3a.1 (J)	-	E3/E1 10 ⁴	471	1489	1078	555	648	528	2274	450	450
A/Albania/289813/2022	2a.3a.1 (J)	2022-12-13	MDCK1/SIAT1 10 ³	197	382	355	225	182	118	311	79	79
A/Sydney/856/2023	2a.3a.1 (J.1)	2024-03-14	SIAT1/SIAT2 10 ³	198	310	337	216	181	125	340	77	77
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	SIAT3 10 ³	60	127	74	128	181	160	302	84	84
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	2024-03-28	E3 (Am1A)2 10 ³	243	522	236	355	458	1165	905	912	752
A/Netherlands/10563/2023	2a.3a.1 (J.2)	2024-03-28	MDCK-MIX2/SIAT3 10 ³	186	250	317	179	240	139	492	110	110
A/Lisboa/216/2023	2a.3a.1 (J.2)	2023-12-15	E3/E1 10 ³	218	416	240	550	243	404	742	846	217
A/Slovenia/49/2024	2a.3a.1 (J.2)	2024-05-20	Mx/SIAT3 10 ³	<	131	<	<	117	<	<	232	2594
TEST VIRUSES												
A/Argentina/1130/2024	2a.3a.1 (J.2)	2024-06-18	SIAT1	117	183	217	89	158	170	436	315	73
A/England/72/2024	2a.3a.1 (J.2)	2024-03-21	SIAT1/SIAT1	95	137	137	68	124	123	251	207	45
A/England/73/2024	2a.3a.1 (J.2)	2024-03-18	MDCK1/SIAT1	43	61	<	64	75	127	236	109	<
A/England/68/2024	2a.3a.1 (J.2)	2024-03-15	MDCK1/SIAT1	49	69	41	65	89	166	311	121	48
A/England/69/2024	2a.3a.1 (J.2)	2024-03-13	MDCK1/SIAT1	82	120	76	116	155	257	555	223	121
A/Lyon/CHUR24.236.48/2024	2a.3a.1 (J.2)	2024-07-29	MDCK2/SIAT2	187	309	320	157	222	232	582	442	85
A/Tarbes/NOMS24.4.9/2024	2a.3a.1 (J.2)	2024-07-27	MDCK2/SIAT2	154	257	267	135	187	213	364	320	70
A/Lyon/RELAB24.17.9/2024	2a.3a.1 (J.2)	2024-07-20	MDCK2/SIAT2	254	526	397	218	248	298	504	530	247
A/Lyon/RELAB/24.16.80/2024	2a.3a.1 (J.2)	2024-06-19	MDCK2/SIAT2	57	73	49	68	97	145	261	122	62

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>2<4-fold

>4<8-fold

>8-fold

not recognised by the antiserum

Fold-reduction table: A/H3N2

Reference Virus	Clade	<4-fold difference		4-fold difference		>4-fold difference		Total
		Number	Percentage	Number	Percentage	Number	Percentage	
A/Thuringen/10/2022 Cell	2b	10	30.3	22	66.7	1	3.0	33
A/Switzerland/28719/2022 Cell	2b	3	9.1	21	63.6	9	27.3	33
A/Stockholm/5/2021 Cell	2a	8	24.3	12	36.4	13	39.4	33
A/Darwin/9/2021 Egg	2a	24	72.7	9	27.3	0	0.0	33
A/Catalonia/NSVH161512067/2022 Cell	2a.1b	3	9.1	22	66.7	8	24.2	33
A/Albania/289813/2022 Cell	2a.3a.1 (J)	236	56.6	114	27.3	67	16.1	417
A/Massachusetts/18/2022 Cell	2a.3a.1 (J)	110	26.4	148	35.5	159	38.1	417
A/Thailand/08/2022 Egg	2a.3a.1 (J)	182	43.6	131	31.4	104	24.9	417
A/Sydney/856/2023 Cell	2a.3a.1 (J.1)	140	33.6	118	28.3	159	38.1	417
A/Croatia/10136/RV/2023 Cell	2a.3a.1 (J.2)	402	96.4	15	3.6	0	0.0	417
A/Croatia/10136/RV/2023 Egg	2a.3a.1 (J.2)	159	41.4	160	41.7	65	16.9	384
A/Netherlands/10563/2023 Cell	2a.3a.1 (J.2)	353	84.7	48	11.5	16	3.8	417
A/Lisboa/216/2023 Egg	2a.3a.1 (J.2)	303	78.9	70	18.2	11	2.9	384
A/Slovenia/49/2024 Cell	2a.3a.1 (J.2)	7	1.8	17	4.4	360	93.8	384
A/Vasteras/SE23-14213/2023 Cell	2a.3a.1 (J.2)	4	100.0	0	0.0	0	0.0	4
A/Switzerland/8649/2023 Egg	2a.3a.1 (J.2)	3	100.0	0	0.0	0	0.0	3
A/France/IDF-IPP29542/2023 Cell	2a.3a.1 (J.4)	234	99.2	1	0.4	1	0.4	236
A/Norway/12374/2023 Cell	2a.3a.1 (J.4)	186	78.8	43	18.2	7	3.0	236
A/BurkinaFaso/3131/2023 Cell	2a.3a.1 (J.4)	62	49.6	5	4.0	58	46.4	125

A/H3N2: Egg isolates

Virus	Status	Genetic group (subclade) [Additional subst.]	Comment
A/Brandenburg/15/2022	sent out recently	2a.3a.1 (J)	Egg adaptation: (D186G)
A/Albania/290243/2022	sent out recently	2a.3a.1 (J) [N122D(-CHO), S91N]	Egg adaptation: (F195Y)
A/Albania/290270/2022	NYMC X-411 (2-way pass) NYMC X-411B (1-way pass)	2a.3a.1 (J) [N122D(-CHO), S91N]	Egg adaptation: (N190T, S228G)
A/Albania/289813/2022	NIB-138 (2-way pass)	2a.3a.1 (J) [N122D(-CHO), S91N]	Egg adaptation: (F195Y, S228T)
A/Switzerland/8649/2023	SAN-032 (2-way pass) X-421 (2-way pass) IVR-253 (2-way pass)	2a.3a.1 (J.2)	Egg adaptation: (X92S, S228T)
A/Croatia/10136/RV/2023	NYMC X-425 (1-way pass, 2-way fail) NYMC X-425A (1-way pass, 2-way in progress) NIB-146 (1-way pass, 2-way in progress)	2a.3a.1 (J.2) [S145N]	Egg adaptation: (D186A)
A/Lisboa/216/2023	NYMC X-423 (2-way pass) NYMC X-423A (1-way pass) NIB-145 (1-way pass, 2-way in progress) IVR-255 (2-way pass)	2a.3a.1 (J.2) [S124N, S228T]	Egg adaptation: (S228T)

A/H3N2: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Albania/289813/2022	2a.3a.1 (J)	MDCK1	F21/23
A/Massachusetts/18/2022	2a.3a.1 (J)	SIAT3/SIAT1	F36/23
A/Thailand/08/2022	2a.3a.1 (J)	E3/E1	F34/23
A/Sydney/856/2023	2a.3a.1 (J.1)	SIAT1/SIAT2	F01/24
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	SIAT3	F06/24
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	E3 (Am1Al2)	F16/24
A/Netherlands/10563/2023	2a.3a.1 (J.2)	MDCK-MIX2/SIAT2	F08/24
A/Slovenia/49/2024	2a.3a.1 (J.2)	MDCKx/SIAT2	F11/24
A/Lisboa/216/2023	2a.3a.1 (J.2)	E3 (Am1Al2)	F15/24
A/Norway/12374/2023	2a.3a.1 (J.4)	SIAT4	F21/24
A/France/IDF-IPP29542/2023	2a.3a.1 (J.4)[K189R]	MDCK1/SIAT2	F22/24
A/BurkinaFaso/3131/2023	2a.3a.1 (J.4)[K189R]	SIAT3	F32/24

A/H3N2: MN reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Albania/289813/2022	2a.3a.1 (J)	MDCK1/SIAT1	F21/23
A/Massachusetts/18/2022	2a.3a.1 (J)	SIAT3/SIAT1	F36/23
A/Thailand/08/2022	2a.3a.1 (J)	E3/E1	F34/23 EGG
A/Sydney/856/2023	2a.3a.1 (J.1)	SIAT1/SIAT2	F01/24
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	SIAT3	F06/24
A/Croatia/10136/RV/2023	2a.3a.1 (J.2)	E3 (Am1Al2)	F16/24
A/Netherlands/10563/2023	2a.3a.1 (J.2)	MDCK-MIX2/SIAT3	F08/24
A/Slovenia/49/2024	2a.3a.1 (J.2)	Mx/SIAT3	F11/24
A/Lisboa/216/2023	2a.3a.1 (J.2)	E3 (Am1Al2)	F15/24

Summary: A/H3N2

Genetic analyses

Clade 3C.2a1b.2a.2 (renamed as **2**) predominated since February 2023 in all geographic regions where A/H3N2 circulated.

During this reporting period, the great majority of H3 viruses detected belong to clade 2a.3a.1, which share substitution E50K with clade 2a.3a and present additional substitutions I140K and I223V (subclade J, vaccine reference A/Thailand/8/2022). Subclade J split into 4 further subclades: of these, subclade J.2 (reference A/Sydney/878/2023) characterised by N122D (-CHO) and K276E became the dominant subclade, predominating in the majority of continents with >90% frequency. Subclade J.1 (reference A/Sydney/856/2023) characterised by I25V, V347M with additional I418V (in some viruses) was detected in minor frequencies (<10%) in Europe, South-East Asia, Africa and Oceania. Minor subclade J.4 characterised by Q173R, K276E and some viruses with K189R predominated in West Africa earlier during the season.

Clade 2a.3a (subclade G.1.3.1, reference A/Finland/402/2023) with substitutions K276E and V347M was detected in West Africa earlier during the season.

Antigenic analyses

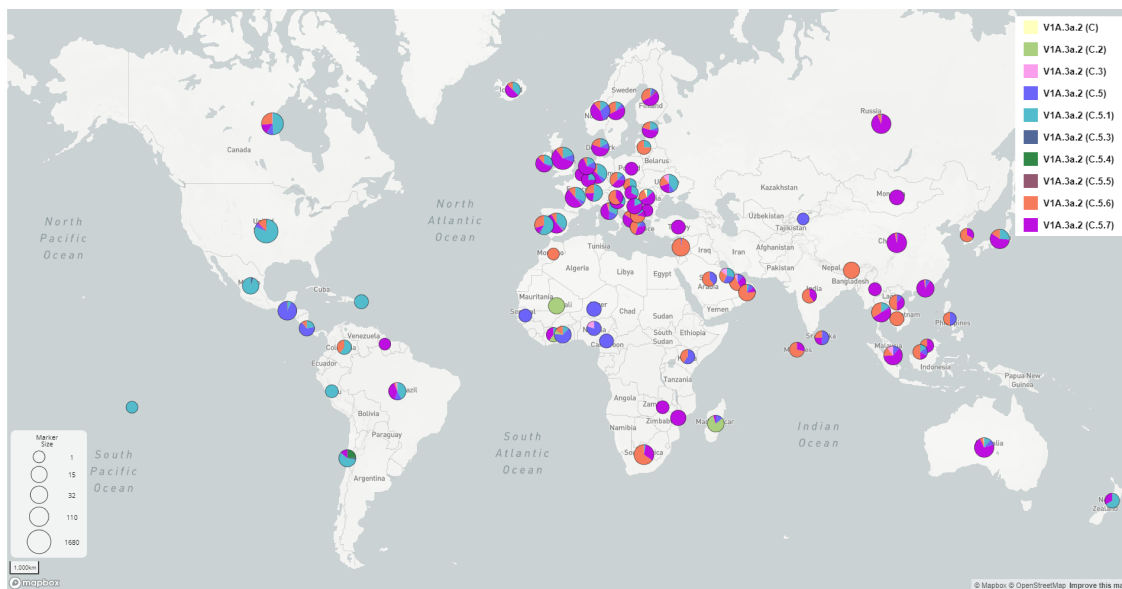
HI titres show that the SH 2024 and NH 2024-25 vaccine strains, egg-based A/Thailand/08/2022 (2a.3a.1 J9J)) and cell-based A/Massachusetts/18/2022 (2a.3a.1 (J)), demonstrate reduced recognition against a significant number of samples in the J.2 and J.4 subclades. MN titres show that the SH 2024 and NH 2024-25 vaccine strain, egg-based A/Thailand/08/2022 (2a.3a.1 (J)) shows reduced reactivity to the vast majority of viruses tested. The cell-based SH 2024 and NH 2024-25 vaccine strain, A/Massachusetts/18/2022 (2a.3a.1 (J)) recognised most viruses tested within 2-fold, but showed significant drop in titre to some J.2 and all J.4 viruses tested to date.

Influenza B

Genetic analyses: B/Victoria

Global distribution of B/Victoria subclades

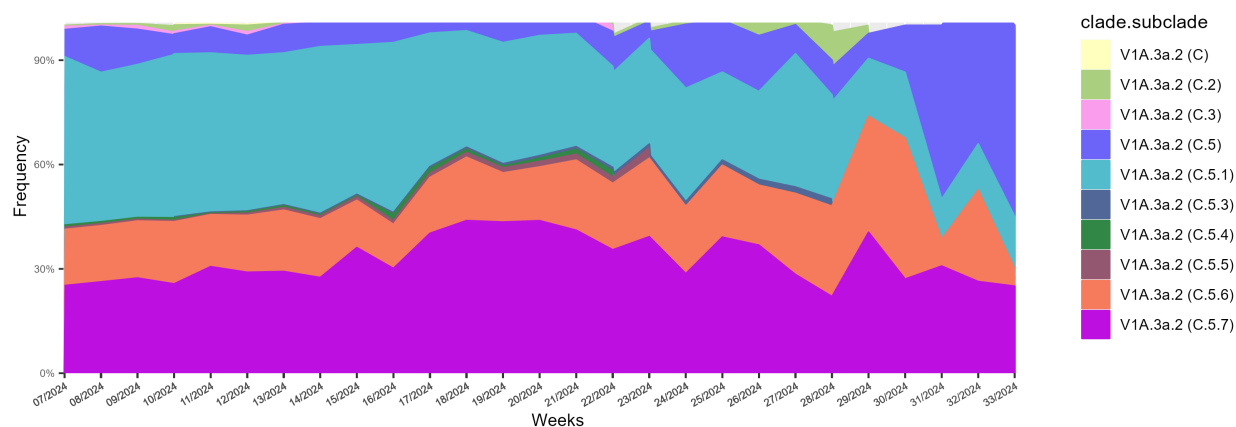
Global geographical distribution of influenza B/Victoria genetic clades (subclades) viruses with collection dates from 1st February 2024 to 31st August 2024, obtained with full HA sequences as deposited in GISAID, downloaded on 12th September 2024 and classified using [Nextclade](#). Map prepared with [Microreact](#). Geographic markers scaled to detection proportions.



Global time-dependent variation of B/Victoria subclades

Global time-dependent variation in frequencies of genetic clades-subclades of B/Victoria viruses collected from 1st February 2024 to 31st August 2024.

Clade-Subclade frequencies over time



Maximum likelihood phylogenetic tree: B/Victoria

Maximum likelihood phylogenetic tree inferred using IQTree2 with HA sequence data obtained from GISAID, collection dates from 1st February 2024 to 31st August 2024, manually curated and then downsampled using Treemmer to retain a representative tree topology of 200 sequences, with emphasis on sequences generated at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

Antigenic analyses: B/Victoria

Haemagglutination inhibition tables: B/Victoria

Table BV-01. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-04-23

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
					Post-infection ferret antiserum					
					B/Bris 60/08	B/Austria 1359417/21	B/Austria 1359417/21	B/Austria 1359417/21	B/Stockholm/3/2022	
					Passage	Egg MDCK	Egg G141	Egg G141R	MDCK	
					Ferret number	Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	
REFERENCE VIRUSES										
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	160	640	320	160	320	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	160	640	320	320	320	
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	80	640	160	1280	160	
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	160	320	160	80	320	
TEST VIRUSES										
242504	B/Bangui/399/2023	V1A.3a.2 (C.2)	2023-10-13	MDCK1	160	320	160	160	320	
242740	B/Schleswig-Holstein/2/2024	V1A.3a.2 (C.5)	2024-02-28	P1/MDCK1	160	320	160	80	320	
242742	B/Brandenburg/1/2024	V1A.3a.2 (C.5.1)	2024-02-15	P1/MDCK1	160	320	160	160	320	
242562	B/Morocco/1847/2024	V1A.3a.2 (C.5.6)	2024-02-16	MDCK1	160	320	160	80	160	
242743	B/Hessen/1/2024	V1A.3a.2 (C.5.7)	2024-01-30	P1/MDCK1	160	320	160	80	320	
242741	B/Bayern/8/2024	V1A.3a.2 (C.5.7)	2024-02-19	P1/MDCK1	160	320	160	80	320	
242739	B/Rheinland-Pfalz/2/2024	V1A.3a.2 (C.5.7)	2024-03-01	P1/MDCK1	160	320	160	160	320	
242737	B/Mecklenburg-Vorpommern/3/2024	V1A.3a.2 (C.5.7)	2024-03-04	P1/MDCK1	160	320	160	160	320	
242738	B/Thuringen/5/2024	V1A.3a.2 (C.5.7)	2024-03-04	P1/MDCK1	160	320	160	160	160	
< relates to the lowest dilution of antiserum used						Vaccine		Vaccine		
1 hyperimmune sheep serum; ND = Not Done						SH 2022		SH 2022		
						NH 2022-23		NH 2022-23		
						SH 2023		SH 2023		
						NH 2023-24		NH 2023-24		
						SH 2024		SH 2024		
						NH 2024-25		NH 2024-25		

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table BV-02. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-06-04

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
					Post-infection ferret antiserum					
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
					Passage	Ferret number				
					Sh 539, 540, 543, 544, 570, 571, 5741	NIB F91/21	F40/21	F44/21	F28/22	
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.9)	
REFERENCE VIRUSES										
B/Brisbane/002/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	320	160	320	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	640	640	320	640	
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640	
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.9)	2022-03-22	SIAT1/MDCK3	320	640	320	160	640	
TEST VIRUSES										
242049	B/Antananarivo/02570/2023	V1A.3a.2 (C.2)	2023-10-10	MDCK1	640	640	320	320	640	
242047	B/Antananarivo/02577/2023	V1A.3a.2 (C.2)	2023-10-12	MDCK1	640	640	640	320	640	
242046	B/Mahajanga/02635/2023	V1A.3a.2 (C.2)	2023-10-16	MDCK1	160	640	160	160	320	
242045	B/Antsirabe/02649/2023	V1A.3a.2 (C.2)	2023-10-18	MDCK1	320	640	320	160	640	
242042	B/Antsirabe/02693/2023	V1A.3a.2 (C.2)	2023-10-26	MDCK1	320	640	320	160	640	
242039	B/Antananarivo/03264/2023	V1A.3a.2 (C.2)	2023-12-11	MDCK1	320	640	320	160	640	
242473	B/SouthSudan/981/2023	V1A.3a.2 (C.5)	2023-09-15	MDCK1	640	640	320	320	640	
242050	B/Antananarivo/02503/2023	V1A.3a.2 (C.5)	2023-09-30	MDCK1	640	640	320	320	640	
242048	B/Antananarivo/02575/2023	V1A.3a.2 (C.5)	2023-10-11	MDCK1	320	640	320	320	640	
242466	B/SouthSudan/985/2023	V1A.3a.2 (C.5)	2023-10-11	MDCK2	640	1280	320	320	1280	
242464	B/SouthSudan/984/2023	V1A.3a.2 (C.5)	2023-10-19	MDCK2	640	640	320	320	640	
242463	B/SouthSudan/982/2023	V1A.3a.2 (C.5)	2023-10-27	MDCK2	320	640	320	160	640	
242458	B/SouthSudan/989/2023	V1A.3a.2 (C.5)	2023-11-29	MDCK2	640	640	320	320	640	
242456	B/SouthSudan/1002/2023	V1A.3a.2 (C.5)	2023-12-11	MDCK1	640	640	320	320	640	
242455	B/SouthSudan/1107/2024	V1A.3a.2 (C.5)	2024-01-05	MDCK1	320	640	320	160	640	
243039	B/Ghana/1128/2024	V1A.3a.2 (C.5)	2024-02-27	MDCK1	640	640	320	320	640	
243038	B/Ghana/1596/2024	V1A.3a.2 (C.5)	2024-03-19	MDCK1	320	640	320	320	640	
243037	B/Ghana/490/2024	V1A.3a.2 (C.5)	2024-03-25	MDCK1	640	640	640	320	640	
243036	B/Ghana/1835/2024	V1A.3a.2 (C.5)	2024-04-02	MDCK1	640	640	320	320	640	
243040	B/Ghana/924/2024	V1A.3a.2 (C.5.6)	2024-02-11	MDCK1	160	320	80	80	160	
242933	B/Albania/303686/2024	V1A.3a.2 (C.5.6)	2024-03-11	MDCK1	640	640	320	320	320	
242929	B/Albania/304012/2024	V1A.3a.2 (C.5.6)	2024-03-20	MDCK1	320	640	320	160	640	
242927	B/Albania/304252/2024	V1A.3a.2 (C.5.6)	2024-04-01	MDCK1	320	640	320	160	640	
243150	B/Ashabat/1/2024	V1A.3a.2 (C.5.7)	2024-01-23	MDCK1/MDCK1	320	640	320	160	640	
242937	B/Albania/303247/2024	V1A.3a.2 (C.5.7)	2024-02-26	MDCK1	640	640	320	320	640	
242936	B/Albania/303289/2024	V1A.3a.2 (C.5.7)	2024-02-28	MDCK1	320	640	320	320	640	
242935	B/Albania/303292/2024	V1A.3a.2 (C.5.7)	2024-02-29	MDCK1	320	640	320	160	320	
242851	B/Bulgaria/2406/2024	V1A.3a.2 (C.5.7)	2024-03-01	MDCK2/MDCK1	320	640	320	160	640	
242934	B/Albania/303490/2024	V1A.3a.2 (C.5.7)	2024-03-04	MDCK1	320	640	320	160	640	
242932	B/Albania/303833/2024	V1A.3a.2 (C.5.7)	2024-03-15	MDCK1	320	640	320	160	640	
242931	B/Albania/303874/2024	V1A.3a.2 (C.5.7)	2024-03-17	MDCK1	320	640	320	160	640	
242930	B/Albania/303949/2024	V1A.3a.2 (C.5.7)	2024-03-20	MDCK1	320	640	160	160	320	
242928	B/Albania/304090/2024	V1A.3a.2 (C.5.7)	2024-03-26	MDCK1	320	320	160	160	320	
242926	B/Albania/304274/2024	V1A.3a.2 (C.5.7)	2024-04-03	MDCK1	320	640	160	160	640	
< relates to the lowest dilution of antiserum used * hyperimmune sheep serum; ND = Not Done					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
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					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
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					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
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					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25					
					Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 20					

Table BV-03. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-06-12

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK
					Passage	Ferret number			
					Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22

Table BV-04. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-06-18

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK
					Passage	Ferret number			
					Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/60/2009		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	320	160	320
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	320	2560	640
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	640
TEST VIRUSES									
243389	B/Kyrgyzstan/20/2024	V1A.3a.2 (C.5)	2024-04-25	MDCK1/MDCK1	320	640	320	160	640
243234	B/Iceland/11923/2024	V1A.3a.2 (C.5.1)	2024-04-16	MDCK1	640	640	320	160	640
243230	B/Iceland/13541/2024	V1A.3a.2 (C.5.1)	2024-04-30	MDCK1	640	1280	320	320	640
243105	B/Lebanon/218/2024	V1A.3a.2 (C.5.6)	2024-02-15	MDCK1	640	640	320	160	640
243104	B/Lebanon/219/2024	V1A.3a.2 (C.5.6)	2024-02-15	MDCK1	640	640	320	160	640
243092	B/Lebanon/623/2024	V1A.3a.2 (C.5.6)	2024-02-21	MDCK1	320	640	320	320	640
243091	B/Lebanon/624/2024	V1A.3a.2 (C.5.6)	2024-02-21	MDCK1	640	640	320	320	640
243084	B/Lebanon/676/2024	V1A.3a.2 (C.5.6)	2024-02-21	MDCK1	320	640	320	320	640
243080	B/Lebanon/678/2024	V1A.3a.2 (C.5.6)	2024-02-26	MDCK1	320	640	320	160	320
243079	B/Lebanon/682/2024	V1A.3a.2 (C.5.6)	2024-02-26	MDCK1	640	640	320	320	640
243236	B/Iceland/10638/2024	V1A.3a.2 (C.5.6)	2024-04-04	MDCK1	320	320	160	160	320
243302	B/Mozambique/694480/2024	V1A.3a.2 (C.5.7)	2024-02-05	MDCK1	640	640	320	320	640
243301	B/Mozambique/694518/2024	V1A.3a.2 (C.5.7)	2024-02-06	MDCK1	640	1280	320	320	1280
243300	B/Mozambique/694529/2024	V1A.3a.2 (C.5.7)	2024-02-09	MDCK1	320	640	320	320	640
243299	B/Mozambique/694532/2024	V1A.3a.2 (C.5.7)	2024-02-09	MDCK1	320	640	320	320	1280
243298	B/Mozambique/694472/2024	V1A.3a.2 (C.5.7)	2024-02-12	MDCK1	640	640	320	320	640
243297	B/Mozambique/694480/2024	V1A.3a.2 (C.5.7)	2024-02-12	MDCK1	640	640	320	320	1280
243296	B/Mozambique/694484/2024	V1A.3a.2 (C.5.7)	2024-02-12	MDCK1	320	1280	320	320	640
243295	B/Mozambique/695872/2024	V1A.3a.2 (C.5.7)	2024-02-14	MDCK1	320	640	320	160	320
243238	B/Iceland/5444/2024	V1A.3a.2 (C.5.7)	2024-02-16	MDCK1	640	640	320	160	640
243293	B/Mozambique/695860/2024	V1A.3a.2 (C.5.7)	2024-02-27	MDCK1	320	640	320	320	640
243237	B/Iceland/7524/2024	V1A.3a.2 (C.5.7)	2024-03-06	MDCK1	640	640	320	320	640
243283	B/Turkey/27248654/2024	V1A.3a.2 (C.5.7)	2024-04-18	MDCK1	320	640	320	320	640
243281	B/Turkey/27255839/2024	V1A.3a.2 (C.5.7)	2024-04-19	MDCK1	320	640	320	320	640
243233	B/Iceland/12512/2024	V1A.3a.2 (C.5.7)	2024-04-20	MDCK1	640	640	320	320	640
243280	B/Turkey/27253933/2024	V1A.3a.2 (C.5.7)	2024-04-24	MDCK1	320	640	320	320	640

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

	Vaccine	Vaccine
	SH 2022	SH 2022
	NH 2022-23	NH 2022-23
	SH 2023	SH 2023
	NH 2023-24	NH 2023-24
	SH 2024	SH 2024
	NH 2024-25	NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table BV-05. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-06-25

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg	B/Austria 1359417/21 Egg	B/Stock 3/22 MDCK
					Passage	Ferret number			
						SH 539, 540, 543, 544, 570, 571, 5741	NIB F91/21	F40/21	F44/21
							V1A	V1A.3a.2 (C)	V1A.3a.2 (C)
							V1A	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/002/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	160	640	320	160	320
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	320	2560	640
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	640
TEST VIRUSES									
243526 B/Ukraine/2733/2024		V1A.3a.2 (C.3)	2024-03-12	MDCK2/MDCK1	320	1280	640	320	640
243521 B/Ukraine/2731/2024		V1A.3a.2 (C.3)	2024-03-18	MDCK2/MDCK1	320	1280	320	320	320
243460 B/Dakar1/2024		V1A.3a.2 (C.5)	2024-03-12	MDCK1/MDCK1	320	640	320	320	640
243640 B/Kyiv/2332/2024		V1A.3a.2 (C.5)	2024-03-15	MDCK1/MDCK1	320	640	320	160	640
243459 B/Dakar2/2024		V1A.3a.2 (C.5)	2024-04-15	MDCK1/MDCK1	320	640	320	160	640
243358 B/Ireland/00016045/2024		V1A.3a.2 (C.5.1)	2024-02-07	MDCK1	320	640	320	160	320
243644 B/Kyiv/1860/2024		V1A.3a.2 (C.5.1)	2024-02-27	MDCK2/MDCK1	640	1280	320	320	1280
243533 B/Ukraine/2634/2024		V1A.3a.2 (C.5.1)	2024-03-04	MDCK1/MDCK1	320	640	320	160	640
243643 B/Kyiv/2384/2024		V1A.3a.2 (C.5.1)	2024-03-06	MDCK2/MDCK1	640	640	320	160	640
243527 B/Ukraine/2780/2024		V1A.3a.2 (C.5.1)	2024-03-11	MDCK1/MDCK1	640	1280	320	320	1280
243525 B/Ukraine/2779/2024		V1A.3a.2 (C.5.1)	2024-03-12	MDCK2/MDCK1	320	640	320	320	640
243641 B/Kyiv/2389/2024		V1A.3a.2 (C.5.1)	2024-03-14	MDCK2/MDCK1	320	640	320	320	640
243523 B/Ukraine/2778/2024		V1A.3a.2 (C.5.1)	2024-03-14	MDCK2/MDCK1	320	640	320	320	640
243522 B/Ukraine/2772/2024		V1A.3a.2 (C.5.1)	2024-03-17	MDCK1/MDCK1	320	640	320	160	640
243520 B/Ukraine/2771/2024		V1A.3a.2 (C.5.1)	2024-03-18	MDCK2/MDCK1	320	640	320	320	640
243518 B/Ukraine/2775/2024		V1A.3a.2 (C.5.1)	2024-03-18	MDCK1/MDCK1	320	640	320	160	640
243517 B/Ukraine/2773/2024		V1A.3a.2 (C.5.1)	2024-03-19	MDCK1/MDCK1	320	640	320	160	640
243515 B/Ukraine/2642/2024		V1A.3a.2 (C.5.1)	2024-03-20	MDCK3/MDCK1	320	640	320	160	640
243514 B/Ukraine/2767/2024		V1A.3a.2 (C.5.1)	2024-03-23	MDCK2/MDCK1	640	1280	320	320	640
243354 B/Ireland/00016364/2024		V1A.3a.2 (C.5.6)	2024-02-21	MDCK2	320	640	320	160	640
243634 B/Ukraine/2633/2024		V1A.3a.2 (C.5.6)	2024-03-04	MDCK1/MDCK1	320	640	320	160	640
243529 B/Ukraine/2734/2024		V1A.3a.2 (C.5.6)	2024-03-11	MDCK2/MDCK1	320	640	320	160	640
243528 B/Ukraine/2735/2024		V1A.3a.2 (C.5.6)	2024-03-11	MDCK2/MDCK1	320	640	320	320	640
243639 B/Kyiv/2336/2024		V1A.3a.2 (C.5.6)	2024-03-15	MDCK1/MDCK1	320	640	160	160	640
243519 B/Ukraine/2774/2024		V1A.3a.2 (C.5.6)	2024-03-18	MDCK1/MDCK1	320	640	320	320	640
243516 B/Ukraine/2776/2024		V1A.3a.2 (C.5.6)	2024-03-19	MDCK1/MDCK1	320	640	320	160	320
243356 B/Ireland/00015975/2024		V1A.3a.2 (C.5.7)	2024-02-13	MDCK1	320	640	320	160	640
243294 B/Mozambique/090907/2024		V1A.3a.2 (C.5.7)	2024-02-15	MDCK2	640	1280	640	640	1280
243645 B/Kyiv/1888/2024		V1A.3a.2 (C.5.7)	2024-02-26	MDCK1/MDCK1	320	640	320	160	640
243642 B/Kyiv/2227/2024		V1A.3a.2 (C.5.7)	2024-03-10	MDCK1/MDCK1	320	640	320	160	640
243352 B/Ireland/00025445/2024		V1A.3a.2 (C.5.7)	2024-03-25	MDCK2	320	1280	640	320	1280
243351 B/Ireland/00028892/2024		V1A.3a.2 (C.5.7)	2024-04-04	MDCK1	640	1280	320	320	1280
243357 B/Ireland/00028716/2024		V1A.3a.2 (C.5.7)	2024-04-06	MDCK1	320	640	320	320	640
243350 B/Ireland/00029580/2024		V1A.3a.2 (C.5.7)	2024-04-09	MDCK2	320	640	320	320	640
243282 B/Turkey/27250026/2024		V1A.3a.2 (C.5.7)	2024-04-18	MDCK2	640	1280	640	320	1280
243349 B/Ireland/00033276/2024		V1A.3a.2 (C.5.7)	2024-04-24	MDCK1	640	1280	320	320	1280
< relates to the lowest dilution of antiserum used					Vaccine SH 2022				
¹ hyperimmune sheep serum; ND = Not Done					NH 2022-23				
					SH 2023				
					NH 2023-24				
					SH 2024				
					NH 2024-25				
					Vaccine SH 2022				
					NH 2022-23				
					SH 2023				
					NH 2023-24				
					SH 2024				
					NH 2024-25				
					≥ 160 (no homologous titre)				
					< 4-fold				
					4-fold				
					8-fold				
					> 8-fold				
					< not recognised by the antiserum				

Table BV-06. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-07-02

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
					Post-infection ferret antiserum					
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stockholm 3/22 MDCK	
					Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22	
					VIA	VIA.3a.2 (C)	VIA.3a.2 (C)	VIA.3a.2 (C)	VIA.3a.2 (C.5)	
REFERENCE VIRUSES										
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	640	320	320	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	640	640	
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640	
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	640	1280	640	320	1280	
TEST VIRUSES										
243699	B/Uppsala/SE24-03429/2024	V1A.3a.2 (C.5.1)	2024-03-13	SIAT1/MDCK1	640	1280	640	320	1280	
243689	B/Linköping/SE24-04374/2024	V1A.3a.2 (C.5.1)	2024-04-04	SIAT1/MDCK1	1280	1280	640	320	1280	
243510	B/Ukraine/2801/2024	V1A.3a.2 (C.5.1)	2024-04-15	MDCK1	640	1280	640	640	1280	
243508	B/Ukraine/2808/2024	V1A.3a.2 (C.5.1)	2024-04-23	MDCK1	640	1280	640	320	1280	
243705	B/Halmstad/SE24-02326/2024	V1A.3a.2 (C.5.6)	2024-02-15	SIAT1/MDCK1	640	1280	640	320	1280	
243496	B/An_Nahdah/5246969/2024	V1A.3a.2 (C.5.6)	2024-03-13	MDCK1	640	640	320	320	640	
243495	B/Qurayyat/7242056/2024	V1A.3a.2 (C.5.6)	2024-03-31	MDCK1	320	640	320	160	640	
243494	B/Qurayyat/5248687/2024	V1A.3a.2 (C.5.6)	2024-04-02	MDCK1	640	1280	640	320	1280	
243687	B/Trollhattan/SE24-04473/2024	V1A.3a.2 (C.5.6)	2024-04-09	SIAT1/MDCK1	640	1280	640	640	1280	
243493	B/Qurayyat/7242476/2024	V1A.3a.2 (C.5.6)	2024-04-21	MDCK1	640	1280	640	320	1280	
243492	B/Qurayyat/7242477/2024	V1A.3a.2 (C.5.6)	2024-04-21	MDCK1	640	1280	640	320	1280	
243676	B/Slovenia/940/2024	V1A.3a.2 (C.5.6)	2024-04-25	MDCKF1/MDCK1	320	640	320	320	640	
243672	B/Slovenia/1043/2024	V1A.3a.2 (C.5.6)	2024-05-13	MDCKF1/MDCK1	640	640	320	320	640	
243670	B/Slovenia/1197/2024	V1A.3a.2 (C.5.6)	2024-06-11	MDCKF1/MDCK1	1280	1280	640	640	1280	
243703	B/Linköping/SE24-03151/2024	V1A.3a.2 (C.5.7)	2024-03-05	SIAT1/MDCK1	640	1280	640	640	1280	
243532	B/Ukraine/2703/2024	V1A.3a.2 (C.5.7)	2024-03-11	MDCK1	1280	1280	640	640	1280	
243701	B/Vasteras/SE24-03684/2024	V1A.3a.2 (C.5.7)	2024-03-11	SIAT1/MDCK1	1280	1280	640	640	1280	
243697	B/Vaxjo/SE24-03874/2024	V1A.3a.2 (C.5.7)	2024-03-14	SIAT1/MDCK1	1280	1280	640	640	1280	
243695	B/Stockholm/SE24-52813/2024	V1A.3a.2 (C.5.7)	2024-03-18	SIAT1/MDCK1	640	1280	640	640	1280	
243678	B/Slovenia/817/2024	V1A.3a.2 (C.5.7)	2024-03-27	MDCKF1/MDCK1	640	1280	640	640	1280	
243693	B/Umea/SE24-04476/2024	V1A.3a.2 (C.5.7)	2024-04-01	SIAT1/MDCK1	640	1280	640	320	1280	
243691	B/Stockholm/SE24-52888/2024	V1A.3a.2 (C.5.7)	2024-04-02	SIAT1/MDCK1	320	640	320	320	640	
243512	B/Ukraine/2810/2024	V1A.3a.2 (C.5.7)	2024-04-09	MDCK1	1280	1280	640	640	1280	
243685	B/Stockholm/SE24-52996/2024	V1A.3a.2 (C.5.7)	2024-04-15	SIAT1/MDCK1	640	1280	640	640	1280	
243491	B/AI_Bidayah/7242565/2024	V1A.3a.2 (C.5.7)	2024-04-21	MDCK1	640	1280	640	320	1280	
243683	B/Stockholm/SE24-53079/2024	V1A.3a.2 (C.5.7)	2024-04-23	SIAT1/MDCK1	640	1280	640	640	1280	
243681	B/Stockholm/SE24-53110/2024	V1A.3a.2 (C.5.7)	2024-04-27	SIAT1/MDCK1	640	1280	640	640	1280	
243674	B/Slovenia/1024/2024	V1A.3a.2 (C.5.7)	2024-05-09	MDCKF1/MDCK1	640	1280	1280	640	1280	
< relates to the lowest dilution of antiserum used						Vaccine SH 2022		Vaccine SH 2022		
1 hyperimmune sheep serum; ND = Not Done						NH 2022-23		NH 2022-23		
						SH 2023		SH 2023		
						NH 2023-24		NH 2023-24		
						SH 2024		SH 2024		
						NH 2024-25		NH 2024-25		
						≥ 160 (no homologous titre)				

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (no homologous titre)

Table BV-07. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-07-10

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stockholm 3/22 MDCK
					Passage				
Ferret number	Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22				
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/60/2008		V1A	2008-08-04	E4/E3	2560	40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	640	1280	640	640	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	1280
TEST VIRUSES									
243756	B/Parma/14/2024	V1A.3a.2 (C.5)	2024-02-21	MDCK2/MDCK1	640	640	320	320	1280
243761	B/Milano/13/2024	V1A.3a.2 (C.5.1)	2024-02-06	MDCK3/MDCK1	640	640	640	640	1280
243758	B/Parma/17/2024	V1A.3a.2 (C.5.1)	2024-02-19	MDCK2/MDCK1	640	640	320	320	1280
243760	B/Parma/11/2024	V1A.3a.2 (C.5.7)	2024-02-12	MDCK2/MDCK1	1280	1280	640	640	1280
243759	B/Parma/13/2024	V1A.3a.2 (C.5.7)	2024-02-19	MDCK2/MDCK1	640	640	320	320	640
243757	B/Perugia/1/2024	V1A.3a.2 (C.5.7)	2024-02-19	MDCK3/MDCK1	640	1280	640	320	1280
243755	B/Perugia/3/2024	V1A.3a.2 (C.5.7)	2024-02-26	MDCK3/MDCK1	640	640	640	320	1280
243753	B/Milano/63/2024	V1A.3a.2 (C.5.7)	2024-04-11	MDCK3/MDCK1	640	640	320	320	640
243752	B/Milano/64/2024	V1A.3a.2 (C.5.7)	2024-04-26	MDCK3/MDCK1	640	640	640	320	1280
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done						Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25			
< 4-fold						Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25			
4-fold									
8-fold									
> 8-fold									
< not recognised by the antiserum									
≥ 160 (no homologous titre)									

Table BV-08. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-07-24

Viruses	Other information						Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 5741	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
							Ferret number	NIB F01/21	F40/21	F44/21	F28/22	
							V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)	
REFERENCE VIRUSES												
B/Brisbane/60/2008			V1A	2008-08-04	E4/E3	1280	40	<40	<40	<40		
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	640	320	320	320	640		
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	640	320	320	320	640		
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	320	2560	640	640		
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	320	640		
TEST VIRUSES												
243958	B/SouthAfrica/R01414/24	V1A.3a.2 (C.5)	2024-02-28	MDCK1/MDCK1	640	640	320	320	640	640		
243786	B/Bratislava/124/2024	V1A.3a.2 (C.5.1)	2024-02-13	MDCKx/MDCK1	640	640	640	320	1280	1280		
242942	B/Albania/302953/2024	V1A.3a.2 (C.5.1)	2024-02-20	MDCK1	320	640	320	320	640	640		
243838	B/Switzerland/68373/2024	V1A.3a.2 (C.5.1)	2024-02-23	MDCK1	640	640	320	320	640	640		
243837	B/Switzerland/35585/2024	V1A.3a.2 (C.5.1)	2024-03-01	MDCK1	640	640	640	320	1280	1280		
243905	B/Serbia/2864/2024	V1A.3a.2 (C.5.1)	2024-04-09	C1/MDCK1	640	640	640	320	640	640		
243833	B/Switzerland/73447/2024	V1A.3a.2 (C.5.1)	2024-05-13	MDCK1/MDCK1	640	640	320	320	1280	1280		
242951	B/Albania/302682/2024	V1A.3a.2 (C.5.6)	2024-02-14	MDCK1	640	640	320	320	640	640		
242941	B/Albania/302974/2024	V1A.3a.2 (C.5.6)	2024-02-21	MDCK1	320	640	320	320	640	640		
243962	B/SouthAfrica/R01271/24	V1A.3a.2 (C.5.6)	2024-02-21	MDCK1/MDCK1	640	640	640	640	1280	1280		
243954	B/SouthAfrica/R01960/24	V1A.3a.2 (C.5.6)	2024-03-13	MDCK1/MDCK1	640	640	640	320	1280	1280		
243835	B/Switzerland/93020/2024	V1A.3a.2 (C.5.6)	2024-04-23	MDCK2/MDCK1	640	640	320	320	640	640		
243944	B/SouthAfrica/R04772/24	V1A.3a.2 (C.5.6)	2024-05-31	MDCK1/MDCK1	640	640	640	320	1280	1280		
243911	B/Serbia/349/2024	V1A.3a.2 (C.5.7)	2024-01-18	C1/MDCK1	640	640	640	320	640	640		
242962	B/Albania/301481/2024	V1A.3a.2 (C.5.7)	2024-01-30	MDCK1	640	640	320	320	640	640		
242957	B/Albania/302354/2024	V1A.3a.2 (C.5.7)	2024-02-09	MDCK1	320	640	320	320	640	640		
242952	B/Albania/302538/2024	V1A.3a.2 (C.5.7)	2024-02-12	MDCK1	320	640	320	320	640	640		
243909	B/Serbia/1420/2024	V1A.3a.2 (C.5.7)	2024-02-20	C1/MDCK1	320	640	320	320	640	640		
243964	B/SouthAfrica/R01183/24	V1A.3a.2 (C.5.7)	2024-02-20	MDCK1/MDCK1	320	640	320	320	640	640		
242938	B/Albania/303193/2024	V1A.3a.2 (C.5.7)	2024-02-26	MDCK1	320	640	320	320	640	640		
243960	B/SouthAfrica/R01627/24	V1A.3a.2 (C.5.7)	2024-02-27	MDCK1/MDCK1	320	640	320	320	640	640		
243907	B/Serbia/2011/2024	V1A.3a.2 (C.5.7)	2024-03-04	C1/MDCK1	640	640	640	320	1280	1280		
243956	B/SouthAfrica/R01800/24	V1A.3a.2 (C.5.7)	2024-03-05	MDCK1/MDCK1	640	640	320	320	1280	1280		
243836	B/Switzerland/29051/2024	V1A.3a.2 (C.5.7)	2024-04-15	MDCK2/MDCK1	640	640	320	320	640	640		
243952	B/SouthAfrica/R03139/24	V1A.3a.2 (C.5.7)	2024-04-18	MDCK1/MDCK1	640	640	320	320	1280	1280		
243950	B/SouthAfrica/R03083/24	V1A.3a.2 (C.5.7)	2024-04-22	MDCK1/MDCK1	640	1280	640	320	1280	1280		
243834	B/Switzerland/45049/2024	V1A.3a.2 (C.5.7)	2024-04-26	MDCK1	640	640	640	320	1280	1280		
243948	B/SouthAfrica/R03535/24	V1A.3a.2 (C.5.7)	2024-05-07	MDCK1/MDCK1	640	640	640	320	1280	1280		
243903	B/Serbia/3517/2024	V1A.3a.2 (C.5.7)	2024-05-12	C1/MDCK1	640	1280	640	320	1280	1280		
243946	B/SouthAfrica/R04533/24	V1A.3a.2 (C.5.7)	2024-05-24	MDCK1/MDCK1	640	640	640	640	1280	1280		

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24
SH 2024
NH 2024-25

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24
SH 2024
NH 2024-25

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table BV-09. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-07-30

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK
					Passage				
Ferret number	Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22				
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/60/2008		V1A	2008-08-04	E4/E3	2560	40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	640	320	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	640	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	320	2560	640
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	1280
TEST VIRUSES									
244021	B/Ghana/2476/2024	V1A.3a.2 (C.5)	2024-04-29	MDCK1	640	640	640	320	1280
244017	B/Ghana/3095/2024	V1A.3a.2 (C.5)	2024-06-05	MDCK1	640	1280	640	640	1280
244018	B/Ghana/825/2024	V1A.3a.2 (C.5)	2024-06-05	MDCK1	1280	1280	640	320	1280
244016	B/Ghana/3226/2024	V1A.3a.2 (C.5)	2024-06-10	MDCK1	640	1280	640	320	640
244015	B/Ghana/3297/2024	V1A.3a.2 (C.5)	2024-06-12	MDCK1	640	640	640	320	1280
244014	B/Ghana/907/2024	V1A.3a.2 (C.5)	2024-06-17	MDCK1	640	640	320	320	640
244020	B/Ghana/729/2024	V1A.3a.2 (C.5.1)	2024-05-20	MDCK1	640	640	320	320	640
244019	B/Ghana/828/2024	V1A.3a.2 (C.5.1)	2024-06-04	MDCK1	640	640	640	320	1280
244013	B/Ghana/3427/2024	V1A.3a.2 (C.5.6)	2024-06-19	MDCK1	640	640	640	320	1280
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done						Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25			
						Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25			
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)									

Table BV-10. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-08-07

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 5741	B/Austria 1359417/21 MDCK NIB F01/21	B/Austria 1359417/21 Egg G141 F40/21	B/Austria 1359417/21 Egg G141R F44/21	B/Stock 3/22 MDCK F28/22
					Passage	Ferret number			
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/60/2008		V1A	2008-08-04	E4/E3	2560	40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	320	2560	640
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	640	640	320	320	1280
TEST VIRUSES									
244097	B/Moramanga/01434/2024	V1A.3a.2 (C.2)	2024-05-08	MDCK1	640	1280	640	640	1280
244096	B/Ambositra/01599/2024	V1A.3a.2 (C.2)	2024-05-22	MDCK1	320	640	320	320	640
244095	B/Antananarivo/01672/2024	V1A.3a.2 (C.2)	2024-06-04	MDCK1	320	640	320	320	640
244092	B/Ambositra/01883/2024	V1A.3a.2 (C.2)	2024-06-17	MDCK1	640	1280	640	640	1280
244091	B/Ambositra/01884/2024	V1A.3a.2 (C.2)	2024-06-17	MDCK1	640	1280	640	320	1280
244089	B/Fianarantsoa/01887/2024	V1A.3a.2 (C.2)	2024-06-18	MDCK1	640	640	320	320	640
244086	B/Antananarivo/01940/2024	V1A.3a.2 (C.2)	2024-06-24	MDCK1	640	1280	640	320	1280
244085	B/Antananarivo/02029/2024	V1A.3a.2 (C.2)	2024-07-01	MDCK1	320	640	640	320	1280
244083	B/Antananarivo/02054/2024	V1A.3a.2 (C.2)	2024-07-04	MDCK1	640	640	320	320	640
244082	B/Antananarivo/02078/2024	V1A.3a.2 (C.2)	2024-07-05	MDCK1	640	640	320	320	640
244080	B/Antananarivo/02104/2024	V1A.3a.2 (C.2)	2024-07-07	MDCK1	640	640	320	320	640
244079	B/Antananarivo/02109/2024	V1A.3a.2 (C.2)	2024-07-07	MDCK1	640	1280	640	320	1280
244078	B/Antananarivo/50010/2024	V1A.3a.2 (C.2)	2024-07-10	MDCK1	640	1280	640	640	1280
244098	B/Fianarantsoa/02052/2024	V1A.3a.2 (C.2)		MDCK1	640	640	640	320	640
244088	B/Antsirabe/01917/2024	V1A.3a.2 (C.5)	2024-06-19	MDCK1	640	640	320	320	1280
244087	B/Fianarantsoa/01905/2024	V1A.3a.2 (C.5)	2024-06-20	MDCK1	640	640	320	320	640
244084	B/Antananarivo/02041/2024	V1A.3a.2 (C.5.7)	2024-07-04	MDCK1	640	1280	640	320	1280

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

Vaccine	Vaccine
SH 2022	SH 2022
NH 2022-23	NH 2022-23
SH 2023	SH 2023
NH 2023-24	NH 2023-24
SH 2024	SH 2024
NH 2024-25	NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table BV-11. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-08-13

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg Sh 539, 540, 543, 571, 5741	B/Austria 1359417/21 MDCK NIB F01/21	B/Austria 1359417/21 Egg G141 F40/21	B/Austria 1359417/21 Egg G141R F44/21	B/Stock 3/22 MDCK F28/22
					Passage	Ferret number			
						V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/60/2008		V1A	2008-08-04	E4/E3	2560	<40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	320	640
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	640	1280	640	320	1280
TEST VIRUSES									
244093	B/Antananarivo/01779/2024	V1A.3a.2 (C.2)	2024-06-17	MDCK2	640	1280	640	640	1280
244170	B/England/57/2024	V1A.3a.2 (C.5)	2024-02-12	SIAT1/MDCK1	640	1280	640	320	1280
244177	B/England/39/2024	V1A.3a.2 (C.5.1)	2024-01-20	SIAT1/MDCK1	640	1280	640	320	1280
244176	B/England/48/2024	V1A.3a.2 (C.5.1)	2024-02-02	MDCK1/MDCK1	640	640	320	320	1280
244174	B/England/45/2024	V1A.3a.2 (C.5.1)	2024-02-05	SIAT1/MDCK1	640	1280	640	320	640
244171	B/England/55/2024	V1A.3a.2 (C.5.1)	2024-02-06	MDCK1/MDCK1	640	1280	640	320	1280
244198	B/Hungary/195/2024	V1A.3a.2 (C.5.1)	2024-02-13	MDCK1/MDCK1	640	640	320	320	1280
244175	B/England/44/2024	V1A.3a.2 (C.5.7)	2024-02-05	SIAT1/MDCK1	640	1280	640	320	1280
244173	B/England/51/2024	V1A.3a.2 (C.5.7)	2024-02-06	MDCK1/MDCK1	320	640	320	320	640
244172	B/England/52/2024	V1A.3a.2 (C.5.7)	2024-02-06	SIAT1/MDCK1	320	640	320	320	640
244169	B/England/58/2024	V1A.3a.2 (C.5.7)	2024-02-15	MDCK1/MDCK1	320	640	320	320	640
244197	B/Hungary/225/2024	V1A.3a.2 (C.5.7)	2024-02-23	MDCK1/MDCK1	640	1280	640	640	1280
< relates to the lowest dilution of antiserum used 1 hyperimmune sheep serum; ND = Not Done						Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25		Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24 SH 2024 NH 2024-25	
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)									

Table BV-12. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-09-04

Viruses	Other information	Haemagglutination inhibition titre									
		Post-infection ferret antiserum									
		Passage	B/Bris/60/09	B/Austria/13594/17/21	B/Austria/13594/17/21	B/Austria/13594/17/21	B/Stock/22				
			Egg	MDCK	Egg	Egg	Egg	MDCK			
		Sh 835, 840, 843, 844, 875, 871, 8741	NIB F01/21	F40/21	F44/21	F28/22					
		Genetic group	Collection Date	Genetic group	VIA	VIA.3a.2 (C)	VIA.3a.2 (C)	VIA.3a.2 (C)	VIA.3a.2 (C.8)	VIA.3a.2 (C.8)	
REFERENCE VIRUSES											
B/Brisbane/60/2009		VIA	2008-06-04	EAE3	2500	40	<40	<40	<40	<40	
B/Austria/13594/17/2021	A127T, P144L, K302R (S141)	VIA.3a.2 (C)	2021-01-09	SIAT1/MDCK4	640	1280	640	320	640	640	
B/Austria/13594/17/2021		VIA.3a.2 (C)	2021-01-09	E3E4	320	1280	640	640	640	640	
B/Austria/13594/17/2021	A127T, P144L, K302R (G14R)	VIA.3a.2 (C)	2021-01-09	E3E5	640	1280	640	640	2560	1280	
B/Stockholm/3/2022	D129G, E183K, D187E	VIA.3a.2 (C.8)	2022-03-22	SIAT1/MDCK3	640	1280	640	320	1280	1280	
TEST VIRUSES											
244253 B/Cameroon/275/2024		VIA.3a.2 (C.2)	2024-01-08	MDCK1	1280	1280	640	640	1280	1280	
244250 B/Cameroon/143/2024		VIA.3a.2 (C.8)	2024-02-01	MDCK1	640	640	640	320	1280	1280	
244248 B/Cameroon/181/2024		VIA.3a.2 (C.8)	2024-02-26	MDCK1	1280	1280	640	640	640	1280	
244247 B/Cameroon/1879/2024		VIA.3a.2 (C.8)	2024-03-02	MDCK1	1280	1280	640	640	640	1280	
244666 B/Lorraine/95214/2024		VIA.3a.2 (C.8)	2024-03-15	C1/MDCK1	1280	1280	640	640	640	1280	
244602 B/Lisboa/81/2024		VIA.3a.2 (C.8)	2024-03-19	MDCK1/MDCK1	1280	1280	640	640	640	1280	
244308 B/SaudiArabia/9276/2024		VIA.3a.2 (C.8)	2024-05-21	MDCK1	640	640	320	160	320	640	
244307 B/SaudiArabia/11846/2024		VIA.3a.2 (C.8)	2024-06-05	MDCK1	320	640	320	160	640	640	
244655 B/Ile_de_France/94275/2024		VIA.3a.2 (C.8)	2024-03-25	C1/MDCK1	640	1280	640	320	1280	1280	
244653 B/Ile_de_France/94214/2024		VIA.3a.2 (C.1)	2024-04-19	C1/MDCK1	320	640	320	160	640	640	
244651 B/Ile_de_France/95050/2024		VIA.3a.2 (C.1)	2024-04-19	C1/MDCK1	640	640	320	160	640	640	
244647 B/Ile_de_France/95041/2024		VIA.3a.2 (C.1)	2024-04-19	C1/MDCK1	640	1280	640	320	1280	1280	
244499 B/Lisboa/80/2024		VIA.3a.2 (C.1)	2024-05-24	MDCK1/MDCK1	640	1280	640	640	640	1280	
244490 B/Lisboa/70/2024		VIA.3a.2 (C.1)	2024-06-23	MDCK1/MDCK1	640	1280	640	640	640	1280	
244488 B/Lisboa/79/2024		VIA.3a.2 (C.1)	2024-06-24	MDCK1/MDCK1	640	1280	640	640	640	1280	
244486 B/Lisboa/80/2024		VIA.3a.2 (C.1)	2024-06-24	MDCK1/MDCK1	640	640	320	640	640	1280	
244416 B/Lyon/RELAB24.11.1/2024		VIA.3a.2 (C.1)	2024-06-27	MDCK2/MDCK1	1280	2560	1280	1280	640	2560	
244414 B/Lyon/CHUR24.213.45/2024		VIA.3a.2 (C.1)	2024-07-06	MDCK3/MDCK1	1280	1280	640	640	640	2560	
244348 B/Spain/2801/2024		VIA.3a.2 (C.1)	2024-07-15	MDCK1	640	1280	640	640	640	1280	
244345 B/Lebanon/1916/2024		VIA.3a.2 (C.6)	2024-04-02	MDCK1	640	640	320	320	640	640	
244344 B/Lebanon/1917/2024		VIA.3a.2 (C.6)	2024-04-02	MDCK1	640	1280	640	320	1280	1280	
244359 B/Spain/2540/2024		VIA.3a.2 (C.6)	2024-05-08	MDCK1	640	1280	640	640	640	1280	
244496 B/Lisboa/76/2024		VIA.3a.2 (C.6)	2024-06-31	MDCK1/MDCK1	640	640	640	320	1280	1280	
244418 B/Lyon/CHUR24.181.49/2024		VIA.3a.2 (C.6)	2024-06-05	MDCK2/MDCK1	640	1280	640	640	640	1280	
244306 B/SaudiArabia/9899/2024		VIA.3a.2 (C.6)	2024-06-08	MDCK1	640	1280	640	640	640	1280	
244305 B/SaudiArabia/7963/2024		VIA.3a.2 (C.6)	2024-06-10	MDCK1	640	1280	640	320	1280	1280	
244304 B/SaudiArabia/72899/2024		VIA.3a.2 (C.6)	2024-06-12	MDCK1	640	1280	640	640	640	1280	
244604 B/Lisboa/76/2024		VIA.3a.2 (C.6)	2024-06-15	MDCK1/MDCK1	640	640	320	320	1280	1280	
244413 B/Clermont-F./NM0524.3.92/2024		VIA.3a.2 (C.6)	2024-07-09	MDCK2/MDCK1	640	1280	640	640	640	2560	
244688 B/Pays_de_Loire/3062/2024		VIA.3a.2 (C.7)	2024-02-26	C1/MDCK1	640	1280	640	320	640	640	
244649 B/Ile_de_France/99949/2024		VIA.3a.2 (C.7)	2024-04-19	C1/MDCK1	640	1280	640	320	1280	1280	
244645 B/Ile_de_France/95186/2024		VIA.3a.2 (C.7)	2024-04-19	C1/MDCK1	640	640	320	320	640	640	
244643 B/Ile_de_France/95250/2024		VIA.3a.2 (C.7)	2024-04-19	C1/MDCK1	320	640	320	320	640	640	
244641 B/Ile_de_France/94958/2024		VIA.3a.2 (C.7)	2024-05-06	C1/MDCK1	640	1280	640	320	1280	1280	
244358 B/Spain/2541/2024		VIA.3a.2 (C.7)	2024-05-08	MDCK1	640	1280	640	640	640	1280	
244356 B/Spain/2587/2024		VIA.3a.2 (C.7)	2024-05-09	MDCK1	640	1280	640	320	1280	1280	
244355 B/Spain/2540/2024		VIA.3a.2 (C.7)	2024-05-10	MDCK1	1280	1280	640	640	640	1280	
244500 B/Lisboa/60/2024		VIA.3a.2 (C.7)	2024-05-18	MDCK1/MDCK1	1280	1280	640	640	640	2560	
244354 B/Spain/2580/2024		VIA.3a.2 (C.7)	2024-05-23	MDCK1	640	1280	640	640	640	1280	
244498 B/Lisboa/62/2024		VIA.3a.2 (C.7)	2024-05-24	MDCK1/MDCK1	1280	1280	640	640	640	1280	
244353 B/Spain/2582/2024		VIA.3a.2 (C.7)	2024-05-26	MDCK1	640	1280	640	320	1280	1280	
244352 B/Spain/2584/2024		VIA.3a.2 (C.7)	2024-06-01	MDCK1	640	1280	640	640	640	1280	
244351 B/Spain/2585/2024		VIA.3a.2 (C.7)	2024-06-01	MDCK1	1280	1280	640	640	640	2560	
244350 B/Spain/2586/2024		VIA.3a.2 (C.7)	2024-06-05	MDCK1	1280	1280	640	640	640	2560	
244639 B/Ile_de_France/95049/2024		VIA.3a.2 (C.7)	2024-06-10	C1/MDCK1	640	640	320	320	640	640	
244602 B/Lisboa/77/2024		VIA.3a.2 (C.7)	2024-06-17	MDCK1/MDCK1	640	1280	640	640	640	1280	
244572 B/HongKong/193/2024		VIA.3a.2 (C.7)	2024-06-22	MDCK1/MDCK1	1280	1280	640	640	640	1280	
244417 B/Grenoble/24.3.73/2024		VIA.3a.2 (C.7)	2024-06-23	MDCK2/MDCK1	640	2560	640	640	640	2560	
244570 B/HongKong/196/2024		VIA.3a.2 (C.7)	2024-07-03	MDCK1/MDCK1	1280	1280	640	640	640	1280	
244568 B/HongKong/196/2024		VIA.3a.2 (C.7)	2024-07-04	MDCK1/MDCK1	1280	1280	640	640	640	1280	
244415 B/Si-Estienne/M0524.3.95/2024		VIA.3a.2 (C.7)	2024-07-06	MDCK2/MDCK1	640	1280	640	640	640	1280	
244566 B/HongKong/197/2024		VIA.3a.2 (C.7)	2024-07-10	MDCK2/MDCK1	1280	1280	640	640	640	1280	
244349 B/Spain/2589/2024		VIA.3a.2 (C.7)	2024-07-10	MDCK1	640	640	320	320	640	640	
244654 B/HongKong/198/2024		VIA.3a.2 (C.7)	2024-07-21	MDCK1/MDCK1	640	1280	640	320	1280	1280	
244652 B/HongKong/199/2024		VIA.3a.2 (C.7)	2024-07-22	MDCK1/MDCK1	1280	1280	640	640	640	2560	

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not Done

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Vaccine
 SH 2022-23
 SH 2023
 NH 2023-24
 SH 2024
 NH 2024-25

Vaccine
 SH 2022
 SH 2023
 NH 2023-24
 SH 2024
 NH 2024-25

Table BV-13. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-09-11

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
					Post-infection ferret antiserum					
					B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 5741	B/Austria 1359417/21 MDCK NIB F01/21	B/Austria 1359417/21 Egg G141 F40/21	B/Austria 1359417/21 Egg G141R F44/21	B/Stock 3/22 MDCK F28/22	
					Passage					
					Ferret number					
						V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES										
B/Brisbane/60/2008		V1A	2008-08-04	E4/E3	2560	40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	640	1280	640	320	640	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	320	1280	640	320	640	
B/Austria/1359417/2021	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640	
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	640	320	640	
TEST VIRUSES										
244813	B/Estonia/179145/2024	V1A.3a.2 (C.5)	2024-04-03	MDCK1	320	640	320	160	640	
244580	B/HongKong/190/2024	V1A.3a.2 (C.5)	2024-06-14	MDCK 1/MDCK1	640	640	320	160	640	
244574	B/HongKong/192/2024	V1A.3a.2 (C.5)	2024-06-21	MDCK 1/MDCK1	640	1280	640	320	640	
244864	B/Luxembourg/3510796/2024	V1A.3a.2 (C.5.1)	2024-04-25	MDCK1	320	640	320	320	640	
244622	B/Croatia/3095/2024	V1A.3a.2 (C.5.1)	2024-06-10	MDCK1	640	640	320	320	640	
244625	B/Croatia/1694/2024	V1A.3a.2 (C.5.6)	2024-03-08	MDCK1	640	640	320	320	1280	
244867	B/Luxembourg/9513091/2024	V1A.3a.2 (C.5.7)	2024-03-05	MDCK1	320	640	320	320	640	
244812	B/Estonia/179138/2024	V1A.3a.2 (C.5.7)	2024-04-05	MDCK1	320	640	320	320	640	
244811	B/Estonia/179136/2024	V1A.3a.2 (C.5.7)	2024-04-07	MDCK1	320	640	320	320	640	
244809	B/Estonia/178996/2024	V1A.3a.2 (C.5.7)	2024-04-11	MDCK1	640	640	320	320	640	
244624	B/Croatia/2798/2024	V1A.3a.2 (C.5.7)	2024-05-17	MDCK1	320	640	320	160	640	
244623	B/Croatia/2825/2024	V1A.3a.2 (C.5.7)	2024-05-20	MDCK1	640	1280	640	320	1280	
244578	B/HongKong/191/2024	V1A.3a.2 (C.5.7)	2024-06-15	MDCK 1/MDCK1	640	640	320	320	640	
244576	B/HongKong/194/2024	V1A.3a.2 (C.5.7)	2024-06-20	MDCK 1/MDCK1	640	1280	640	640	1280	

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24
SH 2024
NH 2024-25

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24
SH 2024
NH 2024-25

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Fold-reduction table: B/Victoria

Reference Virus	Clade	<4-fold difference		4-fold difference		>4-fold difference		Total
		Number	Percentage	Number	Percentage	Number	Percentage	
B/Austria/1359417/2021 Cell	V1A.3a.2 (C)	266	98.9	3	1.1	0	0	269
B/Austria/1359417/2021 Egg G141	V1A.3a.2 (C)	263	97.8	5	1.8	1	0.4	269
B/Austria/1359417/2021 Egg G141R	V1A.3a.2 (C)	1	0.4	58	21.5	210	78.1	269
B/Stockholm/3/2022 Cell	V1A.3a.2 (C.5)	267	99.3	2	0.7	0	0	269

B/Victoria: Egg isolates

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Virus	Status	Genetic group (subclade) [Additional subst.]	Comment
B/Catalonia/3514402NS/2023	BX-129 (1-way fail)	V1A.3a.2 (C.5)	Egg adaptation: (no change)
B/Catalonia/2279261NS/2023	BX-123 (1-way pass) BX-123A (1-way fail)	V1A.3a.2 (C.5.1)	Egg adaptation: (no change)
B/Netherlands/10335/2023	BX-127 (1-way pass, 2-way fail)	V1A.3a.2 (C.2)	Egg adaptation: (no change)
B/Norway/5216/2023	available to send out	V1A.3a.2 (C.3)	Egg adaptation: (no change)
B/Norway/3749/2023	available to send out	V1A.3a.2 (C.3)	Egg adaptation: (no change)
B/Switzerland/53639/2023	available to send out	V1A.3a.2 (C.3)	Egg adaptation: (no change)
B/Switzerland/329/2024	sent out recently	V1A.3a.2 (C.5.6) [T199A]	Egg adaptation: (G141R)

B/Victoria: Reagents and references

Virus	Genetic group	Virus passage	Ferret ID
B/Brisbane/60/2008	V1A	E4/E3	sheep pool
B/Austria/1359417/2021	V1A.3a.2 (C)	SIAT1/MDCK4	NIB F01/21
B/Austria/1359417/2021 G141	V1A.3a.2 (C)	E3/E4	F40/21
B/Austria/1359417/2021 G141R	V1A.3a.2 (C)	E3/E5	F44/21
B/Stockholm/3/2022	V1A.3a.2 (C.5)	SIAT1/MDCK3	F28/22

Summary: B/Victoria

Genetic analyses

Clade V1A.3a.2 viruses characterised by substitutions A127T, P144L, N150K, G184E, N197D (-CHO), K203R and R279K (B/Austria/1359417/2021, subclade C) predominated since February 2023 in geographic regions where B/Victoria-lineage viruses were detected.

During this reporting period, only a minority of B/Victoria viruses were detected and characterised in Europe. Within V1A.3a.2, subclade C.5 with D197E represents the majority of influenza B viruses characterised during the current influenza season. The most frequent subclades observed within C.5 are: C.5.1 with E183K represented by B/Catalonia/2279261NS/2023, as the dominant subclade predominating in the Americas, Portugal, Ukraine, New Zealand and detected in several other countries in minor proportions; C.5.6 (B/Norway/08717/2023) with D129N predominating in South East Asia, Middle East, South Africa and some European countries; C.5.7 with E183K and E128G predominating in China, Japan, Australia, Russian Federation, Mongolia and some countries in Europe. Minor subclades such as C.5.4 (B/Slovenia/924/2023) with V117I, E128K, A154T and K326R were detected in Chile and the US, and C.5.5 (B/Paraguay/2102/2023) with R80G, E184K were detected in North America in very low proportions.

Outside of C.5 viruses, other subclades such as C.2 (T182A, D197E, B/Netherlands/10335/2023) and C.3 (E128K, A154E, S208P, B/Norway/5216/2023) were detected in low frequency in West Africa and Madagascar earlier during the season.

No Clade V1A.3 viruses were detected since 1st February 2023.

No B/Yamagata lineage viruses have been detected since March 2020.

Antigenic analyses

All V1A.3a.2 viruses tested were well-recognised by antisera raised against B/Austria/1359417/2021 and -like viruses.

Susceptibility to antivirals

Antiviral (NAI) testing of isolates with collection dates after 2024-01-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2024									
February	35	7	31		18		0		91
March	25		31		24		0		80
April	12	2	10		26		0		50
May	3	3	1		1		0		8
June	1		4		1		0		6
July	0		0		1		0		1
Totals	76	12	77	0	71	0	0	0	236

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-09-13

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Albania/302024/2024	H1pdm	05-Feb-24	435.79	89.08	H275Y
A/Netherlands/10463/2024	H1pdm	12-Feb-24	11.51		I223V+S247N
A/Netherlands/10481/2024	H1pdm	12-Feb-24	11.14		I223V+S247N
A/Parma/45/2024	H1pdm	16-Feb-24			Q136K
A/Torino/294/2024	H1pdm	02-Apr-24	574.52		H275Y
A/Torino/295/2024	H1pdm	02-Apr-24	690.52		H275Y
A/Iceland/13604/2024	H1pdm	01-May-24	3196.91		H275Y +S247N
A/Iceland/13962/2024	H1pdm	04-May-24	3271.52		H275Y +S247N
A/Iceland/13991/2024	H1pdm	05-May-24	3379.19		H275Y +S247N
A/Arad/565483/2024	H1pdm	05/02/2024	3127.07		H275Y +S247N
A/Netherlands/10529/2024	H1pdm	22/02/2024	11.53		S247N
A/Lebanon/250/2024	H1pdm	27/02/2024	412.9		H275Y

Antiviral (Bal) testing of isolates with collection dates after 2024-01-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2024									
February	37		31		17				85
March	23		30		18				71
April	7		9		18				34
May	4		1		1				6
June			3						
July									0
Totals	71	0	74	0	54	0	0	0	199

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-09-13

Virus name	Subtype/ Lineage	Collection date	Fold difference	NA substitution
			Baloxavir	

[§] Clinical sample not available

Influenza viruses received at the WIC with collection dates from 1st February 2024 were assessed for phenotypic and/or genotypic susceptibility to antivirals. Of these, 88 A/H1N1, 77 A/H3N2 and 71 B/Victoria viruses were phenotypically assessed against oseltamivir and zanamivir. Most viruses showed Normal Inhibition (NI) by both NAIs except for eight A/H1N1 showing HRI with substitutions H275Y alone or with S247N, three A/H1N1 showing borderline RI with substitutions I223V and/or S247N and one virus showing HRI to zanamivir with substitution Q136K.

Phenotypic testing for susceptibility to baloxavir marboxil was performed for 71 A/H1N1, 74 A/H3N2 viruses and 54 B/Victoria viruses, with all of them showing Normal Inhibition.

Genotypic assessment of 423 A/H1N1, 448 A/H3N2 and 333 B/Victoria neuraminidase (NA) gene sequences found some viruses with markers associated with reduced susceptibility to NAI, in addition to those described above: eight A/H1N1 viruses with substitution H275Y, 19 A/H1N1 with S247N, two A/H1N1 viruses with I223V, two A/H1N1 with I223V+S247N (double mutants), one A/H3N2 virus with substitution E119V/E (NI) and one B/Victoria virus with substitution R150K (virus isolation and phenotypic tests in process).

For 379 A/H1N1, 447 A/H3N2 and 327 B/Victoria viruses where PA gene sequencing was successful, no markers associated with reduced inhibition by baloxavir marboxil were identified, except for one A/H1N1 virus with substitution I38T, one A/H3N2 and one B/Victoria virus, both with substitution I38V (virus isolation and phenotypic test in process).